

Chiral Color Ice: Exact Local Handedness Constraints and Möbius Zero Modes in Frustrated Magnets

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Local constraints govern the low-energy physics of frustrated matter, but familiar ice-type rules constrain flux-like quantities and are insensitive to handedness. Here we show that handedness itself can be imposed as an exact local quantum constraint without selecting an axis in spin space. We construct positive-semidefinite, $SU(2)$ -invariant parent Hamiltonians whose complete zero-energy space on a tetrahedron has a prescribed chirality sign, rather than selecting a particular chiral wave function. For spin $1/2$ the local term is a rank-one projector onto a chiral tetrahedral singlet, while for arbitrary spin it factorizes as $\mathcal{B}^\dagger \mathcal{B}$ through a singlet-annihilation operator, with a completely characterized kernel given by the span of the globally rotated chiral color-ice states. For coherent states, the same zero-energy condition becomes an S -independent nonlinear constraint in which three spin directions determine the fourth through a Möbius transformation; compositions of these maps define constraint holonomies on extended lattices. Connecting the same local constraint in different ways produces qualitatively different collective regimes: corner-sharing lattices retain exponentially large quantum ground-state kernels, with rigorous lower bounds already exceeding conventional ice benchmarks; edge-sharing lattices support subdimensional plane or line zero modes; while triangular constructions suppress nonuniform coherent deformations and contain the complete Anderson tower of tetrahedral magnetic order at exactly zero energy. Two inequivalent triangular coverings further show that harmonic zero-mode counting does not determine the size of the quantum kernel. These results establish a tractable setting in which local handedness, nonlinear constraint geometry, and quantum degeneracy can be disentangled and related directly to the connectivity of the constraint network.

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I. INTRODUCTION

Frustrated magnets show how simple local rules can generate collective behavior far richer than conventional magnetic order. In these systems the low-energy physics is organized not around a unique ordered configuration but around an extensive manifold of states selected by a local constraint. Spin ice is the paradigm: the two-in–two-out rule on each tetrahedron of the pyrochlore lattice produces a macroscopically degenerate manifold with power-law correlations, an emergent gauge field, and effective magnetic monopole excitations [1, 2]. Related constraint counting underlies much of the modern understanding of kagome and pyrochlore antiferromagnets, where corner-sharing motifs impose local conditions that remain visible in the long-distance physics [3, 4]. Conventional ice-type rules, however, are achiral: the constraint fixes a flux-like, time-reversal-even quantity, so the constrained manifold is invariant under time reversal and handedness plays no role. Chiral ice-type constructions have so far relied on Ising-like variables, external fields, or purely classical interactions [5–7], and chiral correlations can also emerge spontaneously on a quantum ice background [8]. Whether an intrinsically *chiral* local constraint—one that selects a handedness on every tetrahedron—can be formulated exactly and enforced by a local, spin-rotation-invariant quantum Hamiltonian has remained an open question.

Chirality is the natural route to time-reversal symmetry breaking without magnetic order. Its local quantum measure is the scalar spin chirality

$$\hat{\chi}_{ijk} = \hat{\mathbf{S}}_i \cdot (\hat{\mathbf{S}}_j \times \hat{\mathbf{S}}_k), \quad (1.1)$$

defined on an ordered triple (i, j, k) of sites. This operator is invariant under global spin rotations but changes sign under time reversal and under odd permutations of the three sites: it detects the handedness of a non-coplanar spin triad without singling out any direction in spin space. Scalar chirality lies at the heart of the theory of chiral spin states and chiral spin liquids, from the Kalmeyer–Laughlin wave function to the Wen–Wilczek–Zee order parameter and its realizations in frustrated and itinerant magnets [9–13]. Exact parent Hamiltonians are known for individual chiral wave functions—for the Kalmeyer–Laughlin state and its lattice descendants [14–16]—and explicit scalar-chirality interactions are known

to stabilize chiral spin-liquid phases in kagome Mott insulators [17]. In all of these settings, however, chirality characterizes a *single* correlated ground state, and the parent Hamiltonians that pin it exactly require long-ranged or finely structured multispin couplings [18]. Here we pursue a complementary question: can scalar chirality be built into the very *constraint* that defines an extensively degenerate manifold—in the way the ice rule builds in a local flux condition—using only a local interaction on a single tetrahedron?

The minimal object that carries chirality without magnetization is a regular tetrahedral frame in spin space. As illustrated in Fig. 1, four spins pointing from the center to the vertices of a regular tetrahedron sum to zero and single out no axis; instead they define a genuinely three-dimensional frame whose four oriented faces all carry a nonzero scalar chirality. The chirality changes sign under time reversal and under any improper transformation that reverses the orientation of the frame, so the two handednesses of the tetrahedral configuration are related by time reversal but cannot be connected by any proper global spin rotation. Tetrahedral (“all-in–all-out” in spin space) configurations of this kind appear as regular magnetic orders on triangular and pyrochlore geometries [19–21], as the cyclic phase of spin-2 condensates [22], and, most importantly for our purposes, as local building blocks of the *color ice* manifold: assigning a color A, B, C, D to each of the four tetrahedral spin directions, the classical ground states of the bilinear-biquadratic pyrochlore antiferromagnet are exactly the four-colorings of the lattice in which every tetrahedron contains each color once [23]. Recently it was shown that a purely chiral classical interaction on the pyrochlore lattice selects the chiral subset of these colorings—those with a fixed handedness on every tetrahedron—realizing a classical chiral spin liquid [7]. We refer to this constrained manifold as *chiral color ice* (CCI). It combines a discrete color constraint, extensive degeneracy, and local time-reversal symmetry breaking, and it differs in an essential way from spin ice: the colors are not Ising labels but the legs of a chiral frame in spin space.

These observations raise a sharp question. Can one construct a *local quantum* spin Hamiltonian for which the chiral color-ice states are exact zero-energy ground states? The problem is more subtle than it may appear. In the absence of spin anisotropy, the constraint must be enforced covariantly: the Hamiltonian cannot pin a particular orientation of the tetrahedral frame, so its local kernel must contain the twelve product states of one chiral color sector *together with all their global $SU(2)$ rotations*. The task is thus to build a local, positive-semidefinite, spin-rotation-invariant operator that distinguishes a handedness—a time-reversal-odd property—without introducing any spin-space axis.

Parent-Hamiltonian constructions are the natural tool for this purpose. They have produced some of the sharpest exact statements in quantum many-body physics, from the Majumdar–Ghosh and Klein points to

the AKLT models and the pseudopotential Hamiltonians of the fractional quantum Hall effect, and they continue to organize the search for exactly solvable points in frustrated and topological matter [24–31]. Particularly close in spirit are the coloring-based constructions on kagome and related triangular-motif lattices, where special spin-1/2 Hamiltonians possess macroscopically degenerate ground-state manifolds spanned by 120° three-coloring states [32–36], with an extensive entropy famously computed by Baxter [37]; in parallel work, fully packed valence-bond loop manifolds have been obtained as the exact kernels of local spin-1 models on the same checkerboard and pyrochlore geometries considered here [38]. Those manifolds, however, are achiral—and the coloring ones are built from coplanar states and exist only at points of XXZ *anisotropy*, which singles out the plane of the colors in spin space. A parent-Hamiltonian framework for finite-color, noncoplanar, time-reversal-breaking constraints—enforced without any spin-space anisotropy—has not been available. Supplying it is the purpose of this paper.

Our construction starts from the chiral color-ice configurations on a single tetrahedron, promotes the classical directions to spin- S coherent states, and identifies the local subspace spanned by the allowed product states and their global $SU(2)$ rotations. The orthogonal complement of this subspace defines a local positive-semidefinite parent term, and summing it over the tetrahedra of a lattice yields a frustration-free Hamiltonian whose zero-energy kernel contains every chiral color-ice state. The spin-1/2 case exposes the mechanism with particular clarity: on one tetrahedron the $SU(2)$ -rotated chiral four-coloring states span fifteen of the sixteen dimensions of the Hilbert space, and the single excluded state is one of the two time-reversal-conjugate *chiral tetrahedral singlets*, distinguished only by the sign of its scalar chirality. The local parent Hamiltonian is the projector onto this singlet: it selects a handedness, yet no direction in spin space. For arbitrary S the same structure is encoded algebraically. We construct a Schwinger-boson operator $\mathcal{B}$ that annihilates precisely one chiral singlet shared among the four spins, and show that the local parent Hamiltonian takes the manifestly positive-semidefinite form $\mathcal{B}^\dagger \mathcal{B}$. In this representation each spin S is composed of $2S$ Schwinger bosons, and $\mathcal{B}$ removes one boson from each of the four sites, combined into the four-site spin-1/2 chiral singlet of the forbidden handedness. The zero-energy condition $\mathcal{B}|\Psi\rangle = 0$ therefore states that no four spin-1/2 constituents, one drawn from each site, are found in that chiral singlet—a singlet-annihilation constraint that generalizes the ice rule to a chirality-selecting rule, uniformly in S . The selection is exact at the quantum level, not merely for the classical solutions: every state of the local zero-energy kernel—entangled superpositions included—carries tetrahedral chirality of the selected sign, in the operator sense that the projected chirality is sign-semidefinite on the kernel, and the same inequality descends tetrahedron by tetrahedron to the

many-body common kernel of every lattice model built from these terms.

The kernel obtained in this way is larger than the discrete set of colorings from which the construction departs: on a single tetrahedron its dimension is exactly $(2S+1)^4 - (2S)^4$, to be compared with the twelve colorings of one chiral sector. A useful way to see why is that the constraint fixes the *sign* of the tetrahedral chirality but not its magnitude: the regular tetrahedral configuration and the fully polarized, zero-chirality configurations belong to one and the same family of coherent zero modes. What makes this enlargement tractable—indeed, what organizes the entire many-body problem—is a rigid geometric structure hidden in the constraint. In stereographic coordinates z_i on the Bloch sphere, the coherent-state zero-energy condition—a quadratic polynomial equation $p_-(z_1, z_2, z_3, z_4) = 0$ derived in Sec. V—is affine in each z_i separately, so fixing three spins determines the fourth uniquely on the Riemann sphere: the local constraint acts as a *Möbius completion rule*. The completion map $z_3 \mapsto z_4$ at fixed z_1, z_2 is an elliptic Möbius transformation of order six whose two fixed points are z_1 and z_2 themselves; equivalently, four distinct spins solve $p_- = 0$ exactly when their coordinates form an equianharmonic quadruple on the sphere. Propagating the rule through corner-sharing lattices converts ground-state counting into the combinatorics of composite Möbius transformations: around any closed loop, consistency selects the fixed points of the accumulated map, producing loop zero modes whose count we obtain in closed form for all S . Together with an exact mapping of the polarized sector onto the monomer–dimer problem of the dual (square or diamond) lattice, these modes yield rigorous exponential lower bounds on the ground-state degeneracy of the checkerboard and pyrochlore models—bounds that already exceed the Pauling and Lieb residual entropies of conventional ice. Chiral color ice is thus *more* degenerate than ice, even though its constraint is stronger than a naive count would suggest: the extra entropy is the price exacted by spin-rotation invariance, which forces the ferromagnet, its magnon descendants, and, more generally, every product state with at most one deviated spin per tetrahedron into the kernel, and places the parent Hamiltonians on the boundary of the ferromagnetic phase.

The same local term can be assembled on lattices whose tetrahedra share edges or faces rather than corners, and there the physics changes qualitatively: the overlapping constraints become restrictive enough to single out chiral four-sublattice states. This is the organizing principle of the paper. One local rule, propagated through simplex networks of increasing connectivity, produces

$$\begin{aligned} \text{corner sharing} &\longrightarrow \text{extensive degeneracy,} \\ \text{edge sharing} &\longrightarrow \text{subdimensional zero modes,} \\ \text{face sharing} &\longrightarrow \text{rigid chiral order,} \end{aligned} \quad (1.2)$$

so that the lattices treated below are not a catalogue

of examples but successive stages of a single mechanism. The relative chirality of the two inversion-related families of tetrahedra now matters decisively. With an *alternating* assignment—opposite chirality on the two families, the pattern inherited from pyrochlore color ice—edge sharing on the fcc lattice eliminates local zero modes and singles out the chiral four-sublattice coloring, up to a residual subextensive family of planar deformations; flattening fcc bilayers produces square- and honeycomb-lattice models whose residual deformations are line-like; and on the triangular lattice, where the flattened tetrahedra share faces, the ordered state is rigid. With a *uniform* assignment, by contrast, the coloring rule is frustrated—no four-coloring can satisfy all tetrahedra simultaneously—and no four-sublattice coloring survives, as we verify by exhaustive enumeration on the fcc, square, and honeycomb lattices. In the quantum spin-1/2 models on the ordered branches of these lattices, exact diagonalization shows that the zero-energy kernel contains the complete set of multiplets carrying the quantum numbers of the Anderson tower of states of the chiral four-sublattice order. At the frustration-free point these multiplets are all exactly degenerate; a degeneracy-lifting perturbation that favors the tetrahedrally ordered component would resolve them into the Anderson tower from which the symmetry-broken chiral order emerges [39], although which part of the larger kernel a given perturbation actually selects is left open here. The two triangular coverings—edge-sharing and face-sharing assemblies of the same flattened tetrahedra—provide the sharpest control on this hierarchy: they support identical harmonic zero modes around the ordered state, yet exact diagonalization finds markedly different zero-energy kernels, so harmonic zero-mode counting and quantum degeneracy are independent characteristics of the constraint network.

The models constructed here therefore serve a double purpose. In both roles they are genuinely quantum, SU(2)-symmetric spin models rather than classical vertex systems: the classical language we use throughout refers to their coherent-state (product-state) solutions, which form only part of the kernel. On corner-sharing lattices the models define exactly solvable, extensively degenerate points—chiral analogs of the ice manifold—whose product-state degeneracy is organized by Möbius geometry, in contrast with the divergence-free condition that organizes ice. On edge- and face-sharing lattices they provide exact parent Hamiltonians whose zero-energy kernels contain the chiral four-sublattice states together with the full Anderson-tower content of the corresponding order. In both cases the constraint is enforced by a local, SU(2)-invariant, time-reversal-breaking term built solely from scalar chiralities and products of Heisenberg exchanges. Its time-reversal-even multispin part is of the kind generated by ring exchange in Mott insulators [19], whereas the signed scalar-chirality part requires an explicitly time-reversal-breaking ingredient, such as orbital magnetic flux, an applied magnetic field, or circular driving [10, 40–42]; interactions of this multispin type are

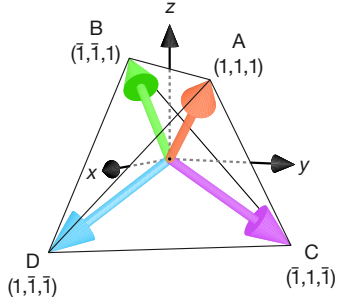


FIG. 1. The elementary motif of chiral color ice: four spins pointing from the center to the vertices of a regular tetrahedron in spin space. A color is associated with each direction, red for $\mathbf{n}^A = \frac{1}{\sqrt{3}}(1, 1, 1)$, green for $\mathbf{n}^B$, magenta for $\mathbf{n}^C$, and cyan for $\mathbf{n}^D$ of Eq. (2.1). The configuration carries no net magnetization and selects no axis in spin space, but each oriented face of the tetrahedron carries a scalar spin chirality of magnitude $4/(3\sqrt{3})$, so the frame has a definite handedness: the two enantiomeric assignments are related by time reversal and cannot be connected by any proper spin rotation.

also the ones that inverse quantum simulation methods are beginning to target [43].

The paper is organized as follows. Section II defines chiral color ice at the classical level and its quantum coherent-state counterpart. Section III constructs the local parent Hamiltonian on a tetrahedron, first for spin 1/2 and then for arbitrary S , and proves the exact chirality-sign constraint throughout its complete local kernel. Section IV develops the Schwinger-boson singlet-annihilation formulation and derives the resulting local kernel dimensions. Section V analyzes the coherent zero-mode variety of a single tetrahedron, derives the Möbius completion rule, and establishes the corresponding bounds on the tetrahedral chirality. Section VI assembles these local constraints into lattice Hamiltonians, defines the quantum common kernel and its handedness constraint, and introduces Möbius propagation and loop holonomy. Section VII establishes the extensive ground-state degeneracy on the checkerboard and pyrochlore lattices and compares it with ice. Section VIII treats the fcc, square, honeycomb, and triangular lattices, whose zero-energy kernels carry the full Anderson-tower content of chiral four-sublattice order, and closes by examining the chirality spectrum of the quantum common kernel; a linearized coherent-state analysis (Appendix G) makes the connectivity hierarchy of these constructions quantitative. Section X collects the principal results in three reference tables, one for each level of the kernel hierarchy, and Section XI discusses the emerging picture and the avenues it opens. Technical derivations, special cases, and counting arguments are collected in Appendixes A–J.

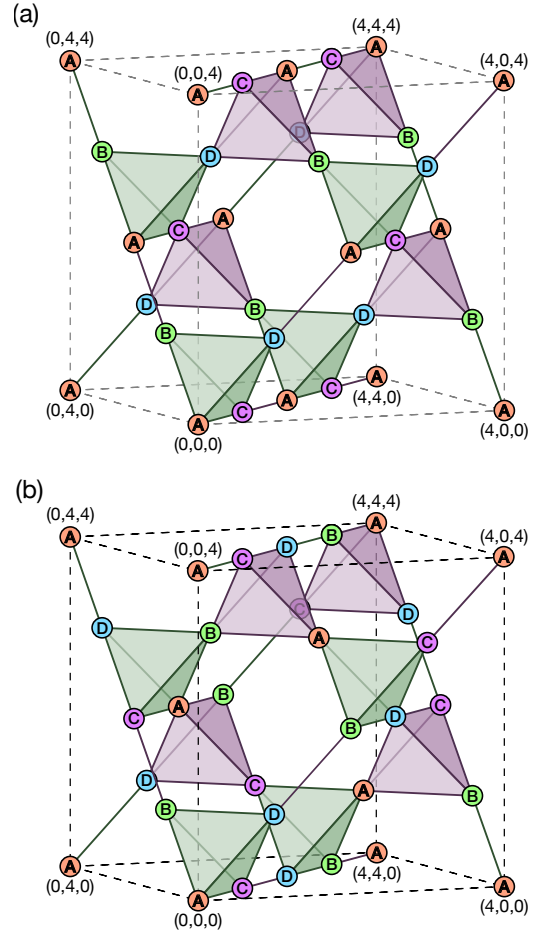


FIG. 2. Chiral color-ice states compatible with the 16-site cubic unit cell of the pyrochlore lattice. (a) The uniform state with a 4-site unit cell, in which the color is constant on each fcc sublattice. (b) An analog of the R state of Ref. [44], generated from (a) by combining the translation $(2, 2, 0)$ with the color permutation $(AD)(BC)$. Both states satisfy the same fixed-ordering CCI constraint on every tetrahedron and hence belong to the same chiral sector. In the geometric convention of Sec. II A, the tetrahedral chirality nevertheless has opposite signs on the two orientation classes of tetrahedra (green and purple), which are interchanged by spatial inversion.

II. CHIRAL COLOR ICE

We begin by characterizing the local spin configurations that will form the building blocks of chiral color ice. This section fixes the local motif—four spins pointing along the four tetrahedral directions—shows that such configurations split into two chiral sectors exchanged by time reversal, and defines chiral color ice: the colorings obtained by prescribing one sector on every tetrahedron of a lattice.

A. Color ice and its chiral sectors

A local color-ice configuration on a tetrahedron consists of four classical spins pointing from the center toward the four vertices of a regular tetrahedron in spin space (a tetrahedral state [19, 20]),

$$\begin{aligned}\mathbf{n}^A &= \frac{1}{\sqrt{3}}(1, 1, 1), & \mathbf{n}^B &= \frac{1}{\sqrt{3}}(\bar{1}, \bar{1}, 1), \\ \mathbf{n}^C &= \frac{1}{\sqrt{3}}(\bar{1}, 1, \bar{1}), & \mathbf{n}^D &= \frac{1}{\sqrt{3}}(1, \bar{1}, \bar{1}),\end{aligned}\quad (2.1)$$

as shown in Fig. 1. Assigning a distinct color to each tetrahedral direction, a global color-ice (CI) state on a lattice of tetrahedra is a four-coloring of the sites such that every tetrahedron contains each of the four colors exactly once. Throughout the paper, the labels A, B, C, D denote the four colors, with reference directions given by Eq. (2.1), and generic spin directions are written $\mathbf{n}_i$ or, in stereographic coordinates, z_i . As shown in Sec. VII B 3 and Appendix F, parts of the analysis depend only on the coloring pattern and not on the reference directions themselves; the color labels then stand for any four distinct spin directions compatible with the local constraint. On the pyrochlore lattice, this manifold is the classical ground-state manifold of the bilinear-biquadratic Heisenberg model [3, 23]. In the absence of anisotropies, global $\text{SO}(3)$ rotations generate equivalent ground states, while time reversal reverses the scalar chirality.

Because any three of the four directions in Eq. (2.1) are noncoplanar, every triangular face of a colored tetrahedron carries a nonzero scalar chirality,

$$\chi_{ijk} = \mathbf{n}_i \cdot (\mathbf{n}_j \times \mathbf{n}_k), \quad (2.2)$$

with magnitude $|\chi_{ijk}| = 4/(3\sqrt{3})$ for the normalized directions of Eq. (2.1). A convenient measure of the chirality of a tetrahedron (i, j, k, l) is the tetrahedral chirality, defined as the oriented sum of the four face chiralities,

$$\chi_{\text{tet}} = \chi_{ijk} + \chi_{ilj} + \chi_{ikl} + \chi_{jlk}. \quad (2.3)$$

Let S_4 denote the group of permutations of the four sites of a tetrahedron. Starting from the reference coloring (A, B, C, D) on the ordered sites (i, j, k, l) , the $4! = 24$ site permutations generate all possible color assignments. We denote the parity of $\sigma \in S_4$ by $\text{sgn}(\sigma) = +1$ for even permutations and $\text{sgn}(\sigma) = -1$ for odd permutations. The twelve even permutations form the alternating subgroup $A_4 \subset S_4$.

For a coloring obtained from the reference assignment by the site permutation σ , the tetrahedral chirality is

$$\chi_{\text{tet}} = -\text{sgn}(\sigma) \frac{16}{3\sqrt{3}}, \quad (2.4)$$

for the conventions of Eqs. (2.1) and (2.3). Thus, even and odd site permutations generate the two chiral sectors. Different tetrahedra of a generic CI configuration may belong to opposite chiral sectors.

We define the *chiral color-ice* (CCI) *coloring set* by fixing the chiral sector on every tetrahedron. We choose an ordering of the four sites of each tetrahedron t once and for all and assign a sign $\varepsilon_t = \pm 1$. The coloring set $\text{CCI}(\{\varepsilon_t\})$ consists of all colorings for which the site permutation $\sigma_t \in S_4$ relating the local coloring to the reference assignment satisfies

$$\text{sgn}(\sigma_t) = \varepsilon_t. \quad (2.5)$$

In this fixed-ordering convention, the tetrahedral chirality is therefore

$$\chi_{\text{tet},t} = -\varepsilon_t \frac{16}{3\sqrt{3}}. \quad (2.6)$$

The chirality constraint thus retains twelve of the $4! = 24$ local color arrangements on each tetrahedron. On the pyrochlore lattice, natural choices include the two uniform patterns, $\varepsilon_t = +1$ or $\varepsilon_t = -1$ for all t , as well as alternating patterns in which ε_t changes sign between the two orientations of tetrahedra.

Throughout the paper we quote tetrahedral chiralities in a single *geometric* convention. The permutation parity defined above refers to a fixed *site ordering*; on the pyrochlore lattice, we use the four fcc sublattice labels for this purpose. For the values of χ_{tet} , by contrast, the sites of every tetrahedron are ordered such that each face circuit entering Eq. (2.3), viewed from the center of that tetrahedron, runs clockwise. Since χ_{tet} is invariant under even permutations of the four sites, this prescription fixes its sign unambiguously, and it does so with one and the same rule for every tetrahedron. On a single tetrahedron, where no spatial embedding is specified, we use the reference site ordering.

The tetrahedra of the pyrochlore lattice fall into two orientation classes exchanged by spatial inversion—the up and down tetrahedra—which we denote $\mathcal{O}_1$ and $\mathcal{O}_2$, taking $\mathcal{O}_1$ to be the up class; the corresponding classes of the other lattices are identified below. The fixed sublattice ordering agrees with the geometric ordering on $\mathcal{O}_1$ but differs from it by an odd permutation on $\mathcal{O}_2$: inversion interchanges the two classes and reverses the view from the center, while leaving the spin directions unchanged, because spins are axial vectors. For a CCI coloring, the geometric tetrahedral chirality is therefore $-\varepsilon_t 16/(3\sqrt{3})$ for $t \in \mathcal{O}_1$ and $+\varepsilon_t 16/(3\sqrt{3})$ for $t \in \mathcal{O}_2$.

For example, in the uniform-parity pyrochlore state shown in Fig. 2(a), every tetrahedron carries the same sublattice coloring and therefore the same permutation parity. Nevertheless, the geometric tetrahedral chirality is $-16/(3\sqrt{3})$ on the up tetrahedra and $+16/(3\sqrt{3})$ on the down tetrahedra. Hence, a CCI state with uniform parity in the fixed-sublattice convention exhibits an alternating pattern of geometric chirality between the two tetrahedral orientations. This is the convention underlying the classical ground-state manifold selected by the uniform chiral interaction of Ref. [7]. The resulting pattern of geometric tetrahedral chiralities is reversed by

spatial inversion $\mathcal{I}$ and by time reversal $\mathcal{T}$ separately, but is invariant under their product $\mathcal{IT}$. Figure 2 shows two CCI states with this chirality pattern that are compatible with the 16-site cubic unit cell.

B. Quantum chiral color-ice states

We represent the classical CCI configurations quantum mechanically by site-factorized spin coherent states. On each site j , we associate a classical direction $\mathbf{n}$ with the spin- S coherent state $|\mathbf{n}_j\rangle$, defined, up to an overall phase, as the maximal-weight eigenstate

$$(\mathbf{n} \cdot \hat{\mathbf{S}}_j)|\mathbf{n}_j\rangle = S|\mathbf{n}_j\rangle. \quad (2.7)$$

Then, for a single tetrahedron with ordered sites $(1, 2, 3, 4)$, the reference CCI product state is

$$|\Psi^{ABCD}\rangle = |\mathbf{n}_1^A\rangle \otimes |\mathbf{n}_2^B\rangle \otimes |\mathbf{n}_3^C\rangle \otimes |\mathbf{n}_4^D\rangle, \quad (2.8)$$

which belongs to the $\varepsilon = +1$ chiral sector in the convention of Sec. II A. The twelve even site permutations $\sigma \in A_4$ generate the twelve local CCI product states in this sector, including the reference state (2.8).

For a global CCI coloring $\{\alpha_j\}$, we define the corresponding quantum product state as

$$|\Psi^{\{\alpha_j\}}\rangle = \bigotimes_j |\mathbf{n}_j^{\alpha_j}\rangle, \quad \alpha_j \in \{A, B, C, D\}, \quad (2.9)$$

where the coloring $\{\alpha_j\}$ satisfies the prescribed CCI constraint on every tetrahedron.

If the parent Hamiltonian is $SU(2)$ invariant, every global spin rotation of a CCI product state must also be a ground state. A global $SU(2)$ transformation rotates all spins simultaneously and induces the corresponding rigid $SO(3)$ rotation of the tetrahedral frame $\{\mathbf{n}^A, \mathbf{n}^B, \mathbf{n}^C, \mathbf{n}^D\}$, while leaving the coloring pattern $\{\alpha_j\}$ unchanged. We therefore require the parent Hamiltonian to annihilate the CCI product states (2.9) together with all their global $SU(2)$ rotations.

III. THE PARENT HAMILTONIAN ON A TETRAHEDRON

We now construct a local operator whose kernel contains the chiral color-ice states of a chosen handedness. Spin-rotation invariance strongly constrains this operator: for spin $1/2$, it fixes the local term essentially uniquely and, as we show below, suggests its natural generalization to arbitrary spin S .

A. Construction principle

The site-factorized structure of the CCI states allows us to impose the parent constraint locally on a single

tetrahedron. For fixed spin length S , we take the state $|\Psi^{ABCD}\rangle$ of Eq. (2.8) as a convenient reference for one of the two chiral sectors and define

$$\mathcal{V}^S = \text{span} \left\{ [\hat{U}^{(S)}(R)]^{\otimes 4} |\Psi^{ABCD}\rangle \mid R \in SO(3) \right\}. \quad (3.1)$$

Here $\hat{U}^{(S)}(R)$ denotes the spin- S rotation associated with R , acting identically on all four spins. For half-integer S , the two $SU(2)$ rotations corresponding to the same $R \in SO(3)$ differ by a sign in the single-spin representation. This sign disappears in the fourfold tensor product, so the operator in Eq. (3.1) is unambiguously defined.

The rotational *orbit* in Eq. (3.1)—the set of states obtained by acting with all global spin rotations on the reference state—already contains all twelve local CCI product states in the same chiral sector. Let $\hat{P}_\sigma$ denote the operator that transports the spin on site i to site $\sigma(i)$. For every even site permutation $\sigma \in A_4$, there exists a proper rotation $R_\sigma \in SO(3)$ of the tetrahedral frame such that

$$\hat{P}_\sigma |\Psi^{ABCD}\rangle = e^{i\phi_\sigma} [\hat{U}^{(S)}(R_\sigma)]^{\otimes 4} |\Psi^{ABCD}\rangle, \quad (3.2)$$

where $e^{i\phi_\sigma}$ is an irrelevant overall phase. Thus the twelve states generated by even site permutations already belong to the rotational orbit in Eq. (3.1). Odd site permutations reverse the handedness and generate the opposite chiral sector.

We then seek a positive-semidefinite local parent term $\mathcal{H}_{\text{CCI}}^{(S)}$ whose kernel contains the subspace $\mathcal{V}^S$ associated with the chosen chiral sector. The parent term for the opposite chirality is obtained by time reversal. We therefore define

$$\mathcal{H}_{\text{CCI}}^{(S), \varepsilon} = \begin{cases} \mathcal{H}_{\text{CCI}}^{(S)}, & \varepsilon = +1, \\ \mathcal{T} \mathcal{H}_{\text{CCI}}^{(S)} \mathcal{T}^{-1}, & \varepsilon = -1. \end{cases} \quad (3.3)$$

In practice, we determine $\mathcal{V}^S$ numerically by sampling global spin rotations of the reference state $|\Psi^{ABCD}\rangle$, including the twelve even site permutations, and orthogonalizing the resulting set. The rank rapidly saturates as the number of sampled rotations increases, yielding $\dim \mathcal{V}^S$.

B. Spin one-half: chiral tetrahedral singlets

The spin- $1/2$ case makes the construction particularly transparent. The local Hilbert space $\mathcal{H}^{1/2}$ has dimension $2^4 = 16$, whereas the construction above yields

$$\dim \mathcal{V}^{1/2} = 15. \quad (3.4)$$

Its orthogonal complement is therefore one dimensional. To identify the missing state, we consider the quantum tetrahedral-chirality operator $\hat{\chi}_{\text{tet}}$ corresponding to Eq. (2.3). For four spin- $1/2$ degrees of freedom, $\hat{\chi}_{\text{tet}}$ has

TABLE I. Total-spin decomposition of three subspaces on a single tetrahedron of four spins S : the fixed-chirality subspace $\mathcal{V}^S$ spanned by the globally rotated CCI product states [Eq. (3.1)], the full color-ice subspace $\mathcal{W}^S$ spanned by the globally rotated CCI product states of both chiral sectors, and the full Hilbert space $\mathcal{H}^S$. Each entry gives the multiplicity of the corresponding SU(2) multiplet with total spin S_{tet} ; weighting each entry by $2S_{\text{tet}} + 1$ gives the total dimensions in the last row. For $S_{\text{tet}} < 2S$, the multiplicities are $2S_{\text{tet}} + 1$ in $\mathcal{V}^S$ and $2(2S_{\text{tet}} + 1)$ in $\mathcal{W}^S$, in agreement with the Anderson towers of tetrahedral order derived in Appendix J.

S_{tet}	$S = \frac{1}{2}$	$S = \frac{1}{2}$	$S = \frac{1}{2}$	$S = 1$	$S = 1$	$S = 1$	$S = \frac{3}{2}$	$S = \frac{3}{2}$	$S = \frac{3}{2}$	$S = 2$	$S = 2$	$S = 2$	$S = \frac{5}{2}$	$S = \frac{5}{2}$	$S = \frac{5}{2}$
	$\mathcal{V}_{S_{\text{tet}}}^{1/2}$	$\mathcal{W}_{S_{\text{tet}}}^{1/2}$	$\mathcal{H}_{S_{\text{tet}}}^{1/2}$	$\mathcal{V}_{S_{\text{tet}}}^1$	$\mathcal{W}_{S_{\text{tet}}}^1$	$\mathcal{H}_{S_{\text{tet}}}^1$	$\mathcal{V}_{S_{\text{tet}}}^{3/2}$	$\mathcal{W}_{S_{\text{tet}}}^{3/2}$	$\mathcal{H}_{S_{\text{tet}}}^{3/2}$	$\mathcal{V}_{S_{\text{tet}}}^2$	$\mathcal{W}_{S_{\text{tet}}}^2$	$\mathcal{H}_{S_{\text{tet}}}^2$	$\mathcal{V}_{S_{\text{tet}}}^{5/2}$	$\mathcal{W}_{S_{\text{tet}}}^{5/2}$	$\mathcal{H}_{S_{\text{tet}}}^{5/2}$
0	1	2	2	1	2	3	1	2	4	1	2	5	1	2	6
1	3	3	3	3	6	6	3	6	9	3	6	12	3	6	15
2	1	1	1	5	6	6	5	10	11	5	10	16	5	10	21
3				3	3	3	7	10	10	7	14	17	7	14	24
4				1	1	1	5	6	6	9	14	15	9	18	24
5							3	3	3	7	10	10	11	18	21
6							1	1	1	5	6	6	9	14	15
7										3	3	3	7	10	10
8										1	1	1	5	6	6
9													3	3	3
10													1	1	1
Total dim.	15	16	16	65	80	81	175	240	256	369	544	625	671	1040	1296

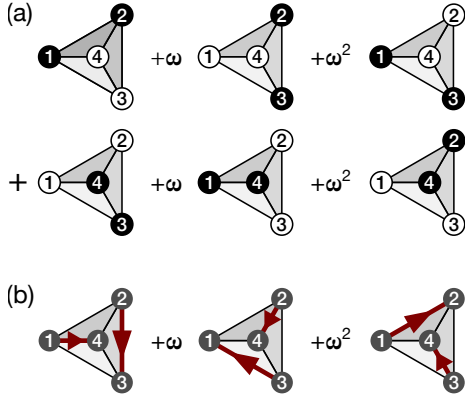


FIG. 3. The chiral tetrahedral singlet $|+\rangle$, Eq. (3.6a), shown on a tetrahedron projected as a triangle with one interior vertex. (a) Superposition of the six S^z -basis states with two up and two down spins, with relative phases 1, ω , and ω^2 . Filled and open circles denote the two spin states. (b) The same state represented as a superposition of the three oriented valence-bond coverings with relative phases 1, ω , and ω^2 [Eq. (3.11a)], up to an overall phase. An arrow from i to j denotes the oriented singlet valence bond $[ij]$.

eigenvalues 0 and $\pm\sqrt{3}$. The two nonzero eigenvalues belong to a time-reversal pair of total-spin singlets [21],

$$\hat{\chi}_{\text{tet}}|\pm\rangle = \pm\sqrt{3}|\pm\rangle, \quad (3.5)$$

where

$$|+\rangle = \frac{1}{\sqrt{6}} [|\downarrow\downarrow\uparrow\uparrow\rangle + |\uparrow\uparrow\downarrow\downarrow\rangle + \omega(|\downarrow\uparrow\uparrow\downarrow\rangle + |\uparrow\downarrow\downarrow\uparrow\rangle) + \omega^2(|\downarrow\uparrow\downarrow\uparrow\rangle + |\uparrow\downarrow\uparrow\downarrow\rangle)], \quad (3.6a)$$

$$|-\rangle = |+\rangle^*, \quad (3.6b)$$

with $\omega = e^{2\pi i/3}$. The arrows label the spin states on sites 1–4 from left to right.

From the viewpoint of site permutations, the two-dimensional singlet sector carries the $[2, 2]$ irreducible representation of S_4 , denoted E in Table IX. Upon restriction to the proper tetrahedral subgroup A_4 , this representation decomposes as

$$E \downarrow_{A_4} = E_+ \oplus E_-, \quad (3.7)$$

with $|+\rangle$ and $|-\rangle$ spanning the two complex-conjugate one-dimensional components.

The chiral singlets take a particularly transparent form in the valence-bond representation. We define a singlet on bond (ij) by

$$[ij] = -[ji] = \frac{1}{\sqrt{2}} (|\uparrow_i\downarrow_j\rangle - |\downarrow_i\uparrow_j\rangle). \quad (3.8)$$

The three valence-bond coverings satisfy

$$[14][23] + [24][31] + [34][12] = 0. \quad (3.9)$$

Let $\hat{P}_{(123)}$ denote the cyclic site permutation that transports the spins as $1 \rightarrow 2 \rightarrow 3 \rightarrow 1$. It acts on the three coverings as

$$[14][23] \mapsto [24][31] \mapsto [34][12] \mapsto [14][23]. \quad (3.10)$$

Consequently,

$$|+\rangle = \frac{\sqrt{2}}{3} e^{i\pi/6} ([14][23] + \omega[24][31] + \omega^2[34][12]), \quad (3.11a)$$

$$|-\rangle = \frac{\sqrt{2}}{3} e^{-i\pi/6} ([14][23] + \omega^2[24][31] + \omega[34][12]), \quad (3.11b)$$

which are eigenstates of $\hat{P}_{(123)}$ with eigenvalues ω^2 and ω , respectively. Thus the two chiralities are the two nontrivial cyclic modes of the three valence-bond coverings, with their handedness encoded in the relative phases $1, \omega, \omega^2$, as illustrated in Fig. 3. They provide a minimal four-site analog of the chiral spin states of Refs. [9, 10].

For any local CCI state in the same chiral sector as the reference state of Eq. (2.8), including all even site permutations and all global $SU(2)$ rotations, one finds

$$\langle + | \Psi_{\text{CCI}} \rangle = 0, \quad (3.12a)$$

$$|\langle - | \Psi_{\text{CCI}} \rangle| = \frac{\sqrt{2}}{3}. \quad (3.12b)$$

The first relation shows that the entire set of rotated fixed-chirality CCI states is orthogonal to the $\chi_{\text{tet}} = +\sqrt{3}$ singlet $|+\rangle$. Since $\dim \mathcal{V}^{1/2} = 15$, this set spans the full orthogonal complement of $|+\rangle$ in the 16-dimensional local Hilbert space. The local parent term is therefore the rank-one projector

$$\mathcal{H}_{\text{CCI}}^{(1/2)} = |+\rangle\langle +|. \quad (3.13)$$

The projector selects the chosen CCI handedness while preserving full spin-rotation invariance, since $|+\rangle$ is a total-spin singlet.

Since $\hat{\chi}_{\text{tet}}$ has eigenvalues 0 and $\pm\sqrt{3}$, and $|+\rangle$ is the unique eigenstate with eigenvalue $+\sqrt{3}$, the projector can equivalently be written as

$$\mathcal{H}_{\text{CCI}}^{(1/2)} = \frac{1}{6} (\hat{\chi}_{\text{tet}} + \sqrt{3}) \hat{\chi}_{\text{tet}}. \quad (3.14)$$

Expanding this polynomial in $\hat{\chi}_{\text{tet}}$ in terms of spin operators gives

$$\begin{aligned} \mathcal{H}_{\text{CCI}}^{(1/2)} &= \frac{1}{16} + \frac{1}{2\sqrt{3}} \sum_{\langle ijk \rangle} \hat{\mathbf{S}}_i \cdot (\hat{\mathbf{S}}_j \times \hat{\mathbf{S}}_k) \\ &\quad - \frac{1}{12} \sum_{\langle ij \rangle} \hat{\mathbf{S}}_i \cdot \hat{\mathbf{S}}_j + \frac{1}{3} \sum_{\langle ijkl \rangle} (\hat{\mathbf{S}}_i \cdot \hat{\mathbf{S}}_j) (\hat{\mathbf{S}}_k \cdot \hat{\mathbf{S}}_l). \end{aligned} \quad (3.15)$$

Here $\langle ijk \rangle$ runs over the four oriented faces of the tetrahedron with the handedness convention of Eq. (2.3), $\langle ij \rangle$ over its six bonds, and $\langle ijkl \rangle$ over the three pairings of opposite bonds, (12, 34), (13, 24), and (14, 23). The parent term thus involves only scalar chiralities, Heisenberg exchanges, and products of Heisenberg exchanges. Useful identities relating these operators for $S = 1/2$ are collected in Appendix A.

C. Arbitrary spin

We now turn to $S > 1/2$. For $S = 1$, the local Hilbert space has dimension $3^4 = 81$, whereas the globally rotated CCI product states span a 65-dimensional subspace. Repeating the construction for $S = 3/2, 2$, and

$5/2$ gives the dimensions and total-spin decompositions collected in Table I. For all spins studied, the dimensions follow the simple pattern

$$\dim \mathcal{V}^S = (2S+1)^4 - (2S)^4. \quad (3.16)$$

We show the origin of this relation for arbitrary S in Sec. IV.

Guided by the spin-1/2 parent term (3.15), we search within the same operator basis—bilinear exchanges, scalar chiralities, and products of bilinear exchanges—for a positive-semidefinite operator that annihilates $\mathcal{V}^S$. A particularly simple solution, valid for arbitrary S , is

$$\begin{aligned} \mathcal{H}_{\text{CCI}}^{(S)} &= \frac{S}{\sqrt{3}} \sum_{\langle ijk \rangle} \hat{\mathbf{S}}_i \cdot (\hat{\mathbf{S}}_j \times \hat{\mathbf{S}}_k) \\ &\quad + \frac{1}{3} \sum_{\langle ijkl \rangle} (S^2 - \hat{\mathbf{S}}_i \cdot \hat{\mathbf{S}}_j) (S^2 - \hat{\mathbf{S}}_k \cdot \hat{\mathbf{S}}_l). \end{aligned} \quad (3.17)$$

We derive the general result analytically in Appendix B. Its classical large- S limit and complete coherent zero-mode variety are discussed in Appendix C1.

D. Exact quantum handedness of the local kernel

For $S = 1/2$, Eq. (3.13) shows directly that the local parent Hamiltonian projects out the chiral singlet of the opposite handedness. The form of Eq. (3.17) allows this chirality-sign selection to be extended to arbitrary spin S and to the complete local quantum kernel.

For the reference chirality, write the local parent term as

$$\mathcal{H}_{\text{CCI}}^{(S)} = \frac{S}{\sqrt{3}} \hat{\chi}_{\text{tet}} + \frac{1}{3} \mathcal{R}, \quad (3.18)$$

where

$$\mathcal{R} = \sum_{\langle ijkl \rangle} (S^2 - \hat{\mathbf{S}}_i \cdot \hat{\mathbf{S}}_j) (S^2 - \hat{\mathbf{S}}_k \cdot \hat{\mathbf{S}}_l), \quad (3.19)$$

and the sum runs over the three pairings of opposite bonds, (12, 34), (13, 24), and (14, 23).

Each factor appearing in Eq. (3.19) is positive semidefinite. Indeed, for two spin- S sites in a sector of pair total spin $J = 0, \dots, 2S$,

$$\begin{aligned} S^2 - \hat{\mathbf{S}}_i \cdot \hat{\mathbf{S}}_j &= \frac{1}{2} [2S(2S+1) - J(J+1)] \\ &= \frac{1}{2} (2S - J)(2S + J + 1) \geq 0. \end{aligned} \quad (3.20)$$

The two factors in each term of Eq. (3.19) act on disjoint pairs of sites and therefore commute. Their product is consequently positive semidefinite, and hence

$$\mathcal{R} \succeq 0. \quad (3.21)$$

Let $|\Psi\rangle$ be any normalized state in the local zero-energy kernel, $\mathcal{H}_{\text{CCI}}^{(S)}|\Psi\rangle = 0$. Taking the expectation value of Eq. (3.18) gives

$$\frac{S}{\sqrt{3}}\langle\Psi|\hat{\chi}_{\text{tet}}|\Psi\rangle = -\frac{1}{3}\langle\Psi|\mathcal{R}|\Psi\rangle. \quad (3.22)$$

Since $\mathcal{R} \succeq 0$,

$$\langle\Psi|\hat{\chi}_{\text{tet}}|\Psi\rangle \leq 0 \quad (3.23)$$

Thus, for four spins of arbitrary length S , the complete quantum kernel of the local CCI parent Hamiltonian obeys the same chirality-sign constraint as its coherent-product states. Zero chirality is allowed, and states in the kernel need not themselves be eigenstates of $\hat{\chi}_{\text{tet}}$.

For the time-reversed parent term, $\mathcal{T}\mathcal{H}_{\text{CCI}}^{(S)}\mathcal{T}^{-1}$, the operator $\mathcal{R}$ is unchanged while $\hat{\chi}_{\text{tet}} \rightarrow -\hat{\chi}_{\text{tet}}$, and the inequality in Eq. (3.23) is therefore reversed.

IV. SINGLET-ANNIHILATION FORMULATION

In this section, we show that the parent Hamiltonian takes a particularly simple form in the Schwinger-boson representation: the local term factorizes as $\mathcal{H}_{\text{CCI}}^{(S)} = \mathcal{B}^\dagger \mathcal{B}$, making its positive-semidefinite character explicit. Here $\mathcal{B}$ is a four-site singlet-annihilation operator, so that the zero-energy condition is simply $\mathcal{B}|\Psi\rangle = 0$. This formulation also provides a direct route to the kernel's dimension and total-spin content, as well as the exact local excitation gap.

A. The operator $\mathcal{B}$

The homogeneous scaling of all the terms in Eq. (3.17) with the spin length makes the Schwinger-boson representation particularly natural, since S enters through the local boson-number constraint. Introducing two bosons a_i and b_i at each site, the spin operators are represented as

$$\hat{S}_i^x = \frac{1}{2} (a_i^\dagger b_i + b_i^\dagger a_i), \quad (4.1a)$$

$$\hat{S}_i^y = \frac{1}{2i} (a_i^\dagger b_i - b_i^\dagger a_i), \quad (4.1b)$$

$$\hat{S}_i^z = \frac{1}{2} (a_i^\dagger a_i - b_i^\dagger b_i), \quad (4.1c)$$

with the local constraint

$$a_i^\dagger a_i + b_i^\dagger b_i = 2S. \quad (4.1d)$$

We introduce the $\text{SU}(2)$ -invariant singlet pair-annihilation operators

$$F_{ij} = a_i b_j - b_i a_j, \quad (4.2)$$

which remove one boson from each of sites i and j in the singlet channel [45] and satisfy $F_{ij} = -F_{ji}$. Next,

we define the three four-site singlet-annihilation operators corresponding to the three perfect matchings of the tetrahedron,

$$X = F_{14}F_{23}, \quad Y = F_{24}F_{31}, \quad Z = F_{34}F_{12}. \quad (4.3)$$

These correspond to the singlet coverings appearing in Eqs. (3.11a) and (3.11b) and satisfy

$$X + Y + Z = 0, \quad (4.4)$$

in analogy with Eq. (3.9).

Motivated by the chiral phase patterns of the $S = 1/2$ singlets, we form the two complex-conjugate combinations

$$\mathcal{B} = \frac{e^{-i\pi/6}}{\sqrt{18}} (X + \omega^2 Y + \omega Z), \quad (4.5a)$$

$$\mathcal{B}^* = \frac{e^{i\pi/6}}{\sqrt{18}} (X + \omega Y + \omega^2 Z). \quad (4.5b)$$

Here the star denotes complex conjugation of the coefficients. Their Hermitian conjugates create the two $S = 1/2$ chiral singlets,

$$\mathcal{B}^\dagger|\text{vac}\rangle = |+\rangle, \quad \mathcal{B}^{*\dagger}|\text{vac}\rangle = |-\rangle, \quad (4.6)$$

shown in Eqs. (3.11) and Fig. 3(a). Correspondingly,

$$\mathcal{B}|-\rangle = 0, \quad \mathcal{B}^*|+\rangle = 0. \quad (4.7)$$

Under the cyclic permutation $\hat{P}_{(123)}$, for which $X \rightarrow Y \rightarrow Z \rightarrow X$ [cf. Eq. (3.10)], the annihilation operator transforms as

$$\hat{P}_{(123)} \mathcal{B} \hat{P}_{(123)}^{-1} = \omega \mathcal{B}. \quad (4.8)$$

Consequently, $\mathcal{B}^\dagger|\text{vac}\rangle = |+\rangle$ transforms with eigenvalue ω^2 , while $\mathcal{B}^{*\dagger}|\text{vac}\rangle = |-\rangle$ transforms with eigenvalue ω , in agreement with Eq. (3.11). Thus, with the convention of Table IX, the singlet belonging to the CCI kernel transforms as E_+ under this three-cycle.

For $S = 1/2$, the projector in Eq. (3.13) suggests the generalization

$$\mathcal{H}_{\text{CCI}}^{(S)} = \mathcal{B}^\dagger \mathcal{B} \quad (4.9)$$

to arbitrary S . Remarkably, this factorization is exact. Expanding Eq. (4.5) gives

$$\begin{aligned} \mathcal{B} = \frac{1}{\sqrt{6}} [& a_4 a_3 b_2 b_1 + b_4 b_3 a_2 a_1 + \omega (a_4 b_3 a_2 b_1 + b_4 a_3 b_2 a_1) \\ & + \omega^2 (a_4 b_3 b_2 a_1 + b_4 a_3 a_2 b_1)]. \end{aligned} \quad (4.10)$$

Substituting this expression into $\mathcal{B}^\dagger \mathcal{B}$ and using the local constraint in Eq. (4.1) reproduces Eq. (3.17). The positive-semidefinite character of the local Hamiltonian is therefore explicit.

Since each of X , Y , and Z is an $SU(2)$ scalar and removes exactly one boson from each site, the same is true of $\mathcal{B}$ and $\mathcal{B}^*$. In particular,

$$[\hat{S}_{\text{tet}}^\alpha, \mathcal{B}] = 0, \quad \hat{S}_{\text{tet}}^\alpha = \sum_{i=1}^4 \hat{S}_i^\alpha, \quad \alpha = x, y, z. \quad (4.11)$$

Thus, $\mathcal{B}$ transforms as a scalar under global spin rotations and maps the Hilbert space with spin S on every site to that with spin $S - \frac{1}{2}$,

$$\mathcal{B} : \mathcal{H}^S \longrightarrow \mathcal{H}^{S-1/2}, \quad (4.12)$$

while preserving the total spin. Its Hermitian conjugate $\mathcal{B}^\dagger$ acts in the reverse direction. Furthermore, since $\mathcal{B}$ and $\mathcal{B}^*$ contain only boson annihilation operators,

$$[\mathcal{B}, \mathcal{B}^*] = 0. \quad (4.13)$$

For arbitrary S , the factorization in Eq. (4.9) implies that a state $|\Psi\rangle$ has zero local energy if and only if

$$\mathcal{B}|\Psi\rangle = 0. \quad (4.14)$$

This *singlet-annihilation constraint* provides the algebraic characterization of the local zero-energy states.

The transformation of $\mathcal{B}$ and $\mathcal{B}^*$ under permutations of the four sites makes the chirality selection explicit. An even permutation maps $\mathcal{B} \rightarrow \omega^k \mathcal{B}$, whereas an odd permutation maps $\mathcal{B} \rightarrow \omega^k \mathcal{B}^*$, with $k \in \{0, 1, 2\}$ depending on the permutation. Consequently, $\mathcal{H}_{\text{CCI}}^{(S)}$ in Eq. (4.9) is invariant under the alternating group A_4 of even permutations, whereas an odd permutation maps it to the parent Hamiltonian of the opposite chirality, $\mathcal{B}^{*\dagger} \mathcal{B}^*$. The construction therefore selects one of the two time-reversal-related chiral sectors while preserving full spin-rotation symmetry.

B. Parent Hamiltonian for the full color-ice subspace

Table I also gives the decomposition of the full color-ice subspace $\mathcal{W}^S$, generated by all 24 site permutations and global spin rotations. Since its two chiral sectors lie in the kernels of $\mathcal{B}$ and $\mathcal{B}^*$, respectively, and $[\mathcal{B}, \mathcal{B}^*] = 0$, the product

$$\mathcal{C} = \mathcal{B}^* \mathcal{B} = \mathcal{B} \mathcal{B}^* \quad (4.15)$$

annihilates the entire subspace $\mathcal{W}^S$. $\mathcal{C}$ removes two Schwinger bosons from each site and is an $SU(2)$ scalar, since both $\mathcal{B}$ and $\mathcal{B}^*$ are scalars. A positive-semidefinite parent Hamiltonian for the full color-ice subspace is consequently

$$\mathcal{H}_{\text{CI}}^{(S)} = \mathcal{C}^\dagger \mathcal{C} = (\mathcal{B}^* \mathcal{B})^\dagger (\mathcal{B}^* \mathcal{B}). \quad (4.16)$$

The absence of chirality selection becomes particularly transparent in the valence-bond representation. Using Eqs. (4.4) and (4.5), one obtains

$$\mathcal{C} = \frac{1}{12} (X^2 + Y^2 + Z^2). \quad (4.17)$$

Thus, the complex phase pattern distinguishing $\mathcal{B}$ from $\mathcal{B}^*$ drops out. A permutation of the four sites merely permutes X , Y , and Z , up to signs from $F_{ij} = -F_{ji}$ that disappear upon squaring. Consequently, $\mathcal{C}$, and hence $\mathcal{H}_{\text{CI}}^{(S)}$, is invariant under the full permutation group S_4 .

The lowest-spin cases illustrate the construction directly. For $S = 1/2$, the two chiral sectors together span the full tetrahedron Hilbert space, $\mathcal{W}^{1/2} = \mathcal{H}^{1/2}$. Hence $\mathcal{C}$ vanishes identically in the physical Hilbert space, and no nontrivial parent Hamiltonian exists for the full color-ice subspace.

For $S = 1$, by contrast, Table I gives $\dim \mathcal{W}^1 = 80$ in the $3^4 = 81$ -dimensional Hilbert space. The one-dimensional complement is the achiral singlet $\mathcal{C}^\dagger |\text{vac}\rangle$. Since the three-dimensional total-spin-singlet sector consists of this state, with $\hat{\chi}_{\text{tet}} = 0$, and two chiral singlets with $\hat{\chi}_{\text{tet}} = \pm 4\sqrt{3}$ belonging to $\mathcal{W}^1$, we can write

$$\mathcal{H}_{\text{CI}}^{(1)} \propto \mathcal{P}_0 (48 - \hat{\chi}_{\text{tet}}^2) \mathcal{P}_0, \quad (4.18)$$

where $\mathcal{P}_0$ projects onto the total-spin-singlet subspace. Thus $\mathcal{H}_{\text{CI}}^{(1)}$ annihilates the two chiral singlets and assigns positive energy only to the achiral one.

C. Zero-energy states and total-spin multiplets

The factorization $\mathcal{H}_{\text{CCI}}^{(S)} = \mathcal{B}^\dagger \mathcal{B}$ also gives a simple way to count the local zero-energy states. The operator $\mathcal{B}$ removes one Schwinger boson from each site, and therefore maps four spin- S sites onto four spin- $(S - \frac{1}{2})$ sites. Here and throughout, S_{tet} denotes the total spin of the four spins of a single tetrahedron, while J is reserved for the total spin of a finite lattice cluster. Since $\mathcal{B}$ commutes with the total spin, it acts separately in each S_{tet} sector,

$$\mathcal{B} : \mathcal{H}_{S_{\text{tet}}}^S \longrightarrow \mathcal{H}_{S_{\text{tet}}}^{S-\frac{1}{2}}, \quad (4.19)$$

where $\mathcal{H}_{S_{\text{tet}}}^S$ is the full subspace with total spin S_{tet} .

As shown in Appendix D, this map is surjective in every total-spin sector, meaning that every state in the target space is obtained from some state in the original space. The number of states annihilated by $\mathcal{B}$ is therefore simply the difference between the dimensions of the two spaces,

$$\dim \ker (\mathcal{B}|_{S_{\text{tet}}}) = \dim \mathcal{H}_{S_{\text{tet}}}^S - \dim \mathcal{H}_{S_{\text{tet}}}^{S-\frac{1}{2}}. \quad (4.20)$$

In particular, the largest total spin in the target space is $4S - 2$. Hence the complete $S_{\text{tet}} = 4S$ and $4S - 1$ sectors are automatically annihilated by $\mathcal{B}$.

Since a spin- S_{tet} multiplet contains $2S_{\text{tet}} + 1$ states, dividing Eq. (4.20) by this factor gives the number of such multiplets. Summing over all total spins gives

$$\dim \ker \mathcal{B} = (2S + 1)^4 - (2S)^4. \quad (4.21)$$

We now identify the kernel of the local parent term. Since every globally rotated CCI state has zero energy and

$$\langle \Psi | \mathcal{H}_{\text{CCI}}^{(S)} | \Psi \rangle = \|\mathcal{B}|\Psi\rangle\|^2, \quad (4.22)$$

all states in $\mathcal{V}^S$ are annihilated by $\mathcal{B}$,

$$\mathcal{V}^S \subseteq \ker \mathcal{B}. \quad (4.23)$$

Appendix D 2 shows that there are no additional zero modes: every state orthogonal to $\mathcal{V}^S$ can be generated by acting with $\mathcal{B}^\dagger$. Hence

$$\mathcal{V}^S = \ker \mathcal{B} \quad (4.24)$$

for arbitrary S .

Equation (4.21) therefore proves Eq. (3.16), while the recursion relation (4.20) gives the total-spin multiplicities of $\mathcal{V}^S$ listed in Table I. For $S = 1/2$, this reduces to the rank-one projector discussed above.

The same argument applies to the full color-ice space $\mathcal{W}^S$. The operator $\mathcal{C} = \mathcal{B}^* \mathcal{B}$ removes two bosons from each site and preserves the total spin,

$$\mathcal{C} : \mathcal{H}_{S_{\text{tet}}}^S \longrightarrow \mathcal{H}_{S_{\text{tet}}}^{S-1}. \quad (4.25)$$

For $S \geq 1$, both steps in this map reach their complete target spaces, as shown in Appendix D. Hence

$$\dim \ker(\mathcal{C}|_{S_{\text{tet}}}) = \dim \mathcal{H}_{S_{\text{tet}}}^S - \dim \mathcal{H}_{S_{\text{tet}}}^{S-1}. \quad (4.26)$$

Moreover, $\mathcal{C}$ annihilates all rotated colorings of both handednesses, so that $\mathcal{W}^S \subseteq \ker \mathcal{C}$. The Appendix again shows that there are no additional zero-energy states. Therefore

$$\mathcal{W}^S = \ker \mathcal{C}, \quad S \geq 1. \quad (4.27)$$

It follows that

$$\dim \mathcal{W}^S = (2S + 1)^4 - (2S - 1)^4, \quad S \geq 1, \quad (4.28)$$

and Eq. (4.26) gives the multiplets listed for $\mathcal{W}^S$ in Table I. For $S = 1/2$, $\mathcal{C}$ vanishes identically and $\mathcal{W}^{1/2} = \mathcal{H}^{1/2}$, as discussed above.

D. Spectral pairing and local gap

The factorization $\mathcal{H}_{\text{CCI}}^{(S)} = \mathcal{B}^\dagger \mathcal{B}$ also determines the nonzero local spectrum. Since $\mathcal{B}$ removes one boson from each site,

$$\mathcal{B} : \mathcal{H}^S \longrightarrow \mathcal{H}^{S-\frac{1}{2}}, \quad (4.29)$$

the spectral partner of $\mathcal{B}^\dagger \mathcal{B}$ on $\mathcal{H}^S$ is $\mathcal{B} \mathcal{B}^\dagger$ on $\mathcal{H}^{S-\frac{1}{2}}$.

Indeed, if $\mathcal{B}^\dagger \mathcal{B}|\Psi\rangle = E|\Psi\rangle$ with $E > 0$, then $\mathcal{B}|\Psi\rangle \neq 0$ and

$$\mathcal{B} \mathcal{B}^\dagger (\mathcal{B}|\Psi\rangle) = E \mathcal{B}|\Psi\rangle. \quad (4.30)$$

Conversely, $\mathcal{B}^\dagger$ maps every $E > 0$ eigenstate of $\mathcal{B} \mathcal{B}^\dagger$ back to an eigenstate of $\mathcal{B}^\dagger \mathcal{B}$ with the same eigenvalue. Thus the two operators have identical nonzero spectra, including multiplicities.

As shown in Appendix D,

$$\mathcal{B}^\dagger : \mathcal{H}^{S-\frac{1}{2}} \longrightarrow \mathcal{H}^S \quad (4.31)$$

is injective. Consequently,

$$\langle \Phi | \mathcal{B} \mathcal{B}^\dagger | \Phi \rangle = \|\mathcal{B}^\dagger | \Phi \rangle\|^2 > 0 \quad (4.32)$$

for every nonzero $|\Phi\rangle \in \mathcal{H}^{S-\frac{1}{2}}$. The entire space $\mathcal{H}^{S-\frac{1}{2}}$ is therefore paired with the nonzero-energy sector of $\mathcal{H}_{\text{CCI}}^{(S)}$. The remaining states in $\mathcal{H}^S$ form $\ker \mathcal{B} = \mathcal{V}^S$, giving

$$\dim \mathcal{V}^S = \dim \mathcal{H}^S - \dim \mathcal{H}^{S-\frac{1}{2}}, \quad (4.33)$$

in agreement with Eq. (3.16).

The same pairing determines the local excitation gap,

$$\Delta_{\text{loc}}(S) = \lambda_{\min}(\mathcal{B} \mathcal{B}^\dagger|_{\mathcal{H}^{S-\frac{1}{2}}}). \quad (4.34)$$

Appendix E shows that $\mathcal{B} \mathcal{B}^\dagger \succeq 4S^2 \mathbb{I}$ on $\mathcal{H}^{S-\frac{1}{2}}$ and that the bound is saturated by its maximal-total-spin multiplet, with $S_{\text{tet}} = 4S - 2$. Hence

$$\Delta_{\text{loc}}(S) = 4S^2. \quad (4.35)$$

This is the energy required to leave the local zero-energy space. It sets a local energy scale, but does not by itself imply a finite many-body gap when the tetrahedral terms are coupled on a lattice.

The multiplet at the gap is obtained directly by spectral pairing. The unique $S_{\text{tet}} = 4S - 2$ multiplet in $\mathcal{H}^{S-\frac{1}{2}}$ is mapped by the $\text{SU}(2)$ scalar $\mathcal{B}^\dagger$ onto an $S_{\text{tet}} = 4S - 2$ multiplet in $\mathcal{H}^S$ with energy $4S^2$. In particular, if $|F\rangle$ is its fully polarized highest-weight state, then $\mathcal{B}^\dagger |F\rangle$ is the highest-weight state of the multiplet at the gap. In the spin basis, this state is a superposition of configurations obtained from the fully polarized spin- S state by lowering two of the four spins by one unit, with relative phases fixed by $\mathcal{B}^\dagger$.

For $S \geq 1$, $\mathcal{H}^S$ contains six $S_{\text{tet}} = 4S - 2$ multiplets. Equation (4.20) shows that five belong to the zero-energy space $\mathcal{V}^S$, while the remaining one is the multiplet at energy $4S^2$ identified above.

V. COHERENT-STATE ZERO MODES AND THE MÖBIUS COMPLETION RULE

As shown above, the globally rotated CCI product states span the exact quantum kernel of the local term

$\mathcal{H}_{\text{CCI}}^{(S)}$. Here we characterize the zero modes that are themselves products of spin coherent states. Their four spin directions obey an S -independent completion rule: once three are specified, the fourth is fixed by a Möbius transformation. We now derive this rule from the singlet-annihilation constraint Eq. (4.14).

We use the stereographic parametrization of the spin-coherent state $|\mathbf{n}\rangle$ defined in Eq. (2.7). In terms of Schwinger bosons,

$$|z; S\rangle = \frac{(a^\dagger + zb^\dagger)^{2S}}{(1 + |z|^2)^S \sqrt{(2S)!}} |0\rangle, \quad (5.1)$$

with the corresponding spin direction

$$\mathbf{n}(z) = \frac{(2\Re z, 2\Im z, 1 - |z|^2)}{1 + |z|^2}. \quad (5.2)$$

The stereographic coordinate $z \in \hat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$ parametrizes the Bloch sphere, with $\hat{\mathbb{C}}$ the Riemann sphere. The action of a single boson annihilation operator is then

$$a|z; S\rangle = \frac{\sqrt{2S}}{\sqrt{1 + |z|^2}} |z; S - \tfrac{1}{2}\rangle, \quad (5.3a)$$

$$b|z; S\rangle = \frac{\sqrt{2S}z}{\sqrt{1 + |z|^2}} |z; S - \tfrac{1}{2}\rangle. \quad (5.3b)$$

It is convenient to introduce the two complex-conjugate quadratic polynomials

$$p_+ = \frac{1}{\sqrt{6}} [z_1 z_2 + z_3 z_4 + \omega^2(z_1 z_3 + z_2 z_4) + \omega(z_1 z_4 + z_2 z_3)], \quad (5.4a)$$

$$p_- = \frac{1}{\sqrt{6}} [z_1 z_2 + z_3 z_4 + \omega(z_1 z_3 + z_2 z_4) + \omega^2(z_1 z_4 + z_2 z_3)], \quad (5.4b)$$

related by $\omega \leftrightarrow \omega^2$.

Since every term in $\mathcal{B}$ contains one boson annihilation operator on each site, its action on a coherent product state preserves the four spin directions while reducing the spin length by $1/2$:

$$\mathcal{B} \bigotimes_{i=1}^4 |z_i; S\rangle = B(z_1, z_2, z_3, z_4) \bigotimes_{i=1}^4 |z_i; S - \tfrac{1}{2}\rangle, \quad (5.5)$$

with

$$B(z_1, z_2, z_3, z_4) = \frac{(2S)^2}{\sqrt{\prod_{i=1}^4 (1 + |z_i|^2)}} p_-(z_1, z_2, z_3, z_4). \quad (5.6)$$

The energy expectation value is therefore

$$\langle z_1 z_2 z_3 z_4 | \mathcal{H}_{\text{CCI}}^{(S)} | z_1 z_2 z_3 z_4 \rangle = |B(z_1, z_2, z_3, z_4)|^2. \quad (5.7)$$

The zero-mode condition is therefore independent of S : for every $S \geq 1/2$, including $S = 1/2$, the same equation

$$p_-(z_1, z_2, z_3, z_4) = 0 \quad (5.8)$$

selects the allowed configurations of the four coherent-state directions.

As a check, the four tetrahedral directions Eq. (2.1) have stereographic coordinates

$$\begin{aligned} z^A &= w_- e^{i\pi/4}, & z^B &= w_- e^{5i\pi/4}, \\ z^C &= w_+ e^{3i\pi/4}, & z^D &= w_+ e^{7i\pi/4}, \\ w_\pm &= \sqrt{2 \pm \sqrt{3}}, \end{aligned} \quad (5.9)$$

and satisfy Eq. (5.8), as do their even color permutations and global spin rotations. Odd permutations exchange ω and ω^2 and therefore satisfy the coefficient-conjugated constraint associated with $\mathcal{B}^*$. In addition, p_- vanishes whenever three coordinates coincide, with the fourth arbitrary. These polarized solutions are the only locus on which the fourth spin is not uniquely fixed by the other three; the degeneracies are classified in Appendix C 2.

A common framework for classical spin liquids describes the ground-state manifold in terms of local constraints that are linear in the spin variables [46–50]. Equation (5.8) has a different structure. Although it is quadratic in the four stereographic coordinates, it is *multi-affine*: it is linear in any one z_i when the other three are held fixed. The local constraint can therefore be solved as an exact completion rule. Solving for the fourth coordinate gives

$$z_4 = -\frac{z_1 z_2 + \omega z_1 z_3 + \omega^2 z_2 z_3}{z_3 + \omega z_2 + \omega^2 z_1}. \quad (5.10)$$

For fixed $z_1 \neq z_2$, this defines the invertible Möbius transformation $z_4 = T_{z_1, z_2}(z_3)$, with

$$T_{z_1, z_2}(z) = -\frac{z(\omega z_1 + \omega^2 z_2) + z_1 z_2}{z + \omega z_2 + \omega^2 z_1}. \quad (5.11)$$

A vanishing denominator simply corresponds to $z_4 = \infty$ and is therefore regular on the Riemann sphere. The exceptional configurations for which the fourth direction is not uniquely determined by the other three occur when spin directions coincide. In fact, any zero mode containing a coincident pair necessarily contains at least three coincident spins, and

$$(z_1, z_2, z_3, z_4) = (z, z, z, w) \quad (5.12)$$

together with its site permutations satisfies Eq. (5.8) for arbitrary w . These special loci are classified in Appendix C 2.

Möbius transformations are familiar in spin physics because the Bloch sphere is naturally identified with $\mathbb{CP}^1$: in stereographic coordinates an $\text{SU}(2)$ rotation acts fractionally linearly on the coherent-state coordinate [51, 52]. More explicitly, Möbius transformations have been used to describe the time evolution of classical spins in stereographic coordinates [53, 54], and they also appear in the Majorana representation of symmetric multiqubit states, where $\text{SL}(2, \mathbb{C})$ transformations act as Möbius maps on the Majorana constellation [55].

Their role here is different. The transformation T_{z_1, z_2} is neither an imposed spin rotation nor a time-evolution operator. Instead, it is generated by the local zero-energy constraint itself: for fixed z_1 and z_2 , solving $p_-(z_1, z_2, z_3, z_4) = 0$ determines the spin at one site from that at another, $z_4 = T_{z_1, z_2}(z_3)$ [Eqs. (5.10) and (5.11)]. The completion maps are generically Möbius transformations in $\text{PSL}(2, \mathbb{C})$ and need not belong to the $\text{SU}(2)$ subgroup corresponding to rigid rotations of the Bloch sphere. They therefore act as spatial propagation rules for the local ground-state constraint. Their compositions along sequences of tetrahedra, and in particular around closed loops, define the corresponding constraint holonomy. In this way the projective geometry of the local zero-energy condition directly controls the existence and dimensionality of coherent zero-mode families on the lattice.

The action of a single completion map is particularly transparent from the identity

$$\frac{T_{z_1, z_2}(z) - z_1}{T_{z_1, z_2}(z) - z_2} = -\omega \frac{z - z_1}{z - z_2}. \quad (5.13)$$

Thus z_1 and z_2 are its two fixed points, while in the projective coordinate $(z - z_1)/(z - z_2)$ the map acts as multiplication by $-\omega = e^{-i\pi/3}$. Hence

$$T_{z_1, z_2}^6(z) = z, \quad (5.14)$$

and every point other than the two fixed points belongs to a six-cycle. Figure 4 illustrates this local propagation for the reference tetrahedral coloring. We emphasize that this order-six property applies to repeated application of the *same* local map with the same fixed pair.

The coherent zero modes obey a sharp handedness constraint. For the sector defined by $\mathcal{B}$, the local zero-mode condition implies

$$\chi_{\text{tet}} \leq 0, \quad (5.15)$$

as shown in Appendix C3. This is the coherent-state counterpart of the exact quantum handedness constraint established for the complete local kernel in Sec. IIID. Independently of the zero-mode constraint, any four unit spin directions satisfy the universal bound

$$|\chi_{\text{tet}}| \leq \frac{16}{3\sqrt{3}}, \quad (5.16)$$

with equality if and only if the four directions form a regular tetrahedral frame; see Appendix C5. Together, these results give the exact chirality window of the coherent zero-mode variety,

$$-\frac{16}{3\sqrt{3}} \leq \chi_{\text{tet}} \leq 0. \quad (5.17)$$

The lower bound is saturated precisely by the negatively oriented regular tetrahedral colorings, up to global spin rotations and even site permutations, while the upper

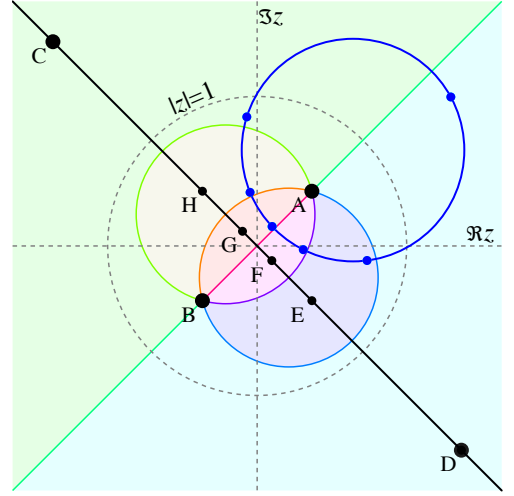


FIG. 4. Local Möbius propagation for the reference tetrahedral coordinates. The points A, B, C, D are the stereographic images of the four color directions given in Eq. (5.9). Keeping the pair (A, B) fixed, the completion map T_{z_A, z_B} leaves A and B invariant and generates the six-cycle $C \rightarrow D \rightarrow E \rightarrow F \rightarrow G \rightarrow H \rightarrow C$, which passes through the two remaining color points and the four additional points $E-H$. Every consecutive pair $X, T_{z_A, z_B}(X)$ of the orbit—whether or not X is a color point—yields an exact coherent zero mode with the four spins at A, B, X , and $T_{z_A, z_B}(X)$. The blue points show a second six-cycle generated from a generic initial point. Its orbit lies on the corresponding Apollonius circle, on which the ratio $|z - z_A|/|z - z_B|$ is constant.

bound is reached on the three-coincident branch, which includes the fully polarized state. Thus the local constraint fixes the *handedness*, but not the magnitude, of the tetrahedral chirality: the coherent zero-mode variety contains continuous paths from the maximally chiral regular tetrahedron to achiral configurations, including polarized ones, without entering the opposite-handedness sector. For the coefficient-conjugated constraint associated with $\mathcal{B}^*$, all chirality inequalities are reversed.

VI. FROM LOCAL CONSTRAINTS TO LATTICE ZERO MODES

We now use the single-tetrahedron results derived above as local building blocks for extended lattices. Assign to each tetrahedron t a chirality label $\varepsilon_t = \pm 1$ and define the lattice Hamiltonian

$$\mathcal{H} \equiv \mathcal{H}[\{\varepsilon_t\}] = \sum_t \mathcal{H}_{\text{CCI}}^{(S), \varepsilon_t}(t), \quad (6.1)$$

where $\mathcal{H}_{\text{CCI}}^{(S), +} = \mathcal{H}_{\text{CCI}}^{(S)}$ denotes the reference local parent term and $\mathcal{H}_{\text{CCI}}^{(S), -} = \mathcal{T} \mathcal{H}_{\text{CCI}}^{(S)} \mathcal{T}^{-1}$ its time-reversed partner.

For lattices in which the tetrahedra fall into two orien-

tation classes $\mathcal{O}_1$ and $\mathcal{O}_2$ (Sec. II A), we take

$$\varepsilon_t = \begin{cases} +1, & t \in \mathcal{O}_1, \\ \varepsilon, & t \in \mathcal{O}_2, \end{cases} \quad \varepsilon = \pm 1. \quad (6.2)$$

Thus $\varepsilon = +1$ gives a uniform assignment of the fixed-site-ordering chirality label, whereas $\varepsilon = -1$ gives an alternating one. This convention will be used below for the checkerboard, pyrochlore, fcc, square, and honeycomb constructions.

For any fixed assignment, time reversal reverses all local chirality labels,

$$\mathcal{T}\mathcal{H}[\{\varepsilon_t\}]\mathcal{T}^{-1} = \mathcal{H}[\{-\varepsilon_t\}]. \quad (6.3)$$

Thus every chirality assignment and its sign-reversed partner define time-reversal-conjugate Hamiltonians with identical spectra. The handedness of the zero-energy states is therefore selected explicitly by the CCI parent Hamiltonian rather than spontaneously. Spontaneous selection would instead require a parent Hamiltonian that treats the two local chiral sectors symmetrically; the full color-ice parent Hamiltonian of Sec. IV B, which is time-reversal invariant, provides such an example.

Since every local term in Eq. (6.1) is positive semidefinite, the zero-energy space of the lattice Hamiltonian is precisely the common kernel

$$\mathcal{K} \equiv \bigcap_t \ker \mathcal{H}_{\text{CCI}}^{(S), \varepsilon_t}(t) = \ker \mathcal{H}. \quad (6.4)$$

Whenever the prescribed chirality assignment admits a compatible global CCI coloring, the corresponding product state and all of its global spin rotations belong to $\mathcal{K}$. The Hamiltonian is then frustration free, with $\mathcal{K}$ as its exact zero-energy ground-state space.

The exact quantum handedness established on a single tetrahedron in Sec. III D immediately carries over to the many-body common kernel. Indeed, every $|\Psi\rangle \in \mathcal{K}$ belongs to the kernel of every local term separately. Hence, for any normalized $|\Psi\rangle \in \mathcal{K}$,

$$\varepsilon_t \langle \Psi | \hat{\chi}_{\text{tet}}(t) | \Psi \rangle \leq 0 \quad (6.5)$$

for every tetrahedron t , where $\hat{\chi}_{\text{tet}}(t)$ is defined in the fixed site ordering of the corresponding local parent term. This is an exact statement about the complete quantum common kernel, including entangled zero-energy states, and is independent of spin length, system size, and lattice connectivity. In the geometric orientation convention of Sec. II A, the alternating assignment gives the same chirality sign on the two tetrahedron orientation classes, whereas for the uniform assignment the geometric chirality changes sign between them.

Equation (6.5) constrains the sign of the tetrahedral chirality throughout the full quantum common kernel, but not its magnitude. In particular, the coherent-state window Eq. (5.17) does not bound the spectrum of the chirality operator within $\mathcal{K}$; projected chirality eigenvalues can exceed the coherent-state values in magnitude, as illustrated in Sec. IX B.

It is useful at this point to distinguish three nested classes of zero-energy states. The smallest consists of the compatible CCI colorings, promoted to coherent product states, together with their global spin rotations. The second is the *coherent zero-mode variety*: the set of coherent product states $\bigotimes_i |z_i; S\rangle$ whose stereographic coordinates satisfy the corresponding local coherent-state constraint on every tetrahedron. For the reference sector this is $p_- = 0$, Eq. (5.8), while the time-reversed sector obeys its coefficient-conjugated counterpart $p_+ = 0$. The largest set is the full quantum common kernel $\mathcal{K}$, regarded here as its set of state vectors. Thus

$$\begin{aligned} & \{\text{rotated CCI colorings}\} \\ & \subset \{\text{coherent zero-mode states}\} \subset \mathcal{K}. \end{aligned} \quad (6.6)$$

The first two sets contain only product states, whereas $\mathcal{K}$ is a linear subspace of the many-body Hilbert space. The first inclusion is already proper on a single tetrahedron because the coherent constraint also admits non-coloring solutions, such as the three-coincident configurations of Eq. (5.12). The second inclusion can likewise be proper on an extended lattice because $\mathcal{K}$ may contain genuinely entangled states.

Here “variety” is used in its mathematical sense as the joint zero set of the local polynomial constraints. It need not be a smooth manifold; in particular, the three-coincident configurations of Eq. (5.12) are singular loci at which the local completion rule ceases to determine the fourth spin uniquely. Throughout the following discussion, completion maps and holonomies characterize the coherent zero-mode variety, whereas exact-diagonalization spectra and quantum degeneracy counts refer to the full common kernel $\mathcal{K}$.

The coherent subset admits a particularly direct local-to-global construction. Suppose a sequence of tetrahedra is traversed such that, at each step, the already specified spin directions determine a local completion map T_i for the remaining spin. Its stereographic coordinate is then propagated by successive Möbius transformations. The order-six property derived above concerns repeated application of a single completion map with the same fixed pair. On an extended lattice the successive maps T_i generally differ, and their composition need not have finite order.

Around a closed loop of length L , the net propagation is described by the Möbius holonomy

$$\mathcal{M}_L = T_L \circ T_{L-1} \circ \cdots \circ T_1. \quad (6.7)$$

A coherent loop configuration closes precisely when its initial stereographic coordinate is a fixed point of $\mathcal{M}_L$. A generic nonidentity Möbius transformation has two distinct fixed points, while a parabolic transformation has a single double fixed point. Such loops therefore admit only isolated coherent solutions. In contrast, when

$$\mathcal{M}_L = \text{id}, \quad (6.8)$$

every initial coordinate closes consistently, producing a continuous family of exact coherent zero modes.

The holonomy therefore measures the coherent flexibility generated by the way in which the local tetrahedral constraints are connected. It does not, by itself, determine the dimension or structure of the full quantum common kernel $\mathcal{K}$. The distinction between these two levels will be important below: different lattice connectivities can produce extensive, subextensive, or only global coherent deformations, while the corresponding quantum kernels may contain additional entangled zero modes.

More generally, the large quantum common kernel can be viewed as the price of allowing families of exact non-collinear coherent product states. For the Heisenberg ferromagnet, an arbitrary global spin direction gives the coherent family $\bigotimes_i |z; S\rangle_i$, whose linear span is the familiar maximal-spin multiplet. Here the local constraint is more flexible: the coherent-state coordinates of a tetrahedron need not coincide but are related through the Möbius completion rule, with three directions determining the fourth. This permits exact product states with noncollinear spin directions. Since every member of such a coherent family is annihilated by the same positive-semidefinite Hamiltonian, the linear span of the entire family necessarily belongs to $\mathcal{K}$. The coherent-state freedom therefore provides a direct mechanism for generating quantum ground-state degeneracy, although it need not exhaust it: the full kernel can be substantially larger and may also contain genuinely entangled zero modes.

We now turn to the corner-sharing checkerboard and pyrochlore lattices, where the overlap of the local constraints produces an extensive zero-energy degeneracy.

VII. CORNER-SHARING LATTICES: CHECKERBOARD AND PYROCHLORE

We now place the local term (4.9) on the checkerboard and pyrochlore lattices. In both cases the tetrahedra are corner-sharing: every spin belongs to two local terms, while tetrahedra of the same family are mutually disjoint. This geometry leaves a large common kernel $\mathcal{K}$ and leads to an extensive ground-state degeneracy. Each crossed plaquette of the checkerboard lattice, or each tetrahedron of the pyrochlore lattice, carries one local parent term with chirality label ε_t . We consider both the uniform assignment and the alternating assignment, in which ε_t changes sign between the two tetrahedron families.

A. Exact diagonalization on the checkerboard lattice

We first examine the many-body kernel by exact diagonalization of the spin-1/2 model on the periodic 16-site checkerboard cluster for both chirality assignments. The zero-energy space is already very large: it contains 38416 states for the alternating assignment and 35714 for the

uniform one, in both cases more than half of the full 2^{16} -dimensional Hilbert space. The multiplet-resolved degeneracies, together with those of the other 16-site geometries, are listed in Table II. For comparison at the level of classical configurations, exhaustive enumeration gives 576 four-colorings on the same cluster, of which 36 satisfy the prescribed alternating chirality and 36 the uniform chirality.

Several features of the checkerboard spectrum are worth emphasizing. For $S = 1/2$, each local term projects onto one of the two chiral four-spin singlets. In any fixed quantization axis, these singlets contain two up and two down spins and are therefore orthogonal to every local basis state with zero or one spin flipped relative to full polarization. Consequently, the fully polarized $J = 8$ multiplet and all $J = 7$ multiplets belong to the common kernel independently of the chirality assignment. Numerically, the alternating and uniform spectra remain identical also in the $J = 6$ and 5 sectors and first differ at $J = 4$, with the alternating pattern producing additional zero modes at lower total spin. Thus the local handedness pattern affects the strongly overlapping part of the many-body kernel but leaves the polarized high-spin sector unchanged. The coexistence of the ferromagnetic multiplet with the chirality-carrying zero-energy states is a special feature of the frustration-free parent-Hamiltonian point. Since the ferromagnetic sector already lies in the local kernel, perturbations favoring polarization can select it without overcoming a local energy cost, making a ferromagnetic phase a natural competitor near the solvable point. A related one-dimensional example is the multicritical point of the zigzag chain with anisotropic Γ exchange, where the Hamiltonian also reduces to a sum of local simplex terms with a common kernel containing the ferromagnet and with a ground-state degeneracy growing polynomially with system size [29]. The analogy is structural: in the present $SU(2)$ -symmetric models, corner sharing produces a much larger, exponentially degenerate kernel.

The periodic 4×4 checkerboard cluster and the 16-site cubic pyrochlore cluster define the same finite constraint problem up to a relabeling of sites. Each contains eight tetrahedra arranged into two families of four. Tetrahedra within a given family are site-disjoint, while every tetrahedron in one family shares exactly one spin with every tetrahedron in the other. The dual graph, whose vertices represent tetrahedra and whose edges represent shared spins, is therefore $K_{4,4}$ in both cases. The two tetrahedral hypergraphs are consequently isomorphic, and the corresponding Hamiltonians have identical spectra and zero-energy multiplet counts for matching uniform or alternating chirality assignments. This equivalence is a special property of these finite clusters.

The remainder of this section explains the origin of the large kernel generated by corner sharing. We first construct exact loop zero modes in polarized backgrounds and then map the polarized product-state sector to matchings on the dual graph. The resulting combinato-

TABLE II. Zero-energy multiplet counts for spin-1/2 models on periodic 16-site clusters. For each total spin J , “full” is the number of multiplets in the complete 16-spin Hilbert space. For the checkerboard, square, and honeycomb lattices, “alt.” and “unif.” denote the alternating and uniform assignments of the fixed-ordering chirality label ε_t , respectively. The triangular face-sharing construction has a single chirality pattern. The checkerboard columns also apply, for the corresponding chirality assignments, to the periodic 16-site cubic pyrochlore cluster, since the two clusters are isomorphic as tetrahedron-incidence structures with dual graph $K_{4,4}$. The final row gives the total number of zero-energy states after weighting each multiplet by $2J + 1$.

J	Checkerboard			Square		Honeycomb		Triangular
	full	alt.	unif.	alt.	unif.	alt.	unif.	
0	1430	662	528	101	56	114	89	1
1	3432	1632	1368	244	118	263	205	3
2	3640	1888	1672	382	312	364	332	5
3	2548	1484	1400	486	476	478	476	7
4	1260	852	840	453	452	452	452	21
5	440	352	352	264	264	264	264	45
6	104	96	96	88	88	88	88	56
7	15	15	15	15	15	15	15	15
8	1	1	1	1	1	1	1	1
Total states	65536	38416	35714	14512	13660	14427	14054	1738

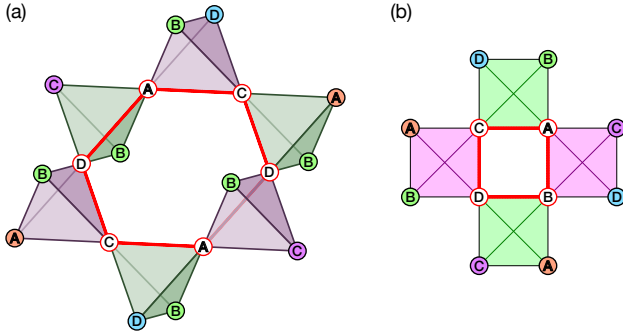


FIG. 5. Embedded loops supporting local zero modes. (a) A hexagonal loop of the pyrochlore lattice (bold red), sharing one site with each of six surrounding tetrahedra. (b) The analogous square loop of the checkerboard lattice.

rial bounds establish an exponential lower bound on the ground-state degeneracy in the thermodynamic limit.

B. Loop zero modes

The local completion rule becomes especially transparent on an embedded one-dimensional loop. Fixing all spins outside the loop turns the constraint on each tetrahedron into a relation between two successive loop spins. The resulting loop problem depends strongly on the background. We distinguish three cases: a uniform polarized background, a generic coherent background, and a four-coloring background.

1. Uniform-color background

Consider first a square loop of the checkerboard lattice or a hexagonal loop of the pyrochlore lattice embedded in a fully polarized environment [Fig. 5]. Each tetrahedron along the loop contains two consecutive loop sites and two environment sites polarized along the same quantization axis. As shown in Appendix F, all but one term in $\mathcal{B}$ then vanish, and the exact zero-mode condition reduces to

$$b_i b_j |\psi\rangle = 0 \quad (7.1)$$

for every loop bond (ij) . Thus two neighboring loop sites cannot both deviate from full polarization.

The transfer-matrix counting is elementary and is left to Appendix F. The resulting loop dimensions are

$$N_4(S) = 1 + 8S + 8S^2, \quad (7.2a)$$

$$N_6(S) = 1 + 12S + 36S^2 + 16S^3. \quad (7.2b)$$

Accordingly, the square loop has $N_4 = 7, 17$, and 31 zero modes for $S = \frac{1}{2}, 1$, and $\frac{3}{2}$, while the hexagon has $N_6 = 18, 65$, and 154 . Direct diagonalization of the embedded loop reproduces these numbers. For $S = \frac{1}{2}$ the same counting is equivalent to the monomer-dimer partition function of the cycle graph, anticipating the matching construction of Sec. VIIC.

2. Generic coherent background

Away from the polarized background, the reduction (7.1) no longer applies and the full Möbius propagation rule of Sec. V must be used. For a loop of length L , the

local completion maps T_i , including the appropriate local site orderings and fixed environment spins, combine into the holonomy $\mathcal{M}_L$ of Eq. (6.7). The loop closes when the initial stereographic coordinate is a fixed point of $\mathcal{M}_L$.

For a generic background, $\mathcal{M}_L$ is a nonidentity Möbius transformation with two distinct fixed points on $\hat{\mathbb{C}}$. The coherent-product problem therefore has two isolated loop solutions rather than a continuous zero mode. A parabolic holonomy is exceptional and has a single double fixed point, whereas identity holonomy gives the continuous family discussed below.

Exact diagonalization of the embedded square and hexagonal problems gives a two-dimensional zero-energy kernel for generic backgrounds at $S = \frac{1}{2}$, 1, and $\frac{3}{2}$, consistent with the two fixed-point solutions. This finite-size agreement provides a check of the Möbius propagation picture. The fixed-point argument itself, however, characterizes only the coherent-product zero modes and does not establish the complete kernel of the loop problem for arbitrary S .

3. Four-coloring background

A four-coloring background is special because every tetrahedron contains all four colors A, B, C, D exactly once. Since the background itself satisfies the local constraint, the colors $c_1, c_2, \dots, c_L$ encountered along a closed loop form an orbit of the corresponding completion maps,

$$T_i(c_i) = c_{i+1}, \quad c_{L+1} \equiv c_1. \quad (7.3)$$

Thus the background color c_1 is always a fixed point of the loop holonomy $\mathcal{M}_L$. Whether this isolated solution extends to a continuous zero mode is determined by the full holonomy.

For an elementary hexagon of the four-sublattice pyrochlore coloring, the spins follow the three-color sequence

$$XYZXYZ, \quad (7.4)$$

as illustrated in Fig. 5(a). For this sequence, the six completion maps have identity holonomy,

$$\mathcal{M}_6 = \text{id}. \quad (7.5)$$

An arbitrary deformation of one spin can therefore be propagated around the hexagon and closes after one circuit, producing a continuous chiral analog of a weathervane mode. The derivation of Eq. (7.5) is given in Appendix F 2 a.

Identity holonomy also has an exact quantum consequence. Consider an embedded loop of length L with its coherent environment fixed, and suppose that every local completion map is well defined and invertible. If

$$\mathcal{M}_L = \text{id}, \quad (7.6)$$

then the complete quantum kernel $\mathcal{K}_{\text{loop}}$ of the reduced loop constraints is spanned by the continuously propagated coherent states,

$$\mathcal{K}_{\text{loop}} = \text{span} \left\{ \left| \tilde{\psi}(z) \right\rangle \mid z \in \hat{\mathbb{C}} \right\}, \quad (7.7)$$

and has dimension

$$\dim \mathcal{K}_{\text{loop}} = 2SL + 1. \quad (7.8)$$

A derivation is given in Appendix F 3. There, identity holonomy allows the loop constraints to be transformed into nearest-neighbor singlet-annihilation constraints, which force the connected loop into its maximal-spin LS multiplet.

Identity holonomy is a property of the specific color sequence around the loop, rather than of four-coloring backgrounds as such. For example, the four-color sequence $ABCD$ around an empty square of the checkerboard lattice, Fig. 5(b), has nonidentity holonomy and therefore admits only isolated coherent fixed-point solutions, whereas the $ABCABC$ sequence around a pyrochlore hexagon has identity holonomy and supports a continuous family.

The linearized coherent-state analysis of Appendix G reveals a further distinction between first-order and finite deformations. Around the four-coloring background, the linearized constraint matrix $\mathbf{R}(\mathbf{k})$ has nullity two at every wavevector on both the checkerboard and pyrochlore lattices, giving $N/2 + 1$ complex infinitesimal zero modes. On the checkerboard lattice these include modes localized on individual empty squares. Their holonomy is parabolic: the linearized multiplier around the square is unity even though the full Möbius transformation is not the identity. The square modes therefore satisfy the constraints to first order but are obstructed at higher order. By contrast, the identity holonomy of the pyrochlore hexagon allows the corresponding infinitesimal modes to extend to exact continuous weathervane families.

Equation (7.8) applies to the embedded-loop problem with the coherent environment fixed. It does not imply that the full checkerboard or pyrochlore many-body kernel is exhausted by coherent states; additional entangled zero modes are present.

C. Matching representation of the polarized sector

For $S = \frac{1}{2}$, a large polarized subset of the common kernel $\mathcal{K}$ has a direct formulation as a matching problem. We choose a fixed quantization axis and introduce the dual graph G , whose vertices represent tetrahedra and whose edges represent the physical spin sites shared by neighboring tetrahedra. The dual graphs of the checkerboard and pyrochlore lattices are, respectively, the square and diamond lattices. In both cases, G is a 4-regular bipartite graph; for a system of N spins, it has N edges and $N/2$ vertices.

We associate a down spin with an occupied edge of G . For $S = \frac{1}{2}$, the operator $\mathcal{B}$ annihilates every four-site basis state containing at most one down spin. Hence any spin configuration in which no two down spins belong to the same tetrahedron is an exact zero-energy state. On the dual graph, this means that no two occupied edges share a vertex. The occupied edges therefore form a *matching* of G .

The number of such polarized product zero modes is consequently the matching partition function at unit activity,

$$\mathcal{D}_{\text{match}}(G) = Z_G(1) = \sum_k m_k(G), \quad (7.9)$$

where $m_k(G)$ denotes the number of matchings containing k edges.

Rigorous results for matchings of 4-regular bipartite graphs then give exponential bounds on this polarized zero-energy sector,

$$1.3919^N \lesssim \mathcal{D}_{\text{match}} \leq 209^{N/16} \simeq 1.3964^N. \quad (7.10)$$

In particular, the lower bound alone establishes an exponentially large CCI ground-state degeneracy. The graph-theoretic derivation of these bounds, including the origin of the constant 209, is given in Appendix H.

The matching construction describes only a polarized subset of the full common kernel. The kernel $\mathcal{K}$ also contains coherent zero modes away from the polarized sector as well as genuinely entangled states. Equation (7.10) therefore provides a rigorous lower bound on the ground-state degeneracy rather than a calculation of the full residual entropy.

D. Comparison of degeneracies

The loop and matching constructions identify explicitly countable subsets of the common kernel $\mathcal{K}$ and therefore provide lower bounds on its ground-state degeneracy. This allows a direct comparison with familiar classical ice manifolds. Here N denotes the number of physical spin sites, and we compare the exponential degeneracy per spin, $N^{-1} \ln \dim \mathcal{K}$, with the corresponding classical quantity $N^{-1} \ln \mathcal{N}_{\text{ice}}$.

1. Checkerboard lattice

On compatible periodic checkerboard clusters, one can choose $N/8$ square loops such that no two share a tetrahedron. Their zero modes can therefore be chosen independently, giving

$$\dim \mathcal{K}_{\text{cb}} \geq [N_4(S)]^{N/8}. \quad (7.11)$$

For $S = \frac{1}{2}$, where $N_4 = 7$, this yields

$$\dim \mathcal{K}_{\text{cb}} \geq 7^{N/8} \simeq 1.2754^N. \quad (7.12)$$

This already exceeds the exact asymptotic degeneracy of square ice [56],

$$W_{\text{sq. ice}} = \left(\frac{4}{3}\right)^{3N/4} \simeq 1.2408^N. \quad (7.13)$$

The matching construction of Sec. VII C gives a much stronger bound,

$$\dim \mathcal{K}_{\text{cb}} \geq \mathcal{D}_{\text{match}} \gtrsim 1.3919^N. \quad (7.14)$$

Thus even explicitly constructed subsets of $\mathcal{K}$ already have a degeneracy density substantially larger than that of square ice.

2. Pyrochlore lattice

On compatible periodic pyrochlore clusters, one can choose $N/12$ elementary hexagons such that their surrounding tetrahedra are disjoint. Their zero modes can therefore be chosen independently, giving

$$\dim \mathcal{K}_{\text{pyr}} \geq [N_6(S)]^{N/12}. \quad (7.15)$$

For $S = 1/2$, where $N_6 = 18$, this yields

$$\dim \mathcal{K}_{\text{pyr}} \geq 18^{N/12} \simeq 1.2723^N. \quad (7.16)$$

This already exceeds the conventional Pauling estimate for pyrochlore spin ice [57],

$$W_{\text{Pauling}} = \left(\frac{3}{2}\right)^{N/2} \simeq 1.2247^N. \quad (7.17)$$

As for the checkerboard lattice, the matching construction of Sec. VII C gives a substantially stronger bound,

$$\dim \mathcal{K}_{\text{pyr}} \geq \mathcal{D}_{\text{match}} \gtrsim 1.3919^N. \quad (7.18)$$

Thus the explicitly constructed CCI subsets already exhibit a degeneracy density substantially larger than the conventional spin-ice benchmark. We emphasize that, unlike the square-ice result above, the Pauling value is an estimate rather than the exact pyrochlore-ice degeneracy.

VIII. EDGE-SHARING TETRAHEDRA: FCC, SQUARE, AND HONEYCOMB LATTICES

We next place the same local parent Hamiltonian, Eq. (3.17), on lattices of edge-sharing tetrahedra. The increased overlap between neighboring constraints qualitatively changes the structure of the zero-energy kernel. On the corner-sharing checkerboard and pyrochlore lattices, neighboring tetrahedra share only one spin, allowing the local completion rule to generate an extensively degenerate family of coherent zero modes. For edge-sharing tetrahedra, two spins are shared and the corresponding constraints are therefore much more strongly

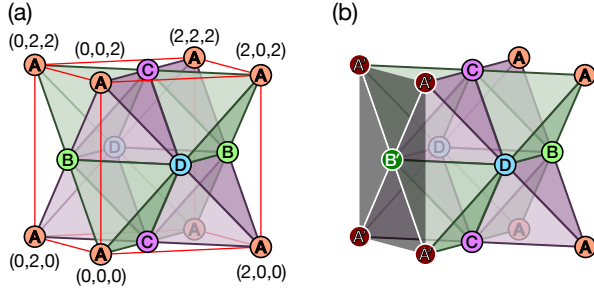


FIG. 6. Tetrahedral order and a planar zero mode on the fcc lattice. (a) The elementary tetrahedra form two orientation classes exchanged by inversion (green and purple). Assigning opposite local chirality labels to the two classes is compatible with the four-sublattice tetrahedral coloring shown. In the coordinate convention of the figure, the cubic magnetic unit cell (red) is generated by $(2, 0, 0)$, $(0, 2, 0)$, and $(0, 0, 2)$. For the alternating chirality assignment, inversion $\mathcal{I}$ and time reversal $\mathcal{T}$ separately exchange the two local chiral sectors, whereas their product $\mathcal{IT}$ is a symmetry. The four-sublattice tetrahedral ordered state shown preserves this combined symmetry. (b) A planar zero-energy deformation of the tetrahedral state. The spin directions A' and B' within a single (100) plane can be varied continuously, with the local Möbius completion rule fixing one in terms of the other, while all spins outside the plane remain at their tetrahedral directions.

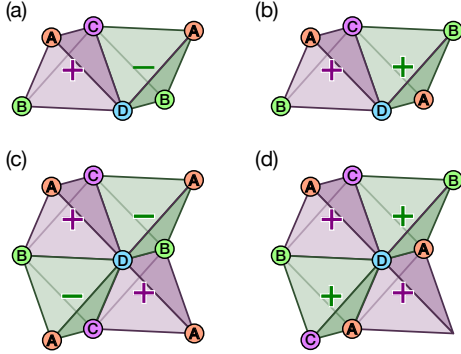


FIG. 7. Propagation of the four-coloring constraint across shared edges for (a) the alternating and (b) the uniform chirality assignments; the corresponding ε_t are indicated in the figure. Once the colors on one tetrahedron are fixed, an edge-sharing neighbor already contains two of them. Its remaining two sites must therefore carry the other two colors, with their ordering uniquely determined by the prescribed chirality parity. (c) For the alternating assignment, repeated propagation around the closed sequence is consistent and reproduces the four-sublattice coloring of Fig. 6(a). (d) For the uniform assignment, the same procedure returns an incompatible color assignment, showing that a global four-coloring is frustrated.

coupled, substantially restricting the available zero-mode deformations. We first discuss the three-dimensional fcc lattice and then its two-dimensional square- and honeycomb-lattice descendants.

A. fcc lattice

The elementary nearest-neighbor tetrahedra of the fcc lattice fall into two orientation classes exchanged by inversion [Fig. 6(a)]. Tetrahedra overlap along edges, and every fcc site belongs to eight elementary tetrahedra. We place one local CCI term on every tetrahedron and consider two choices of the chirality label ε_t : a uniform assignment, with the same ε_t on the two orientation classes, and an alternating assignment, with opposite ε_t .

Here and below, “uniform” and “alternating” refer to the fixed-ordering chirality labels ε_t , rather than directly to the sign of the geometric tetrahedral chirality of Sec. II A; in the geometric convention, the alternating assignment is precisely the one that enforces the same handedness on every tetrahedron. For the alternating assignment, inversion $\mathcal{I}$ exchanges the two tetrahedron classes while time reversal $\mathcal{T}$ exchanges the two local chiral sectors. Neither is a symmetry separately, whereas their product $\mathcal{IT}$ leaves the Hamiltonian invariant. The four-sublattice tetrahedral state shown in Fig. 6(a) also preserves this combined symmetry.

Edge sharing makes the discrete four-coloring constraint particularly restrictive. Once the colors on one tetrahedron are fixed, a neighboring tetrahedron sharing an edge already contains two of the four colors. Its remaining sites must carry the other two, and the prescribed ε_t fixes their ordering. The coloring therefore propagates uniquely from one tetrahedron to the next, as illustrated in Figs. 7(a) and 7(b).

For the alternating assignment, this propagation is globally consistent. As shown in Fig. 7(c), it closes around the elementary sequence and generates the four-sublattice tetrahedral coloring of Fig. 6(a). Since the network of edge-sharing tetrahedra is connected, specifying the coloring on one tetrahedron fixes it throughout the lattice. The discrete CCI four-coloring is therefore unique up to even permutations of the four colors and lattice symmetries. Its global $\text{SO}(3)$ rotations generate the corresponding orbit of tetrahedrally ordered states.

For the uniform assignment, by contrast, the same propagation is inconsistent: following the constraint around the closed sequence in Fig. 7(d) returns a different color to an already fixed site. This provides a local obstruction to a global four-coloring, rather than a finite-size effect. We therefore focus below on the alternating, order-supporting branch.

The uniqueness of the discrete four-coloring does not imply that the coherent zero-mode variety is discrete. Starting from the four-sublattice state, consider a single (100) plane as in Fig. 6(b), keeping all spins outside the plane fixed. The sites in the plane belong to two sublattices. Choosing a common spin direction A' on the A sublattice—here and below, a primed color label denotes the common direction of that sublattice after a continuous deformation away from its reference value—the local Möbius completion rule determines a corresponding direction B' on the B sublattice. The same pair (A', B')

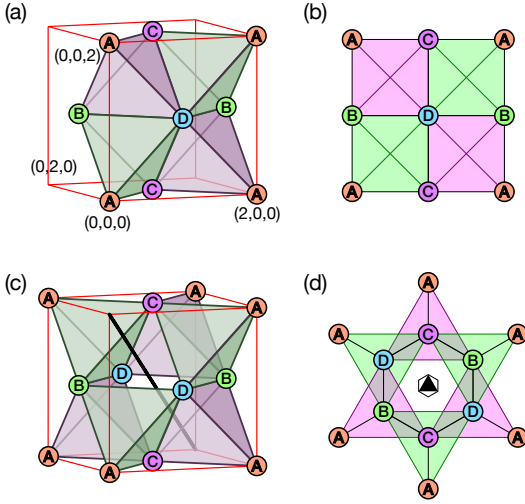


FIG. 8. Square- and honeycomb-lattice models obtained from fcc bilayers. (a,b) Two adjacent (010) planes project onto a square lattice, where the edge-sharing tetrahedra become crossed plaquettes. (c,d) Two adjacent $(\bar{1}\bar{1}1)$ triangular planes project onto a honeycomb lattice, where each tetrahedron becomes a site together with its three nearest neighbors. The two tetrahedron orientation classes are shown in green and magenta. The red frame indicates the projection of the cubic fcc unit cell, and the black line in (c,d) marks the $[\bar{1}\bar{1}1]$ axis of the S_6 roto-reflection.

satisfies the constraint on every tetrahedron intersecting the plane. Varying A' therefore generates a continuous family of exact coherent zero-energy states localized on that plane.

The construction can be repeated on sufficiently separated parallel (100) planes, so that no tetrahedron intersects two deformed planes. The corresponding deformations are then independent. Since a number proportional to the linear system size L of such planes can be chosen, the fcc model has at least $O(L)$ independent continuous zero-mode parameters around the four-sublattice state. These planar modes thus provide a subextensive family of exact zero-energy deformations. The construction gives a lower bound on the dimension of the coherent zero-mode variety, while the linearized analysis of Appendix G shows that it is exhaustive at first order. The kernel of $R(\mathbf{k})$ is nontrivial only on the three $\langle 100 \rangle$ axes of the Brillouin zone, giving $6L - 3$ complex infinitesimal modes on an L^3 torus of magnetic cells. These modes are exhausted by the planar deformations together with the three global Möbius modes. Thus there are no additional infinitesimal coherent deformations around the four-sublattice state.

Restricting the fcc construction to appropriate bilayers turns these planar zero modes into the line defects of the square and honeycomb models discussed next.

B. Square and honeycomb lattices

The square and honeycomb models are two-dimensional descendants of the fcc construction obtained by collapsing the bilayers shown in Fig. 8. In both cases, the two tetrahedron orientation classes are inherited from the fcc parent. Using the fixed-site-ordering convention introduced above, both Hamiltonians take the form of Eq. (6.1), with $\mathcal{O}_1$ and $\mathcal{O}_2$ corresponding to the green and magenta tetrahedra. Thus $\varepsilon = +1$ gives the uniform assignment of chirality labels between the two classes, while $\varepsilon = -1$ gives the alternating assignment.

The projection also makes the distinction between the fixed-ordering chirality label and the apparent planar circulation particularly transparent. Since the two tetrahedron classes are viewed from opposite sides of the collapsed bilayer, the projected circulation acquires an additional orientation-dependent sign. A uniform ε_t assignment therefore appears as alternating planar circulation, whereas an alternating assignment appears as uniform circulation. The corresponding honeycomb symmetry patterns are shown in Fig. 9.

In both the square and honeycomb geometries, only the alternating branch supports the four-sublattice tetrahedral coloring inherited from the fcc parent. For the uniform branch, propagation of the coloring constraint through the edge-sharing tetrahedra leads to an inconsistency, as in Fig. 7. On the honeycomb lattice, the same incompatibility is also reflected directly in the projected chirality pattern of Fig. 9.

Classical four-coloring enumeration provides an independent check. Before imposing the chirality constraint, the 16-site square and honeycomb clusters contain, respectively, 168 and 96 color-ice configurations. In each case, 12 satisfy the alternating CCI assignment, forming the orbit of a single tetrahedral coloring under the proper tetrahedral color permutations, whereas none satisfy the uniform assignment. For the square lattice, the absence of a uniform-branch CCI coloring was also verified on periodic tori up to 8×6 sites.

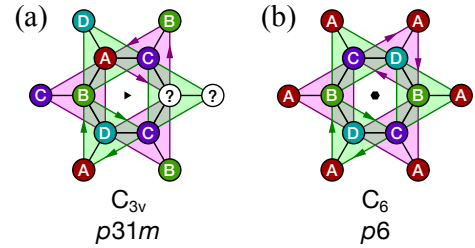


FIG. 9. Wallpaper symmetries of the chirality-decorated honeycomb models. Arrows indicate the circulation of the projected chirality term with respect to a fixed normal to the plane. (a) A uniform assignment of the fixed-ordering chirality labels ε_t produces alternating planar circulation, with point group C_{3v} and wallpaper group $p31m$. (b) An alternating assignment produces uniform planar circulation, with point group C_6 and wallpaper group $p6$.

The alternating branch also inherits the subdimensional zero modes of the fcc construction. The planar fcc deformation descends to the line modes shown in Fig. 10. Mutually separated parallel lines can be deformed independently, giving at least $O(L)$ continuous zero-mode parameters on an $L \times L$ sample. These line modes therefore form a subextensive family of exact coherent zero-energy deformations. As for the planar fcc modes, this construction provides a lower bound. The linearized analysis of Appendix G shows that it is exhaustive at first order: the kernel of $R(\mathbf{k})$ is confined to two lines in the Brillouin zone for the square lattice and three symmetry-related lines for the honeycomb lattice. This gives $4L - 1$ and $6L - 3$ complex infinitesimal modes, respectively, including the three global Möbius modes, in one-to-one correspondence with the line deformations of Fig. 10.

Exact diagonalization on the 16-site clusters further illustrates the distinction between the polarized and chirality-sensitive parts of the zero-energy kernel (Table II). The uniform and alternating branches have identical zero-mode counts in the high-total-spin sectors but separate at lower total spin.

The simplest chirality-blind states are the polarized 4:0 and 3:1 configurations, as they are annihilated by either local chiral parent term. In the coherent-state description, they belong to the three-coincident locus in Eq. (5.12), which is common to the conditions $p_- = 0$ and $p_+ = 0$ and therefore does not distinguish the two local chiralities. On the corner-sharing lattices, this same local 4:0/3:1 condition underlies the matching construction of Sec. VII C.

The high-spin kernel is, however, larger than this polarized product-state subset. Starting in the two-down-spin sector, zero-energy states can contain amplitudes for configurations with two down spins on the same tetrahedron, provided that the many-body wave function remains orthogonal to the forbidden chiral singlet on every tetrahedron. As long as the corresponding local con-

straint vectors are linearly independent, their total rank is fixed simply by their number and is therefore independent of the chirality assignment. This explains why the uniform and alternating spectra remain identical near full polarization even beyond the polarized product-state sector.

Farther from saturation, linear dependencies develop among the local constraints and can depend on the relative chiralities of overlapping tetrahedra. The two branches then begin to separate. For the 16-site square cluster the first difference occurs at $J = 4$, whereas for the honeycomb cluster it occurs at $J = 3$. The detailed fixed- J^z counting and its relation to the multiplet-resolved ED spectrum are given in Appendix I.

IX. FACE-SHARING TETRAHEDRA: THE TRIANGULAR LATTICE AND CHIRAL MAGNETIC ORDER

The triangular-lattice construction provides the strongest overlap of local tetrahedra considered here: neighboring four-site terms can share three spins, corresponding to an entire face of the tetrahedron. This strongly constrains the propagation of local zero modes and suppresses the extensive degeneracies found on the corner-sharing lattices. The zero-energy kernel nevertheless preserves a four-sublattice chiral ordered component, together with additional polarized zero modes.

A. Ground-state structure and Anderson tower

The local constraints remain compatible with a four-sublattice family of coherent zero modes containing the regular tetrahedral state shown in Fig. 11(b). With the site ordering of Fig. 11(a), the sublattice labels on every rhombus differ from those of a reference rhombus only by an even permutation. Consequently, any four-sublattice configuration satisfying the local CCI constraint on one reference rhombus satisfies it on every rhombus.

The face-sharing geometry strongly restricts deformations of this ordered family. In contrast to the fcc, square, and honeycomb constructions, we find no continuous local, line, or plane zero modes. The linearized analysis of Appendix G makes this statement exhaustive at first order: $R(\mathbf{k})$ has no kernel at nonzero momentum, while its kernel at Γ consists only of the three global Möbius deformations. Thus the four-sublattice state has no nonuniform infinitesimal coherent deformation. Additional exact product zero modes nevertheless survive in the polarized sector: since a local term annihilates every configuration with at most one down spin on a rhombus, any global configuration satisfying this condition is an exact zero mode. A maximal-density example is shown in Fig. 11(c).

To identify the ordered component of the full quantum kernel, we performed exact diagonalization on periodic

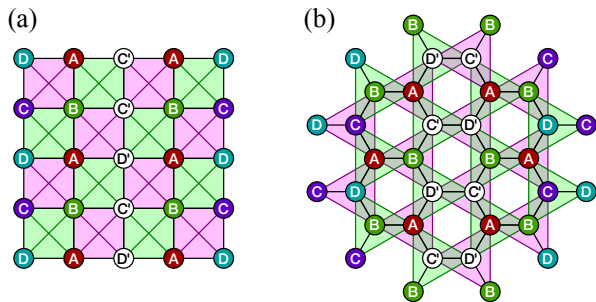


FIG. 10. Exact zero-energy line deformations on (a) the square and (b) the honeycomb lattice for the alternating, order-supporting chirality assignment. The green and magenta tetrahedra belong to the two orientation classes and carry opposite ε_t . Along the indicated line, the spin directions C' and D' can be varied continuously according to the Möbius completion rule, while all spins outside the line remain fixed.

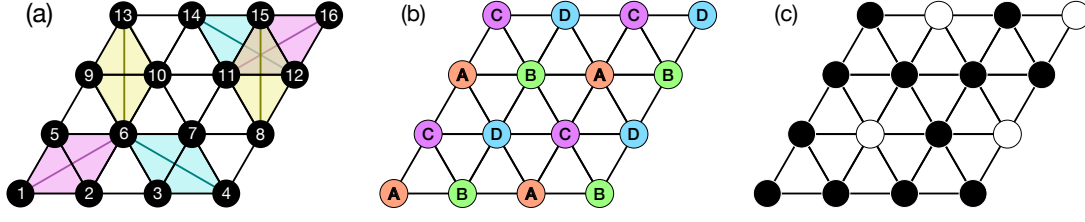


FIG. 11. Triangular-lattice construction and representative zero-energy states. (a) A 16-site cluster. The colored rhombi, interpreted as flattened tetrahedra, are the four-site supports of the local Hamiltonian in Eq. (3.15). They occur in three orientations (cyan, magenta, and yellow), and the full Hamiltonian sums the local term over all translations of all three orientations. Consequently, every elementary triangle belongs to three tetrahedra; for example, the triangle (11, 12, 15) is shared by (14, 11, 12, 15), (8, 12, 15, 11), and (11, 12, 16, 15). (b) A four-sublattice coherent zero-energy state. The regular tetrahedral configuration is one member of this family; more generally, the four sublattice spin directions may be continuously deformed subject to the local Möbius constraint $p_- = 0$, Eq. (5.8). (c) A maximal-density polarized zero mode built from local $\uparrow\uparrow\downarrow$ motifs (filled and open circles denote $\uparrow$ and $\downarrow$). At lower densities, the same local constraint admits disordered polarized zero-energy configurations.

TABLE III. Zero-energy multiplet content of the $N = 12$ and $N = 16$ triangular-lattice clusters, resolved by total spin J and space-group irreducible representation. For point group C_6 , the little co-groups at Γ , K , and M are C_6 , C_3 , and C_2 . We choose the counterclockwise sixfold rotation $\mathbf{a}_1 \rightarrow \mathbf{a}_2$, $\mathbf{a}_2 \rightarrow \mathbf{a}_2 - \mathbf{a}_1$; reversing its sense, or the chirality, complex-conjugates the labels. The irreps Γ_m , $K_{0,\pm}$, and $M_{\pm}$ are labeled by eigenvalues $e^{im\pi/3}$, $1, e^{\pm 2\pi i/3}$, and ± 1 , respectively. Their momentum stars contain 1, 2, and 3 points; the generic Q points belong to six-point stars with trivial little co-group. Blank entries denote zero multiplicity, and a dash denotes a momentum absent from the finite-cluster Brillouin zone. “Total” gives the number of zero-energy spin multiplets after accounting for the space-group irrep dimension. The final column gives the number $\mathcal{V}_J^{\tilde{S}}$ of spin- J multiplets in the tetrahedral Anderson tower, with $\tilde{S} = 3/2$ for $N = 12$ and $\tilde{S} = 2$ for $N = 16$; see Table X.

N	J	Γ_0	Γ_{+1}	C_6 ($d=1$)				Γ_3	C_3 ($d=2$)			C_2 ($d=3$)		C_1 ($d=6$)			Total	$\mathcal{V}_J^{\tilde{S}}$
				Γ_{-1}	Γ_{+2}	Γ_{-2}			K_0	K_+	K_-	M_+	M_-	Q_{12}	$Q_{16,1}$	$Q_{16,2}$		
12	0	1															1	1
	1											1					3	3
	2				1	1						1					5	5
	3	1								1		2		1			15	7
	4	1			1	1			1	1	1	1		2			24	5
	5								1			1		1			11	3
	6	1															1	1
16	0					1			–	–	–						1	1
	1								–	–	–	1					3	3
	2	1			1				–	–	–	1					5	5
	3					1			–	–	–	2					7	7
	4	1			1	1			–	–	–	2			1	1	21	9
	5	2		1	1		2		–	–	–	4	1		2	2	45	7
	6	1			2	2			–	–	–	3	2		3	3	56	5
	7								–	–	–	1			1	1	15	3
	8	1							–	–	–						1	1

$N = 12$ and $N = 16$ clusters preserving the full C_6 point symmetry. The zero modes were resolved by total spin and by the irreducible representations of the chiral wallpaper group $p6$; the results are summarized in Table III.

A characteristic finite-size signature of four-sublattice magnetic order is its Anderson tower of states [58–62]. Grouping the $N/4$ spin-1/2 sites of each sublattice into

a collective spin of maximal length

$$\tilde{S} = \frac{N}{8}, \quad (9.1)$$

the ordered-state subspace is the finite-spin realization $\mathcal{V}^{\tilde{S}}$ of four collective spins introduced for a single tetrahedron.

For a rigid tetrahedral rotor, the symmetry content at fixed total spin J is obtained by restricting the spin- J

representation of $SO(3)$ to the proper tetrahedral group,

$$\mathcal{M}_J = D^J \downarrow_{A_4}. \quad (9.2)$$

Its decomposition into A , $E_{\pm}$, and T irreducible representations is derived in Appendix J. Primitive translations map the one-dimensional tetrahedral irreps to the Γ point, whereas the triplet T carries the three momenta in the M -point star. With the rotation convention of Table III,

$$A \leftrightarrow \Gamma_0, \quad E_+ \leftrightarrow \Gamma_{-2}, \quad E_- \leftrightarrow \Gamma_{+2}, \quad T \leftrightarrow M_+. \quad (9.3)$$

Reversing the sense of C_6 , or equivalently the chiral sector, interchanges Γ_{+2} and Γ_{-2} .

At finite $\tilde{S}$, the rotor can additionally acquire a one-dimensional stabilizer character under the discrete tetrahedral identifications of the order-parameter space. As shown in Appendix J,

$$\eta_{\tilde{S}} = E_+^{\otimes N/4}, \quad \mathcal{V}_{\tilde{S}} \simeq \mathcal{M}_J \otimes \eta_{\tilde{S}}, \quad 0 \leq J \leq 2\tilde{S}. \quad (9.4)$$

The twist leaves the rotor multiplicities unchanged but cyclically permutes the one-dimensional Γ -point irreps. Thus the $N = 12$ cluster is untwisted, $\eta_{3/2} = A$, whereas the $N = 16$ cluster carries $\eta_2 = E_+$. In particular, the tower singlet changes from Γ_0 for $N = 12$ to Γ_{-2} for $N = 16$, while the $J = 1$ tower remains in the M_+ sector.

The exact spectra reproduce both the rotor multiplicities and these cluster-dependent symmetry labels. For $N = 12$, the tower exhausts the zero-energy sectors for $J = 0, 1, 2$, with multiplicities 1, 3, 5; at $J = 3$ the expected tower states remain present together with additional zero modes. For $N = 16$, the tower similarly exhausts the $J = 0, 1, 2, 3$ sectors, with multiplicities 1, 3, 5, 7, while additional states first appear at $J = 4$. The agreement of both multiplicities and space-group quantum numbers provides a finite-size symmetry fingerprint of the tetrahedral ordered component of the zero-energy kernel.

For $J > 2\tilde{S}$, the finite-spin space $\mathcal{V}_{\tilde{S}}$ no longer follows the rigid-rotor multiplicities, and additional zero modes become increasingly important. In particular, the complete $J = N/2$ and $J = N/2 - 1$ sectors are at zero energy because their highest-weight states contain, respectively, zero and one down spin and are annihilated by every local term. The full kernel is therefore strictly larger than the tetrahedral rotor sector.

For a generic Hamiltonian in a tetrahedrally ordered phase, the Anderson-tower levels acquire finite-size splittings that collapse in the thermodynamic limit. At the present frustration-free point, the entire finite-size realization of this tower is instead pinned exactly at zero energy. This is an exact representation-theoretic embedding of the chiral four-sublattice ordered component in the zero-energy kernel; by itself it does not establish thermodynamic long-range order, since the kernel also contains additional polarized and entangled states.

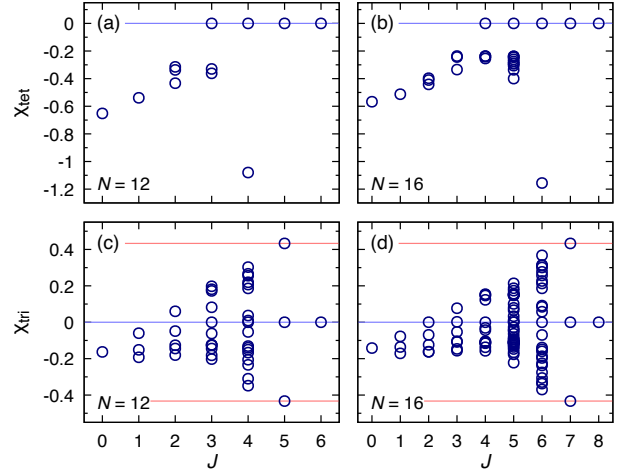


FIG. 12. Chirality spectra within the zero-energy kernels of the (a,c) 12-site and (b,d) 16-site triangular clusters, resolved by total spin. (a,b) Eigenvalues of the projected tetrahedral chirality $\mathcal{P}_0 \hat{\chi}_{\text{tet}}(t) \mathcal{P}_0$ on a single rhombus. All eigenvalues are nonpositive. (c,d) Eigenvalues of $\mathcal{P}_0 \hat{\chi}_{ijk} \mathcal{P}_0$ on a single elementary triangle. In contrast to the tetrahedral chirality, the triangle chirality is not sign definite and reaches the extrema $\pm\sqrt{3}/4$ available to three spin-1/2 degrees of freedom, indicated by the red horizontal lines.

B. Chirality spectra in the quantum common kernel

The local result of Sec. III D implies that every state in the common kernel obeys the chirality-sign constraint in Eq. (6.5). We now examine the spectrum of the local chirality operator within this many-body subspace.

Let $\mathcal{P}_0$ denote the projector onto $\mathcal{K}$. The expectation-value inequality Eq. (6.5), valid for every state in $\mathcal{K}$, is equivalently the operator statement

$$\varepsilon_t \mathcal{P}_0 \hat{\chi}_{\text{tet}}(t) \mathcal{P}_0 \preceq 0. \quad (9.5)$$

For the chirality sector used in the triangular-lattice calculations below, $\varepsilon_t = +1$, so the projected tetrahedral chirality is negative semidefinite on every rhombus.

We consider two local observables. The first is the tetrahedral chirality $\hat{\chi}_{\text{tet}}(t)$ on a rhombus, the direct quantum analogue of the quantity constrained by the parent Hamiltonian. The second is the chirality $\hat{\chi}_{ijk}$ of a single triangular face. Comparing the two determines whether the constraint selects handedness only at the tetrahedral level or, more strongly, also fixes the chirality sign of each individual face.

The finite-cluster spectra in Figs. 12(a,b) therefore illustrate an exact property of the complete common kernel rather than providing finite-size evidence for it; beyond the sign, they reveal how the eigenvalues are distributed within the sign-constrained sector.

The total-spin-resolved spectra of the projected local tetrahedral chirality are shown in Figs. 12(a,b). On the

12-site cluster, the spectrum progressively contracts toward zero with increasing total spin and vanishes identically for $J = 5$ and 6. These sectors lie entirely in the near-polarized part of the zero-energy kernel and represent the quantum counterpart of the $\chi_{\text{tet}} = 0$ end of the classical zero-mode family.

At the opposite end, the unique $J = 0$ zero-energy state carries a large, spatially uniform tetrahedral chirality. For $N = 12$, $\langle \hat{\chi}_{\text{tet}}(t) \rangle_{J=0} = -0.6518$ on every rhombus, compared with $\chi_{\text{tet}}^{\text{coh}} = -2/(3\sqrt{3}) \simeq -0.3849$ for the maximally chiral tetrahedral spin-1/2 coherent-product state. The singlet magnitude is therefore larger by a factor of approximately 1.69. For the 16-site singlet we similarly find $\langle \hat{\chi}_{\text{tet}}(t) \rangle = -0.5672$ on every rhombus, about 1.47 times the coherent-product value. Quantum fluctuations within $\mathcal{K}$ therefore preserve the selected handedness while allowing the local chirality to exceed the maximal magnitude accessible to a spin-1/2 coherent product state.

The chirality of an individual elementary triangle behaves quite differently. The projected local operator $\mathcal{P}_0 \hat{\chi}_{ijk} \mathcal{P}_0$ has eigenvalues of both signs and reaches $\pm\sqrt{3}/4$ [Figs. 12(c,d)]. This mirrors the coherent zero-mode variety, where the chiralities of individual tetrahedral faces are not sign definite and can change sign along a Möbius orbit. The sign-definite handedness imposed by the CCI constraint is therefore a property of the tetrahedrally oriented combination in Eq. (2.3), rather than of its individual triangular faces.

C. Alternative edge-sharing construction on the triangular lattice

The triangular lattice also admits an edge-sharing realization of the CCI constraint that is obtained naturally as a dimensional reduction of the fcc construction. Recall that the fcc lattice viewed along a $[111]$ direction consists of triangular layers in the usual $abcabc\dots$ stacking (lowercase letters denote stacking registries, to avoid confusion with the color labels). We retain one complete abc period and impose periodic boundary conditions in the $[111]$ direction, so that the layer following c is identified with the original a layer. Equivalently, we quotient the fcc lattice by the shortest lattice translation parallel to $[111]$ that returns a triangular layer to the same registry. Collapsing the $[111]$ coordinate then superposes the three a , b , and c registries. Their union forms a triangular lattice whose lattice constant is smaller by a factor $\sqrt{3}$ than that of an individual (111) layer.

An elementary fcc tetrahedron has a $1 + 3$ structure with respect to two neighboring (111) layers: one vertex lies in one layer and the other three form a triangle in the adjacent layer. After the three-layer quotient and projection, such a tetrahedron therefore becomes a four-site motif consisting of one site and three alternating nearest neighbors of the resulting triangular lattice, as shown in Fig. 13. The three bonds connecting the

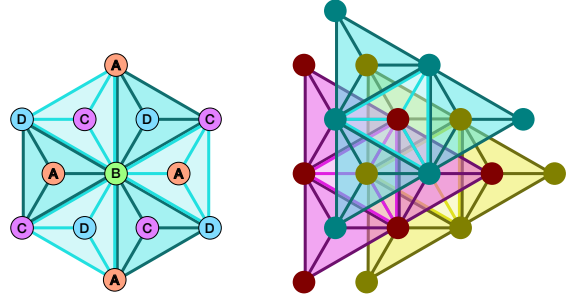


FIG. 13. Alternative edge-sharing construction on the triangular lattice. Each four-site motif consists of a central site and three alternating nearest neighbors. Of the six edges of the associated tetrahedron, three are nearest-neighbor bonds of the triangular lattice and the remaining three connect second neighbors. (a) Representative motifs of the two orientations, which are exchanged by a C_6 rotation. (b) The motifs can be organized into three symmetry-related rhombille families; together they generate the full C_6 -symmetric set of $2N$ four-site terms.

central site to the outer sites become nearest-neighbor bonds, whereas the three edges of the outer triangle connect second neighbors. The two orientations of the fcc tetrahedron project to the two motifs related by a C_6 rotation. Since the periodic three-layer fcc quotient contains two elementary tetrahedra per projected site, the complete construction contains $2N$ local CCI terms, precisely the full C_6 -symmetric set used below.

This fcc origin also clarifies the fate of the planar zero modes of the three-dimensional model. On the fcc lattice, a representative zero mode is obtained by deforming the two spin directions carried by a single (100) plane, while the remaining two sublattice directions are held fixed; the Möbius constraint fixes one of the two deformed directions in terms of the other [Fig. 6(b)]. In the square- and honeycomb-lattice bilayers, the intersection of such a plane with the retained slab is one dimensional, and the planar mode consequently descends to the line defects discussed above.

The $[111]$ reduction leading to the triangular lattice behaves differently because the transverse direction is periodic. A single (100) plane is not invariant under the three-layer identification; the periodic zero mode must therefore contain all of its images under the $[111]$ compactification. After projection, this periodically repeated family occupies two of the four tetrahedral sublattices throughout the two-dimensional system. Thus the fcc planar mode does not become a localized line defect on the triangular lattice. It instead reduces to a spatially periodic four-sublattice deformation,

$$(A, B, C, D) \longrightarrow (A', B', C, D), \quad B' = T_{z_C, z_D}(A'), \quad (9.6)$$

up to permutations of the four sublattice labels. The (010) and (001) planar families give the corresponding deformations of the other pairings of tetrahedral sublattices. These projected modes are therefore contained

in the continuously deformable four-sublattice family of coherent zero modes, rather than producing additional subdimensional defects. At linear order this projected family is precisely a global Möbius deformation—its tangent $\delta z_i \propto (z_i - z_C)(z_i - z_D)$ vanishes on the two fixed sublattices—consistent with the linearized analysis of Appendix G, which finds no nonuniform infinitesimal coherent modes for either triangular covering.

The important difference from the preceding face-sharing construction is the incidence structure of the local terms. Two distinct motifs (tetrahedra) of Fig. 13 share at most two sites in the infinite lattice, and hence are edge-sharing, whereas two rhombi of Fig. 11 can share three sites, corresponding to a common tetrahedral face. The triangular lattice therefore provides both edge- and face-sharing realizations of the same local chiral constraint.

The four-sublattice ordered state remains an exact zero-energy state of the alternative construction. With the site ordering indicated in Fig. 13, the four sublattice labels on each motif are related to those of the reference tetrahedron by an even permutation, so every local term selects the same CCI handedness. The complete finite-size ordered subspace $\mathcal{V}^{\tilde{S}}$, including the twisted Anderson-tower content discussed above, is therefore again contained in the common kernel. What changes is the number of additional zero modes allowed around this ordered component.

We tested this directly by exact diagonalization. The $N = 16$ cluster provides a clean realization of the edge-sharing incidence structure: no two distinct motifs share more than two sites. The corresponding zero-energy multiplet content is given in Table IV. The $N = 12$ torus is exceptional: periodic wrapping causes twelve pairs of motifs to share three sites, so that this cluster acquires an artificial face-sharing component. We therefore use the $N = 16$ results for the comparison below. The decomposition into the three rhombille families is a property of the infinite lattice; on the $N = 16$ torus the Hamiltonian is defined directly from the full C_6 -symmetric set of $2N$ motifs.

The difference from the face-sharing model is already pronounced in the low-spin spectrum. For $N = 16$, the edge-sharing construction has 13, 15, 33, and 52 zero-energy multiplets in the $J = 0, 1, 2, 3$ sectors, respectively. In the face-sharing construction, the corresponding sectors contain only 1, 3, 5, and 7 multiplets and are exhausted by the tetrahedral rotor. Nevertheless, the symmetry sectors required by the twisted ordered component remain present in Table IV: the Γ_{-2} tower singlet occurs at $J = 0$, the M_+ triplet at $J = 1$, and the higher- J tower irreps are contained in the corresponding zero-energy sectors. Thus the four-sublattice ordered component survives, but the weaker overlap leaves a much larger kernel around it.

The difference is also substantial in the total dimension of the kernel. Weighting the multiplets by $2J + 1$, the $N = 16$ edge-sharing model has 3684 zero-energy

states, compared with 1738 for the face-sharing rhombus construction. By contrast, the high-spin polarized sectors are unchanged: the complete $J = 8$ and $J = 7$ sectors remain at zero energy, giving respectively one and 15 multiplets, as required by the zero- and one-down-spin argument.

The linearized analysis sharpens this comparison (Appendix G): around the four-sublattice state, neither triangular construction has a nonuniform infinitesimal coherent zero mode. In both cases $R(\mathbf{k})$ has trivial kernel for every $\mathbf{k} \neq 0$, even though the edge-sharing claw construction has a much larger quantum zero-energy kernel. The number of infinitesimal coherent modes and the dimension of the full quantum kernel can therefore vary independently, and the appearance of line modes is not an automatic consequence of edge sharing.

This comparison separates the effect of constraint connectivity from that of the underlying lattice. The same triangular lattice, the same local CCI projector, and the same four-sublattice chiral order give very different zero-energy kernels depending on how strongly the four-site terms overlap. Face sharing suppresses the additional low-spin zero modes and leaves a rigid ordered component, whereas edge sharing permits a much larger residual kernel. The hierarchy between corner-, edge-, and face-sharing constructions is therefore controlled by the overlap of the local constraints rather than by the lattice geometry alone. At the level of coherent deformations, this hierarchy is particularly transparent: the zero set of $R(\mathbf{k})$ extends over the full Brillouin zone for the corner-sharing checkerboard and pyrochlore lattices, is restricted to lines for the fcc, square, and honeycomb constructions, and reduces to Γ alone for both triangular coverings, as shown in Appendix G.

X. SUMMARY OF RESULTS

For reference, we collect here the principal results established in the preceding sections. The purpose of this section is to provide a compact map of the construction and its consequences; their physical interpretation, relation to other constrained spin systems, and possible extensions are discussed separately in Sec. XI. The results naturally separate into three levels: the local quantum kernel of a single tetrahedron, the corresponding coherent-state zero-mode variety, and the common kernel obtained by imposing the local constraints on extended lattices [cf. Eq. (6.6)]. Tables V–VII summarize these statements, one table per level, and give the sections in which they are derived.

Several distinctions summarized in Tables V–VII are useful in organizing the results. First, the rotated CCI colorings form only a subset of the coherent zero-mode variety, which in turn is contained in the complete quantum common kernel [Eq. (6.6)]. The Möbius completion rule and its holonomies characterize the coherent product-state sector, whereas exact-diagonalization spec-

TABLE IV. Zero-energy multiplet content of the alternative edge-sharing triangular-lattice construction, resolved using the same space-group conventions as Table III. Blank entries denote zero multiplicity and a dash denotes a momentum not contained in the finite-cluster Brillouin zone. The “Total” column gives the number of zero-energy spin multiplets after weighting by the dimension of the corresponding space-group irrep. The $N = 12$ data are included for completeness, but periodic wrapping produces twelve pairs of motifs with a three-site overlap; the $N = 16$ cluster realizes the intended edge-sharing geometry without such overlaps.

N	J	$C_6 \ (d=1)$						$C_3 \ (d=2)$			$C_2 \ (d=3)$		$C_1 \ (d=6)$			Total
		Γ_0	Γ_{+1}	Γ_{-1}	Γ_{+2}	Γ_{-2}	Γ_3	K_0	K_+	K_-	M_+	M_-	Q_{12}	$Q_{16,1}$	$Q_{16,2}$	
12	0	3				1			1					–	–	6
	1		1				1	1	1		1			–	–	9
	2	1			2	2		2	1	1	1			–	–	16
	3	2	1				1	1	2		2		1	–	–	22
	4	2			1	2		2	2		1	1	2	–	–	31
	5							1			1		1	–	–	11
16	6	1												–	–	1
	0	2			2	3		–	–	–		2	–			13
	1							–	–	–	1		–	1	1	15
	2	4			3	2		–	–	–	2	2	–	1	1	33
	3	1				2	1	–	–	–	4		–	3	3	52
	4	4			4	4		–	–	–	5	2	–	4	4	81
	5	3	1	1	1		2	–	–	–	5		–	7	7	107
	6	2			2	3		–	–	–	3	3	–	4	4	73
	7							–	–	–	1		–	1	1	15
	8	1						–	–	–			–			1

tra, kernel dimensions, and matching bounds refer to the full quantum zero-energy space.

Second, the local CCI constraint fixes the handedness of the tetrahedrally oriented combination of face chiralities, but does not fix its magnitude and does not impose a definite chirality sign on each individual triangular face. This distinction persists in the complete many-body quantum kernel, as demonstrated explicitly by the projected chirality spectra in Sec. IX B.

Finally, the lattice constructions separate coherent flexibility from quantum ground-state degeneracy. Around the same four-sublattice tetrahedral state, the support of the linearized zero modes changes from the full Brillouin zone on the corner-sharing lattices, to one-dimensional momentum-space sets in the fcc, square, and honeycomb models, and finally to Γ alone in the triangular constructions. At the same time, the comparison of the two triangular coverings shows that identical harmonic zero-mode content does not imply identical quantum kernel dimensions. The overlap structure of the local tetrahedral constraints therefore controls several distinct properties of the zero-energy space that need not vary together.

XI. OUTLOOK

We set out to answer a concrete question: can a local quantum spin Hamiltonian enforce an ice-type rule that knows about handedness? The models constructed here answer it in the affirmative, and in a stronger form than we had anticipated. A single four-site term, built only from scalar chiralities and products of Heisenberg exchanges, annihilates every chiral four-coloring state and all of its global rotations, for any spin length. Three technical facts carry most of the weight. For $S = 1/2$ the term is the projector onto one member of the time-reversal pair of tetrahedral singlets, so selecting a handedness costs exactly one state in sixteen. For general S the same statement takes the form $\mathcal{B}^\dagger \mathcal{B}$ with a singlet-annihilation condition whose classical solutions obey a Möbius completion rule: three spins fix the fourth, loops fix their own consistency, and the ground-state counting reduces to fixed points and matchings, with residual-entropy bounds that exceed the Pauling and Lieb values for ice. Finally, the chirality window $-16/(3\sqrt{3}) \leq \chi_{\text{tet}} \leq 0$ pins down what the constraint actually enforces: the sign of the local handedness, exactly, and nothing else. In this precise sense the coherent zero-mode variety is the deformable envelope of the rigid classical chiral states of Ref. [7].

The large degeneracy is not a defect of the construction; it is forced. A local, spin-rotation-invariant, positive-semidefinite term whose kernel contains one co-

TABLE V. Summary of the principal results for the single-tetrahedron quantum problem. The second column states the result, and the last column gives the corresponding derivation or numerical characterization in the main text; additional proofs and technical details are given in the appendices cited in the corresponding sections.

Scope	Result	Location
Spin-1/2 parent term	The rotated CCI product states of one handedness span 15 of the 16 dimensions of the four-spin Hilbert space; the local parent Hamiltonian is the rank-one projector onto the complementary chiral tetrahedral singlet, $\mathcal{H}_{\text{CCI}}^{(1/2)} = +\rangle\langle+ $, expressible entirely in terms of Heisenberg exchange, scalar spin chirality, and products of exchanges.	Sec. III B; Eqs. (3.13) and (3.15)
Arbitrary spin	For four spins of arbitrary length S , an $\text{SU}(2)$ -invariant, positive-semidefinite parent Hamiltonian can be constructed in the same local operator basis. It annihilates all global rotations of the CCI product states of the selected handedness.	Sec. III C; Eq. (3.17)
Singlet-annihilation form	The arbitrary- S parent Hamiltonian factorizes exactly as $\mathcal{H}_{\text{CCI}}^{(S)} = \mathcal{B}^\dagger \mathcal{B}$, where $\mathcal{B}$ is a four-site $\text{SU}(2)$ -singlet annihilation operator that removes one Schwinger boson from every site. The local zero-energy condition is therefore exactly $\mathcal{B} \Psi\rangle = 0$.	Sec. IV; Eqs. (4.9) and (4.14)
Complete local kernel	The kernel of $\mathcal{B}$ is exactly the subspace $\mathcal{V}^S$ spanned by the globally rotated CCI product states, $\mathcal{V}^S = \ker \mathcal{B}$. Its dimension is $(2S+1)^4 - (2S)^4$, and its total-spin content follows from a sector-by-sector recursion between four spin- S and four spin- $(S - \frac{1}{2})$ degrees of freedom.	Sec. IV C; Eqs. (4.21) and (4.24); Table I
Quantum handedness	Every normalized state in the complete local quantum kernel, including entangled states, obeys a definite sign constraint on the tetrahedral chirality. In the reference convention, $\langle \hat{\chi}_{\text{tet}} \rangle \leq 0$; the time-reversed parent Hamiltonian reverses this inequality.	Sec. III D; Eq. (3.23)
Both chiral sectors	Combining the two complex-conjugate singlet-annihilation operators gives a positive-semidefinite parent Hamiltonian for the full color-ice subspace containing both handedness sectors. The resulting operator is invariant under the full site-permutation group S_4 . For $S \geq 1$, its complete kernel is the span of the globally rotated color-ice states of both chiralities.	Secs. IV B and IV C; Eq. (4.16)
Local spectrum	The nonzero spectra of $\mathcal{B}^\dagger \mathcal{B}$ (spin S) and $\mathcal{B} \mathcal{B}^\dagger$ [spin $(S - \frac{1}{2})$] are paired, giving the exact local gap $\Delta_{\text{loc}}(S) = 4S^2$ —a local excitation scale that does not by itself imply a many-body spectral gap.	Sec. IV D; Eq. (4.35)

herent product state must annihilate the entire rotated family, and the singlet-annihilation structure then admits the polarized states and their matching descendants as well. Every parent Hamiltonian of this kind therefore sits at a degenerate multiphase point adjacent to a ferromagnetic phase, as the kagome three-coloring point does in its own setting [32]. The productive attitude toward such a point is to perturb away from it and map the surrounding phase diagram; quantum melting of noncoplanar tetrahedral order is indeed known to produce chiral spin liquids [63], and the field theories of such chiral spin-liquid-to-spin-crystal transitions are now being formulated [64]. Here we should be precise about what is and is not in hand. On the corner-sharing lattices we know the product-state sector of the kernel completely—the Möbius coordinates parametrize it—and we know the polarized matching states, but we do not have a closed-form basis for the full kernel of the pyrochlore model; controlled degenerate perturbation theory over the entire kernel is therefore still out of reach in general. What is well posed already is the exploration of the neighborhood: weak antiferromagnetic exchange, biquadratic terms, or

a chirality field $\lambda \hat{\chi}_{\text{tet}}$ each define a definite path away from the solvable point, and one can ask which zero-energy states they favor. It is natural to expect a selection effect—by energy, by fluctuations [65, 66], or by the environment in the pointer-state sense discussed below—and the chiral color-ice states themselves are the obvious candidates to be selected. Whether that happens, or the ferromagnet wins, or an intermediate regime intervenes, is open on both the classical and the quantum side.

What do correlations look like within the constrained ensemble itself? Ice-type manifolds form Coulomb phases: an emergent divergence-free field produces pinch points in the structure factor [67], and the classification of Refs. [47, 48] rests on the linear constraints that generate them. The condition $p_- = 0$ is not linear, so chiral color ice falls outside this classification, and whether its correlations retain any pinch-point-like structure is a sharp open question. The Möbius completion rule offers a concrete route to it: because the fourth spin of every tetrahedron is determined by the other three, ensembles of configurations that are *exactly at zero energy* can be generated by growing the lattice tetrahedron by tetrahe-

TABLE VI. Summary of the principal results for the coherent-state geometry. Columns as in Table V.

Scope	Result	Location
Coherent modes	zero For products of spin-coherent states, the zero-energy condition is independent of S and reduces to the multi-affine polynomial equation $p_-(z_1, z_2, z_3, z_4) = 0$ on the Riemann sphere. Away from the exceptional locus, three spin directions determine the fourth uniquely.	Sec. V; Eq. (5.8)
Möbius completion	Solving the coherent zero-mode constraint for one spin gives an invertible Möbius transformation, $z_4 = T_{z_1, z_2}(z_3)$. For a fixed noncoincident pair (z_1, z_2) , these two points are the fixed points of the map and the remaining points form six-cycles. Unique completion fails only on the three-coincident branch.	Sec. V; Eqs. (5.10) and (5.12); Appendix C 2
Coherent chirality	The coherent zero-mode variety has a definite handedness but a continuously varying chirality magnitude,	Sec. V; Eq. (5.17); Appendices C 3 and C 5
$-\frac{16}{3\sqrt{3}} \leq \chi_{\text{tet}} \leq 0$		
	for the reference sector. The lower endpoint is attained by the regular tetrahedral frame, whereas the achiral endpoint occurs on the three-coincident branch. Independently of the zero-mode constraint, $16/(3\sqrt{3})$ bounds the tetrahedral chirality of any four unit spin directions.	
Loop holonomy	Successive local completion maps around a closed loop define a Möbius holonomy $\mathcal{M}_L$. Generic nonidentity holonomy gives isolated coherent loop solutions, whereas identity holonomy produces a continuous family. For identity holonomy this statement extends to the complete quantum loop problem: the kernel is spanned by the propagated coherent states and has dimension $2SL + 1$.	Secs. VI, VII B, and VII B 3; Eq. (7.8)

dron and completing each new tetrahedron by Eq. (5.10). Turning such ensembles into a statement about the correlations of the coherent zero-mode variety, however, requires care that we have not undertaken here: a declared probability measure (the variety carries no canonical one), a treatment of loop closure and Möbius holonomy on periodic clusters, and controlled system-size and error analyses. We therefore leave the correlation pattern of chiral color ice as an open problem. We caution, moreover, against expecting the correlation pattern of a stable phase. The solvable point sits on the boundary of the ferromagnetic phase, and its closest relative, the multicritical point of the zigzag chain [29], is a critical point rather than a phase; the appropriate question is not the field theory of a chiral-color-ice *liquid*, but the nature of this multiphase point itself—whether it is critical, what gapless structure and entanglement it carries, and what controls the transitions out of it. The completion-rule sampling and the exact zero-energy kernels of the quantum clusters give this question an unusually concrete starting point.

The ordered side of the phase diagram comes with its own exact statement, and with a caveat worth repeating. The four-sublattice coloring survives only when the chirality assignment *alternates* between the two inversion-related families of tetrahedra: on the fcc, square, and honeycomb lattices with this assignment (and on the triangular lattice, whose face-sharing covering admits the coloring directly), the kernel contains the chiral four-

sublattice states and the complete set of multiplets of their Anderson tower, all at zero energy. A weak perturbation that favors this ordered component would split these multiplets into the tower from which the symmetry-broken state is assembled in the thermodynamic limit; which part of the kernel a given perturbation selects is a question we leave open. The kernel is not exhausted by the ordered component—a subextensive family of planar (fcc) or line (square, honeycomb) deformations survives, along with the discrete polarized motifs, and only on the triangular lattice do the continuous deformations disappear entirely—but this residual degeneracy is parametrically smaller than the extensive one of the corner-sharing lattices, and it leaves the tower content intact. For the uniform assignment, the coloring rule is frustrated and this entire structure is absent—we verified by exhaustive enumeration that not a single four-coloring survives on the fcc, square, or honeycomb lattices. This puts the tower-of-states analysis of Ref. [39] into an unusually clean setting. There, the decomposition of classical order into tower eigenstates, and the selection of low-entanglement pointer states by an environment, had to be extracted numerically from split spectra; here the kernel is exactly degenerate, its product-state members are classified completely, and each of them carries a definite handedness. The emergence of a classical chirality from a quantum-degenerate kernel can therefore be studied here without any of the usual finite-size ambiguity. On the technical side, the matrix-product construc-

TABLE VII. Summary of the principal results for the extended lattices. Columns as in Table V.

Scope		Result	Location
Many-body common kernel		Summing the local positive-semidefinite terms gives a lattice Hamiltonian whose ground-state space is their common kernel; whenever a compatible CCI coloring exists, the model is frustration free. The handedness condition extends to the complete many-body kernel: $\varepsilon_t \langle \hat{\chi}_{\text{tet}}(t) \rangle \leq 0$ on every tetrahedron and for every ground state.	Sec. VI; Eqs. (6.4) and (6.5)
Corner sharing		The checkerboard and pyrochlore constructions have extensively degenerate quantum ground-state spaces. Embedded loops generate explicit coherent and quantum zero modes, while for $S = 1/2$ a polarized subset of the common kernel maps onto matchings of the dual square or diamond lattice.	Secs. VII, VII B, and VII C; Table II
Extensive degeneracy		For $S = 1/2$, rigorous bounds on the matching partition function give $1.3919^N \lesssim \mathcal{D}_{\text{match}} \lesssim 1.3964^N$ for an explicitly constructed polarized subset of the corner-sharing ground-state space; the lower bound alone establishes exponential growth of the full quantum degeneracy, and independent loop constructions give complementary bounds for arbitrary S .	Sec. VII D; Eqs. (7.10), (7.11), and (7.15)
fcc lattice		For the alternating assignment, constraint propagation fixes the four-sublattice tetrahedral coloring uniquely once one tetrahedron is specified; the uniform assignment is frustrated. Exact coherent zero-energy deformations localized on (100) planes yield a subextensive family of zero modes.	Sec. VIII A
Square and honeycomb		The alternating branch supports the four-sublattice state, the uniform branch admits no compatible coloring, and the fcc planar modes descend to exact line deformations, with $4L - 1$ (square) and $6L - 3$ (honeycomb) infinitesimal coherent zero modes.	Sec. VIII B; Eqs. (G42) and (G53)
Triangular lattice: face sharing		The face-sharing rhombus construction retains the four-sublattice tetrahedral zero-energy family but has no nonuniform infinitesimal coherent deformation about it: only the three global Möbius directions survive. Exact diagonalization identifies the corresponding finite-size tetrahedral Anderson tower inside the common kernel, together with additional zero modes at higher total spin.	Sec. IX; Eq. (G62); Table III
Tetrahedral Anderson tower	An-	The finite-size ordered component can be identified with the four-collective-spin space $\mathcal{V}^{\bar{S}}$. Its spin- J multiplicities follow from restricting the $\text{SO}(3)$ representation to the proper tetrahedral group A_4 . A finite-spin stabilizer character can shift the discrete lattice quantum numbers without changing the multiplicities. For the triangular clusters, the predicted multiplicities and space-group sectors occur exactly at zero energy.	Sec. IX; Eqs. (9.3) and (9.4); Appendix J
Quantum chirality spectra	chirality	Within the complete triangular-lattice common kernel, the projected tetrahedral chirality operator is negative semidefinite in the selected sector, whereas the chirality of an individual triangular face is not sign definite. Low-spin quantum zero modes can also carry a tetrahedral-chirality magnitude exceeding the maximal value available to a spin-1/2 coherent product state.	Sec. IX B; Eq. (9.5)
Triangular lattice: edge sharing	lattice: edge sharing	An alternative edge-sharing realization on the same triangular lattice contains the same four-sublattice ordered component and has the same harmonic zero-mode content—only the global Möbius modes remain— but a substantially larger full quantum zero-energy kernel. On the $N = 16$ cluster, the edge-sharing and face-sharing constructions contain 3684 and 1738 zero-energy states, respectively.	Sec. IX C; Eq. (G67); Table IV
Constraint-overlap hierarchy		The linearized coherent zero modes occupy the full Brillouin zone for the corner-sharing checkerboard and pyrochlore lattices, one-dimensional momentum-space sets for the fcc, square, and honeycomb constructions, and only Γ for the two triangular constructions. The corresponding numbers of infinitesimal coherent deformations are therefore extensive, subextensive, and $O(1)$, respectively.	Appendix G; Table VIII
Harmonic quantum modes	versus zero	The two triangular-lattice constructions have identical infinitesimal coherent zero-mode spaces around the four-sublattice state but substantially different full quantum kernels; coherent flexibility and quantum kernel dimension are therefore independent characteristics of the constraint problem.	Sec. IX C; Appendix G
Polarized/high-spin sector		Fully polarized states and the one-spin-flip sector are annihilated by every local CCI term independently of the chirality assignment. Consequently, the complete $J = N/2$ and $J = N/2 - 1$ multiplet sectors are at zero energy. Differences between chirality assignments appear only at lower total spin, where linear dependencies can develop among the overlapping local constraints.	Secs. VII A, VIII B, and IX; Appendix I

tion of Ref. [29] for degenerate frustration-free points applies to our models on strips and cylinders. The zero-energy kernel, as the common kernel of all local terms, remains pinned at zero energy under arbitrary reweighting of those terms—the structural prerequisite for protected embeddings inside thermalizing Hamiltonians [68, 69]; whether such embeddings produce genuine scar phenomenology, such as long-lived chirality oscillations [70], is left open here.

We should be honest about where such Hamiltonians might exist. We know of no material whose microscopic couplings realize Eq. (3.17) as written. What exists in nature are the ingredients: scalar-chirality interactions generated by ring exchange and orbital fields, applied fields, or circular driving in Mott insulators [10, 19, 40–42], the tetrahedral triple- $\mathbf{q}$ order that itinerant frustrated magnets select, with its spontaneous topological Hall response [12, 13], and tetrahedral spin frames in the cyclic phase of spin-2 condensates [22]. The most direct route would be engineered, and we state its status carefully. As an always-on analog Hamiltonian, Eq. (3.17) requires a specific $SU(2)$ -invariant combination of three- and four-spin couplings that no current platform provides natively—although programmable three- and four-body interactions have been demonstrated for small trapped-ion registers [71]—and the inverse quantum-simulation framework of Ref. [43], which reconstructs a parent Hamiltonian for a variationally prepared target state, is at present a theoretical proposal. For $S = 1/2$, a digital route is more immediate: each local term is the rank-one projector of Eq. (3.13), so exponentiating or measuring a single tetrahedron term reduces to a four-qubit controlled-phase primitive conjugated by a fixed local basis change, a gate class now demonstrated with high fidelity in neutral-atom arrays [72]. When such capabilities mature, exactly solvable points with fully computable ground-state spaces are the natural benchmarks against which they will be validated. Meanwhile, the solvable point serves the same purpose that the Majumdar–Ghosh and AKLT points have long served: not as a model of a particular compound, but as the fixed reference from which the surrounding phase diagram can be understood.

The construction generalizes in more than one direction. The recipe has three inputs: a motif of sites, a set of colored product states on it, and the symmetry group under whose orbit the target set of states is completed before projecting onto its complement. Nothing ties these inputs to four colors, tetrahedra, or the full spin-rotation group. Keeping the motif and colors but shrinking the symmetry to rotations about a single axis reproduces, by the same steps, anisotropic coloring models of the kagome three-coloring type [32, 33]—which shows that the known XXZ solvable points and the present $SU(2)$ -symmetric ones are members of one family, distinguished only by how much symmetry the constraint is asked to respect. Changing the local motif instead changes the constrained objects themselves: projectors on triangular faces of the same corner-sharing lattices produce fully

packed valence-bond *loop* manifolds as exact spin-1 kernels [38], so colorings, loops, and the chiral constraint studied here are all members of one family of exactly enforceable local rules. In the other direction, three-color chiral constraints on triangles, larger simplices, and $SU(N)$ analogs can be pursued with the same tools. Spin ice demonstrated that a local flux rule generates emergent gauge fields and fractionalization. The models constructed here demonstrate that a local handedness rule is equally consistent, and that its consequences—residual entropy beyond that of ice, a constraint geometry outside the linear Coulomb-phase classification, and exactly protected subspaces of chiral states—can be worked out in full. What grows from this point, under perturbations, at finite temperature, and in other symmetry classes, is now a definite program rather than a speculation.

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Data availability.—The numerical material supporting the conclusions of this article—exact-diagonalization multiplet tables, four-coloring enumerations, matching-count calculations, chirality spectra, and the symbolic-verification and mode-counting scripts—is available from the authors upon reasonable request.

Appendix A: Scalar chirality

This appendix collects operator identities relating scalar chirality to permutation and exchange operators, and used to rewrite the spin-1/2 parent Hamiltonian in terms of chirality operators.

1. Three spin-1/2 sites

For three spin-1/2 sites, the scalar chirality of Eq. (1.1) can be expressed through cyclic permutations as

$$\hat{\chi}_{ijk} = \frac{i}{4} (\mathcal{P}_{ijk} - \mathcal{P}_{ikj}), \quad \mathcal{P}_{ikj} = \mathcal{P}_{ijk}^{-1}, \quad (\text{A1})$$

as noted in Ref. [10]. Its eigenvalues are $\pm\sqrt{3}/4$ on the two total-spin-1/2 doublets and zero on the total-spin-3/2 quartet. Consistently, [11]

$$\begin{aligned} \hat{\chi}_{ijk}^2 &= \frac{3}{32} - \frac{1}{8} (\hat{\mathbf{S}}_i \cdot \hat{\mathbf{S}}_j + \hat{\mathbf{S}}_j \cdot \hat{\mathbf{S}}_k + \hat{\mathbf{S}}_k \cdot \hat{\mathbf{S}}_i) \\ &= \frac{15}{64} - \frac{1}{16} (\hat{\mathbf{S}}_i + \hat{\mathbf{S}}_j + \hat{\mathbf{S}}_k)^2. \end{aligned} \quad (\text{A2})$$

2. Four spin-1/2 sites

On a tetrahedron, the consistently oriented sum of the four face chiralities defines the tetrahedral chirality $\hat{\chi}_{\text{tet}}$ of Eq. (2.3). For four spin-1/2 sites, its eigenvalues are $\pm\sqrt{3}$ on the two chiral singlets of Eq. (3.6) and zero on the remaining fourteen states.

Applying Eq. (A2) to the four faces gives

$$\sum_{\langle ij \rangle} \hat{\mathbf{S}}_i \cdot \hat{\mathbf{S}}_j = \frac{3}{2} - 4 \sum_f \hat{\chi}_f^2, \quad (\text{A3})$$

where f runs over the four faces. Similarly, using the spin-1/2 product rule $S_i^\alpha S_i^\beta = \frac{1}{4} \delta_{\alpha\beta} + \frac{i}{2} \epsilon_{\alpha\beta\gamma} S_i^\gamma$, the symmetric quadrilinear operator

$$\mathcal{Q} = (\hat{\mathbf{S}}_1 \cdot \hat{\mathbf{S}}_2)(\hat{\mathbf{S}}_3 \cdot \hat{\mathbf{S}}_4) + (\hat{\mathbf{S}}_1 \cdot \hat{\mathbf{S}}_3)(\hat{\mathbf{S}}_2 \cdot \hat{\mathbf{S}}_4) + (\hat{\mathbf{S}}_1 \cdot \hat{\mathbf{S}}_4)(\hat{\mathbf{S}}_2 \cdot \hat{\mathbf{S}}_3) \quad (\text{A4})$$

can be written as

$$\mathcal{Q} = \frac{3}{16} - \sum_f \hat{\chi}_f^2 + \frac{1}{2} \hat{\chi}_{\text{tet}}^2. \quad (\text{A5})$$

These operators also have a simple representation in terms of the total spin $\hat{\mathbf{S}}_{\text{tet}} = \sum_{i=1}^4 \hat{\mathbf{S}}_i$. In particular,

$$\mathcal{Q} = \frac{1}{8} (\hat{\mathbf{S}}_{\text{tet}}^2)^2 - \frac{7}{8} \hat{\mathbf{S}}_{\text{tet}}^2 + \frac{15}{16}, \quad (\text{A6})$$

and

$$\hat{\chi}_{\text{tet}}^2 = \frac{1}{4} (\hat{\mathbf{S}}_{\text{tet}}^2 - 2) (\hat{\mathbf{S}}_{\text{tet}}^2 - 6) = 3 \mathcal{P}_{S_{\text{tet}}=0}. \quad (\text{A7})$$

Thus $\hat{\chi}_{\text{tet}}^2$ is three times the projector onto the two-dimensional total-spin singlet sector. The chirality itself is not a function of $\hat{\mathbf{S}}_{\text{tet}}^2$, since it distinguishes the two singlets by their eigenvalues $\pm\sqrt{3}$.

3. Scalar chirality in coherent-state coordinates

For a product of spin-coherent states (5.1) with stereographic coordinates z_i , the S -normalized scalar chirality of an oriented triangle is

$$\chi_{ijk} = \frac{4 \operatorname{Im} [(1 + z_i^* z_j)(1 + z_j^* z_k)(1 + z_k^* z_i)]}{(1 + |z_i|^2)(1 + |z_j|^2)(1 + |z_k|^2)}. \quad (\text{A8})$$

This representation is particularly convenient for evaluating chirality on the coherent zero-mode variety parametrized by the Möbius completion rule (5.10).

Appendix B: Derivation of the spin- S Hamiltonian

Here we derive Eq. (3.17) directly from the spin-1/2 parent Hamiltonian. Let $n = 2S$ and consider the balanced complete four-partite, or Turán, graph $T(4n, 4) = T(8S, 4)$, whose vertex set is partitioned into four independent subsets $V_1, \dots, V_4$, each containing n spin-1/2 degrees of freedom. Selecting one vertex from each subset defines one of the n^4 four-site cliques. We place the spin-1/2 CCI parent term on every such clique and sum,

$$\tilde{\mathcal{H}} = \sum_{v_1 \in V_1} \cdots \sum_{v_4 \in V_4} \mathcal{H}_{\text{CCI}}^{(1/2)}(v_1, v_2, v_3, v_4). \quad (\text{B1})$$

The different terms in Eq. (3.15) acquire simple multiplicity factors in this sum. The constant term occurs n^4 times. For a fixed pair of subsets V_i, V_j , a given bilinear term is accompanied by arbitrary choices of the vertices in the other two subsets and therefore acquires a factor n^2 . Similarly, a scalar-chirality term involving three subsets acquires a factor n , while a product of two bilinears involving all four subsets acquires no additional multiplicity.

Introduce the collective spins

$$\hat{\mathbf{S}}_i = \sum_{v \in V_i} \hat{\mathbf{S}}_v^{(1/2)}, \quad i = 1, \dots, 4. \quad (\text{B2})$$

Using the factorization of the sums over the independent subsets, Eq. (B1) becomes

$$\begin{aligned} \tilde{\mathcal{H}} &= \frac{n^4}{16} + \frac{n}{2\sqrt{3}} \sum_{\langle ijk \rangle} \hat{\mathbf{S}}_i \cdot (\hat{\mathbf{S}}_j \times \hat{\mathbf{S}}_k) \\ &\quad - \frac{n^2}{12} \sum_{\langle ij \rangle} \hat{\mathbf{S}}_i \cdot \hat{\mathbf{S}}_j + \frac{1}{3} \sum_{\langle ijkl \rangle} (\hat{\mathbf{S}}_i \cdot \hat{\mathbf{S}}_j) (\hat{\mathbf{S}}_k \cdot \hat{\mathbf{S}}_l). \end{aligned} \quad (\text{B3})$$

With $n = 2S$, this is

$$\begin{aligned} \tilde{\mathcal{H}} &= S^4 + \frac{S}{\sqrt{3}} \sum_{\langle ijk \rangle} \hat{\mathbf{S}}_i \cdot (\hat{\mathbf{S}}_j \times \hat{\mathbf{S}}_k) \\ &\quad - \frac{S^2}{3} \sum_{\langle ij \rangle} \hat{\mathbf{S}}_i \cdot \hat{\mathbf{S}}_j + \frac{1}{3} \sum_{\langle ijkl \rangle} (\hat{\mathbf{S}}_i \cdot \hat{\mathbf{S}}_j) (\hat{\mathbf{S}}_k \cdot \hat{\mathbf{S}}_l). \end{aligned} \quad (\text{B4})$$

The six bilinear terms can be grouped according to the three pairings of opposite bonds, so that

$$S^4 - \frac{S^2}{3} \sum_{\langle ij \rangle} \hat{\mathbf{S}}_i \cdot \hat{\mathbf{S}}_j + \frac{1}{3} \sum_{\langle ijkl \rangle} (\hat{\mathbf{S}}_i \cdot \hat{\mathbf{S}}_j) (\hat{\mathbf{S}}_k \cdot \hat{\mathbf{S}}_l) = \frac{1}{3} \sum_{\langle ijkl \rangle} (S^2 - \hat{\mathbf{S}}_i \cdot \hat{\mathbf{S}}_j) (S^2 - \hat{\mathbf{S}}_k \cdot \hat{\mathbf{S}}_l). \quad (\text{B5})$$

Restricting each subset V_i to its fully symmetric maximal-spin sector,

$$\hat{\mathbf{S}}_i^2 = S(S+1), \quad (\text{B6})$$

identifies $\hat{\mathbf{S}}_i$ with a physical spin- S operator. Equation (B4) then becomes precisely Eq. (3.17). This is the usual symmetrization construction underlying higher-spin AKLT states [28].

Positive semidefiniteness follows directly from the construction: Eq. (B1) is a sum of positive-semidefinite spin-1/2 parent terms, and its restriction to the maximal-spin sector remains positive semidefinite. The zero modes are inherited just as directly. A spin- S coherent state can be represented as the fully symmetric product of $2S$ identical spin-1/2 coherent states,

$$|\mathbf{n}; S\rangle \simeq \bigotimes_{v=1}^{2S} |\mathbf{n}; \frac{1}{2}\rangle_v, \quad (\text{B7})$$

where the product already lies in the maximal-spin sector. Hence a spin- S CCI product state corresponds on the decorated graph to placing the same CCI direction on every spin-1/2 constituent of a given subset V_i . Every four-site clique in Eq. (B1) then carries a spin-1/2 CCI configuration and is annihilated by its local parent term. The resulting spin- S CCI coherent-product state is therefore an exact zero-energy state of Eq. (3.17).

Appendix C: Geometry of the coherent zero-mode variety

Here we collect the geometric results used in Sec. V: the classical-vector form of the coherent-state energy, the exceptional locus of the Möbius completion rule, and the bounds on the tetrahedral chirality.

1. Classical-vector form

For a product of spin-coherent states, the expectation value of Eq. (3.17) is

$$\langle \{\mathbf{n}_i\} | \mathcal{H}_{\text{CCI}}^{(S)} | \{\mathbf{n}_i\} \rangle = S^4 H_{\text{CCI}}, \quad (\text{C1})$$

where

$$H_{\text{CCI}} = \frac{1}{\sqrt{3}} \sum_{\langle ijk \rangle} \mathbf{n}_i \cdot (\mathbf{n}_j \times \mathbf{n}_k) + \frac{1}{3} \sum_{\langle ijkl \rangle} (1 - \mathbf{n}_i \cdot \mathbf{n}_j)(1 - \mathbf{n}_k \cdot \mathbf{n}_l). \quad (\text{C2})$$

Here $|\mathbf{n}_i| = 1$. This is the Cartesian form of $|B|^2/S^4$ and is therefore nonnegative. Its zero-energy configurations are equivalently described by the stereographic constraint Eq. (5.8).

Away from the three-coincident locus, the fourth spin can be written directly in terms of the first three as

$$\mathbf{n}_4 = \frac{\mathbf{B}_{123}}{\|\mathbf{B}_{123}\|}, \quad (\text{C3})$$

where

$$\mathbf{B}_{123} = (1 - \mathbf{n}_2 \cdot \mathbf{n}_3)\mathbf{n}_1 + (1 - \mathbf{n}_1 \cdot \mathbf{n}_3)\mathbf{n}_2 + (1 - \mathbf{n}_1 \cdot \mathbf{n}_2)\mathbf{n}_3 + \sqrt{3}(\mathbf{n}_1 \times \mathbf{n}_2 + \mathbf{n}_2 \times \mathbf{n}_3 + \mathbf{n}_3 \times \mathbf{n}_1). \quad (\text{C4})$$

Equation (C3) is the unit-vector form of the Möbius completion rule Eq. (5.10). A generic local zero mode therefore has six real parameters, the orientations of three spins. After quotienting by three global spin rotations, the space of inequivalent generic solutions is three dimensional. For comparison, the Heisenberg-tetrahedron constraint $\sum_i \mathbf{n}_i = 0$ leaves a two-dimensional manifold modulo global rotations [3, 73].

2. Exceptional locus of the completion rule

The only configurations for which three spins do not uniquely determine the fourth are those containing at least three coincident spin directions. To see this, suppose for example that $z_1 = z_2 = z$. Then

$$p_-(z, z, z_3, z_4) = \frac{1}{\sqrt{6}}(z - z_3)(z - z_4). \quad (\text{C5})$$

The analogous factorizations for the other coincident pairs differ only by site permutations and phase factors. Thus a zero mode containing a coincident pair necessarily contains a third coincident spin, and the exceptional branch is

$$(z_1, z_2, z_3, z_4) = (z, z, z, w), \quad (\text{C6})$$

together with its site permutations, with arbitrary $w \in \hat{\mathbb{C}}$.

Equivalently, a matrix representative of the completion map Eq. (5.11) is

$$M(z_1, z_2) = \begin{pmatrix} -(\omega z_1 + \omega^2 z_2) & -z_1 z_2 \\ 1 & \omega z_2 + \omega^2 z_1 \end{pmatrix}, \quad (\text{C7})$$

with $\det M = -(z_1 - z_2)^2$. Hence the completion map is invertible for $z_1 \neq z_2$. When $z_1 = z_2 = z$, it gives $z_4 = z$ unless also $z_3 = z$, in which case z_4 is unconstrained. A vanishing denominator in Eq. (5.11) corresponds to $z_4 = \infty$, and the Möbius map extends holomorphically to this point on the Riemann sphere.

3. Equilateral representation and chirality sign

For a solution with four distinct spin directions, perform a global rotation that sends $\mathbf{n}_3$ to the south pole, so that $z_3 = \infty$. Equation (5.8) then reduces to

$$z_4 + \omega z_1 + \omega^2 z_2 = 0, \quad (\text{C8})$$

whose general solution can be written as

$$z_1 = c + \rho, \quad z_2 = c + \rho\omega, \quad z_4 = c + \rho\omega^2, \quad (\text{C9})$$

where $c, \rho \in \mathbb{C}$. Thus the three finite stereographic coordinates form an equilateral triangle centered at c with a fixed orientation.

Substituting Eq. (C9) directly into Eq. (2.3) gives

$$\chi_{\text{tet}} = -\frac{12\sqrt{3}|\rho|^2}{(1+|c+\rho|^2)(1+|c+\omega\rho|^2)(1+|c+\omega^2\rho|^2)}. \quad (\text{C10})$$

The chirality therefore takes values in the interval

$$-\frac{16}{3\sqrt{3}} \leq \chi_{\text{tet}} \leq 0. \quad (\text{C11})$$

$c = 0$ and $|\rho|^2 = 1/2$ realizes the lower bound, which corresponds to the regular tetrahedral configuration, while $\chi_{\text{tet}} = 0$ is reached for $\rho = 0$ on the three-coincident branch.

4. Geometric meaning of the tetrahedral chirality

The tetrahedral chirality has a simple geometric representation. Equation (2.3) can be written as

$$\chi_{\text{tet}} = -\det \begin{pmatrix} 1 & \mathbf{n}_1^\top \\ 1 & \mathbf{n}_2^\top \\ 1 & \mathbf{n}_3^\top \\ 1 & \mathbf{n}_4^\top \end{pmatrix}. \quad (\text{C12})$$

Subtracting the first row from the remaining three gives

$$\chi_{\text{tet}} = -\det(\mathbf{n}_2 - \mathbf{n}_1, \mathbf{n}_3 - \mathbf{n}_1, \mathbf{n}_4 - \mathbf{n}_1). \quad (\text{C13})$$

Thus χ_{tet} equals minus six times the oriented volume of the tetrahedron whose vertices are the four spin directions $\mathbf{n}_i$. Several properties follow immediately. The chirality changes sign under an odd permutation of the four sites and is invariant under an even permutation. It vanishes if and only if the four spin directions are coplanar as points in spin space. Global proper rotations leave χ_{tet} invariant, whereas reversing all spins changes its sign. Maximizing $|\chi_{\text{tet}}|$ for four unit vectors is therefore equivalent to maximizing the volume of a tetrahedron whose vertices lie on the unit sphere. The maximum is attained precisely by a regular tetrahedron, as proved below.

5. Universal bound on the tetrahedral chirality

The magnitude bound does not require the zero-mode condition $p_- = 0$; it holds for arbitrary four unit vectors. Define

$$\bar{\mathbf{n}} = \frac{1}{4} \sum_{i=1}^4 \mathbf{n}_i, \quad \mathbf{u}_i = \mathbf{n}_i - \bar{\mathbf{n}}, \quad Q = \sum_{i=1}^4 \mathbf{u}_i \mathbf{u}_i^\top. \quad (\text{C14})$$

Starting from Eq. (C12), subtract $\bar{\mathbf{n}}$ times the first column from the three spin-coordinate columns. This leaves the determinant unchanged and replaces each row $(1, \mathbf{n}_i^\top)$ by $(1, \mathbf{u}_i^\top)$. Since $\sum_i \mathbf{u}_i = 0$, squaring the determinant and using the corresponding Gram matrix gives

$$\chi_{\text{tet}}^2 = \det \begin{pmatrix} 4 & 0 \\ 0 & Q \end{pmatrix} = 4 \det Q. \quad (\text{C15})$$

Moreover,

$$\text{tr } Q = \sum_{i=1}^4 |\mathbf{n}_i - \bar{\mathbf{n}}|^2 = 4(1 - |\bar{\mathbf{n}}|^2) \leq 4. \quad (\text{C16})$$

If $\lambda_{1,2,3} \geq 0$ are the eigenvalues of the positive-semidefinite matrix Q , the arithmetic-geometric mean inequality gives

$$\chi_{\text{tet}}^2 = 4\lambda_1\lambda_2\lambda_3 \leq 4 \left(\frac{\lambda_1 + \lambda_2 + \lambda_3}{3} \right)^3 \leq \frac{256}{27}, \quad (\text{C17})$$

and hence

$$|\chi_{\text{tet}}| \leq \frac{16}{3\sqrt{3}}. \quad (\text{C18})$$

Equality requires saturation of both inequalities. The trace bound is saturated only when $\bar{\mathbf{n}} = 0$, while equality in the arithmetic-geometric mean requires $\lambda_1 = \lambda_2 = \lambda_3 = 4/3$. Thus

$$Q = \frac{4}{3} \mathbb{I}. \quad (\text{C19})$$

To identify the corresponding spin configuration, introduce the 3×4 matrix

$$U = (\mathbf{u}_1 \ \mathbf{u}_2 \ \mathbf{u}_3 \ \mathbf{u}_4). \quad (\text{C20})$$

Then $UU^\top = Q = (4/3)\mathbb{I}$, so $U^\top U$ has nonzero eigenvalues $\{4/3, 4/3, 4/3\}$. Moreover, $U\mathbf{1} = 0$, and since U has rank three, $\mathbf{1}$ spans the kernel of $U^\top U$. Therefore

$$U^\top U = \frac{4}{3} \left(\mathbb{I}_4 - \frac{1}{4} \mathbf{1}\mathbf{1}^\top \right). \quad (\text{C21})$$

At equality $\bar{\mathbf{n}} = 0$, so $\mathbf{u}_i = \mathbf{n}_i$. The off-diagonal matrix elements therefore give

$$\mathbf{n}_i \cdot \mathbf{n}_j = -\frac{1}{3}, \quad i \neq j. \quad (\text{C22})$$

Thus equality is attained only by a regular tetrahedral frame, with the sign of χ_{tet} fixed by its orientation. Combining Eq. (C18) with Eq. (C10) proves the window Eq. (5.17) and identifies the chiral colorings as its unique lower-bound configurations, up to global rotations and even site permutations.

6. Apollonius circles and the six-cycle

For fixed $z_1 \neq z_2$, introduce the projective coordinate

$$\varsigma(z) = \frac{z - z_1}{z - z_2}. \quad (\text{C23})$$

Equation (5.13) then takes the simple form

$$\varsigma(T_{z_1, z_2}(z)) = -\omega \varsigma(z) = e^{-i\pi/3} \varsigma(z). \quad (\text{C24})$$

Thus each application of the completion map rotates ς by 60° while preserving its modulus. Consequently, the six-cycle generated by a point z lies on the Apollonius circle

$$\frac{|z - z_1|}{|z - z_2|} = \text{const.} \quad (\text{C25})$$

The complementary family $\arg[(z - z_1)/(z - z_2)] = \text{const}$ consists of generalized circles through z_1 and z_2 ; the completion map sends each member of this family to the one whose projective angle differs by $-\pi/3$. The two families form orthogonal pencils of generalized circles.

Since T_{z_1, z_2} fixes z_1 and z_2 , the update

$$(z_1, z_2, z_3, z_4) \mapsto (z_1, z_2, z_4, T_{z_1, z_2}(z_4)) \quad (\text{C26})$$

is equivalently the simultaneous Möbius action of T_{z_1, z_2} on the constrained quadruple. For a generic fixed pair this transformation is not a physical rotation of the Bloch sphere. It preserves generalized circles, but not spherical distances, and therefore neither the individual face chiralities nor the magnitude of the tetrahedral chirality are preserved along the orbit. The sign of χ_{tet} , however, remains fixed by Eq. (5.15). For the six-cycle through the reference coloring shown in Fig. 4, one finds

$$\chi_{\text{tet}} = -\frac{16}{3\sqrt{3}} \left(1, \frac{1}{2}, \frac{1}{6}, \frac{1}{9}, \frac{1}{6}, \frac{1}{2}\right). \quad (\text{C27})$$

Thus the Möbius propagation preserves the handedness selected by the constraint, but not the magnitude of the chirality.

A special simplification occurs when the two fixed spins are antipodal,

$$z_2 = -\frac{1}{z_1^*}. \quad (\text{C28})$$

In this case $T_{z_1, z_2} \in PSU(2)$ and the Möbius transformation becomes a physical rotation of the Bloch sphere.

Rotating the fixed pair to the poles, $z_1 = 0$ and $z_2 = \infty$, gives

$$z_4 = T_{0, \infty}(z_3) = -\omega z_3 = e^{-i\pi/3} z_3, \quad (\text{C29})$$

which is a rotation through $-\pi/3$ about the axis defined by the fixed antipodal spins. Thus the generic six-cycle is a genuinely Möbius orbit, becoming an ordinary rigid sixfold rotation only in the antipodal case.

Appendix D: Counting the zero-energy states

1. States annihilated by $\mathcal{B}$

A convenient way to characterize the states annihilated by $\mathcal{B}$ is to use the polynomial representation of spin states [22]. At each site we identify

$$(a_i^\dagger)^{2S-n} (b_i^\dagger)^n |0\rangle \longleftrightarrow z_i^n, \quad 0 \leq n \leq 2S. \quad (\text{D1})$$

A four-site spin- S state is therefore represented by a polynomial $f(z_1, z_2, z_3, z_4)$ whose degree in each variable is at most $2S$.

Consider now

$$\mathcal{B}^\dagger : \mathcal{H}^{S-\frac{1}{2}} \longrightarrow \mathcal{H}^S. \quad (\text{D2})$$

A state in $\mathcal{H}^{S-\frac{1}{2}}$ has degree at most $2S-1$ in each variable. In the polynomial representation, $\mathcal{B}^\dagger$ acts by multiplication with the polynomial p_+ defined in Eq. (5.4a),

$$\mathcal{B}^\dagger f = p_+ f. \quad (\text{D3})$$

Since the polynomial ring has no zero divisors and p_+ is not the zero polynomial, $p_+ f = 0$ is possible only for $f = 0$. Hence multiplication by p_+ , and with it $\mathcal{B}^\dagger$, is injective on the full space $\mathcal{H}^{S-\frac{1}{2}}$.

It follows that $\mathcal{B}$ is surjective onto $\mathcal{H}^{S-\frac{1}{2}}$: in finite dimensions $\mathcal{B}$ and $\mathcal{B}^\dagger$ have equal rank, and injectivity gives $\text{rank } \mathcal{B}^\dagger = \dim \mathcal{H}^{S-\frac{1}{2}}$. Surjectivity holds separately in every total-spin sector as well. Let $\mathcal{P}_{S_{\text{tet}}}$ denote the projector onto the sector with total spin S_{tet} , which commutes with the $SU(2)$ scalar $\mathcal{B}$. Given $|w\rangle$ in that sector of $\mathcal{H}^{S-\frac{1}{2}}$, full-space surjectivity provides $|v\rangle$ with $\mathcal{B}|v\rangle = |w\rangle$, and the projected state satisfies $\mathcal{B}\mathcal{P}_{S_{\text{tet}}}|v\rangle = \mathcal{P}_{S_{\text{tet}}}\mathcal{B}|v\rangle = |w\rangle$. This sector-resolved surjectivity is what enters the dimension count and proves Eq. (4.20).

The same argument applies to $\mathcal{B}^*$. Applying first $\mathcal{B}$ and then $\mathcal{B}^*$ therefore shows that

$$\mathcal{C} = \mathcal{B}^* \mathcal{B} : \mathcal{H}^S \longrightarrow \mathcal{H}^{S-1} \quad (\text{D4})$$

reaches the whole target space for $S \geq 1$. This gives Eq. (4.26).

2. Why there are no additional zero-energy states

We now prove that the states annihilated by $\mathcal{B}$ are exactly those in $\mathcal{V}^S$. Equivalently, there are no additional states in the local zero-energy kernel beyond the linear span of the globally rotated CCI states.

Let $|f\rangle$ be orthogonal to $\mathcal{V}^S$, and denote its polynomial representation by $f(z_1, z_2, z_3, z_4)$. With the normalization of Eq. (D1), the overlap with a product of four unnormalized coherent states is

$$\langle w_1 w_2 w_3 w_4 | f \rangle = [(2S)!]^4 f(\bar{w}_1, \bar{w}_2, \bar{w}_3, \bar{w}_4). \quad (\text{D5})$$

Thus $|f\rangle \perp \mathcal{V}^S$ implies that the polynomial f vanishes on the complex conjugates of all globally rotated CCI configurations.

The space $\mathcal{V}^S$ is a complex vector space invariant under global $\text{SU}(2)$ rotations. Invariance under the $\text{su}(2)$ generators therefore extends to their complex linear span $\text{sl}(2, \mathbb{C})$, and hence to the connected group $\text{SL}(2, \mathbb{C})$ [74]. The $\text{SL}(2, \mathbb{C})$ orbit of the reference state consequently remains inside $\mathcal{V}^S$. In stereographic coordinates this complexified action is the usual $\text{PSL}(2, \mathbb{C})$ Möbius action. Using Eq. (D5), and noting that complex conjugation maps $\text{SL}(2, \mathbb{C})$ onto itself, we conclude that f vanishes on the full $\text{PSL}(2, \mathbb{C})$ orbit of the conjugated reference coloring.

To identify this orbit, consider first the polynomial p_+ associated with $\mathcal{B}^\dagger$. Up to its normalization, it is the quadratic form

$$p_+ = \frac{1}{2\sqrt{6}} \mathbf{z}^\top C_+ \mathbf{z}, \quad \mathbf{z} = (z_1, z_2, z_3, z_4)^\top, \quad (\text{D6})$$

with

$$C_+ = \begin{pmatrix} 0 & 1 & \omega^2 & \omega \\ 1 & 0 & \omega & \omega^2 \\ \omega^2 & \omega & 0 & 1 \\ \omega & \omega^2 & 1 & 0 \end{pmatrix}. \quad (\text{D7})$$

Because $1 + \omega + \omega^2 = 0$, every row of C_+ sums to zero, so

$$C_+(1, 1, 1, 1)^\top = 0. \quad (\text{D8})$$

On the other hand,

$$\det \begin{pmatrix} 0 & 1 & \omega^2 \\ 1 & 0 & \omega \\ \omega^2 & \omega & 0 \end{pmatrix} = 2, \quad (\text{D9})$$

and therefore

$$\text{rank } C_+ = 3. \quad (\text{D10})$$

If a homogeneous quadratic polynomial factors into two linear forms, its quadratic-form matrix has rank at most two. Hence p_+ is irreducible over $\mathbb{C}$.

The conjugated reference coloring lies on the hypersurface $p_+ = 0$. For four distinct coordinates this condition

has a simple projective interpretation. Direct expansion gives

$$(z_1 - z_3)(z_2 - z_4) - e^{-i\pi/3}(z_1 - z_4)(z_2 - z_3) = \sqrt{6} e^{i\pi/3} p_+(z_1, z_2, z_3, z_4), \quad (\text{D11})$$

so that $p_+ = 0$ is equivalent, for four distinct points, to the equianharmonic cross-ratio condition

$$\frac{(z_1 - z_3)(z_2 - z_4)}{(z_1 - z_4)(z_2 - z_3)} = e^{-i\pi/3}. \quad (\text{D12})$$

For p_- the corresponding value is the complex conjugate $e^{+i\pi/3}$.

A Möbius transformation is uniquely determined by the images of three distinct points and preserves their cross ratio. Consequently, every ordered quadruple of distinct points satisfying Eq. (D12) is obtained from the conjugated reference coloring by a global $\text{PSL}(2, \mathbb{C})$ transformation. The complexified CCI orbit therefore coincides with the distinct-point part of the hypersurface

$$V(p_+) = \{(z_1, z_2, z_3, z_4) \in \mathbb{C}^4 : p_+(z_1, z_2, z_3, z_4) = 0\}. \quad (\text{D13})$$

Since p_+ is irreducible, $V(p_+)$ is an irreducible hypersurface. Its distinct-point locus is nonempty and Zariski open, and is therefore dense in $V(p_+)$. Since f vanishes on the complexified CCI orbit, it follows that f vanishes on a dense subset of $V(p_+)$ and hence on the entire hypersurface. We now use the standard algebraic-geometric fact that, for an irreducible polynomial over $\mathbb{C}$, every polynomial vanishing identically on its zero set is divisible by that polynomial [75]. Therefore

$$f = p_+ q. \quad (\text{D14})$$

Since p_+ has degree exactly one in every coordinate and the polynomial ring has no zero divisors,

$$\deg_{z_i} q = \deg_{z_i} f - 1 \leq 2S - 1 \quad (\text{D15})$$

whenever $q \neq 0$. Thus q represents a state $|q\rangle \in \mathcal{H}^{S-\frac{1}{2}}$. Moreover, as shown above, multiplication by p_+ is precisely the polynomial representation of $\mathcal{B}^\dagger$, and therefore

$$|f\rangle = \mathcal{B}^\dagger |q\rangle. \quad (\text{D16})$$

Thus every state orthogonal to $\mathcal{V}^S$ belongs to the image of $\mathcal{B}^\dagger$.

Now let $|\psi\rangle$ satisfy

$$\mathcal{B}|\psi\rangle = 0. \quad (\text{D17})$$

For every $|f\rangle \perp \mathcal{V}^S$, Eq. (D16) gives

$$\langle f | \psi \rangle = \langle q | \mathcal{B} | \psi \rangle = 0. \quad (\text{D18})$$

Hence $|\psi\rangle$ is orthogonal to the entire orthogonal complement of $\mathcal{V}^S$ and must itself belong to $\mathcal{V}^S$. Together with

the inclusion $\mathcal{V}^S \subseteq \ker \mathcal{B}$ established in Eq. (4.23), this proves

$$\mathcal{V}^S = \ker \mathcal{B}. \quad (\text{D19})$$

The same reasoning applies to the full color-ice space $\mathcal{W}^S$. If $|f\rangle \perp \mathcal{W}^S$, its polynomial vanishes on the complexified orbits of both chiralities. It is therefore divisible by both p_+ and p_- . The two polynomials are irreducible by the same rank argument and are not proportional, so they are relatively prime. Hence

$$f = p_+ p_- q. \quad (\text{D20})$$

For $S \geq 1$, the polynomial q has degree at most $2S - 2$ in each coordinate and therefore represents a state $|q\rangle \in \mathcal{H}^{S-1}$. Since multiplication by $p_+ p_-$ is the polynomial representation of $\mathcal{C}^\dagger$, we obtain

$$|f\rangle = \mathcal{C}^\dagger |q\rangle. \quad (\text{D21})$$

Repeating the orthogonality argument then gives

$$\mathcal{W}^S = \ker \mathcal{C}. \quad (\text{D22})$$

Appendix E: Exact gap of the local parent Hamiltonian

Here we prove Eq. (4.35) for the exact gap of a single tetrahedral parent term. As discussed in Sec. IV D, $\mathcal{B}^\dagger \mathcal{B}$ on $\mathcal{H}^S$ and $\mathcal{B} \mathcal{B}^\dagger$ on $\mathcal{H}^{S-\frac{1}{2}}$ have identical nonzero spectra. It is therefore sufficient to determine the lowest eigenvalue of $\mathcal{B} \mathcal{B}^\dagger$ on $\mathcal{H}^{S-\frac{1}{2}}$.

On this Hilbert space the Schwinger-boson number at every site is fixed,

$$N_i = a_i^\dagger a_i + b_i^\dagger b_i = 2S - 1 \equiv m. \quad (\text{E1})$$

Since $\mathcal{B}$ contains one annihilation operator from each site, it is linear in the boson operators at any given site. We define

$$D_{i,a} = [\mathcal{B}, a_i^\dagger], \quad D_{i,b} = [\mathcal{B}, b_i^\dagger]. \quad (\text{E2})$$

Normal ordering $\mathcal{B} \mathcal{B}^\dagger$ gives contributions with zero through four contractions. Using $a_i a_i^\dagger = 1 + a_i^\dagger a_i$ and $b_i b_i^\dagger = 1 + b_i^\dagger b_i$, one obtains

$$\begin{aligned} \mathcal{B} \mathcal{B}^\dagger &= \mathcal{B}^\dagger \mathcal{B} + \sum_i \left(D_{i,a}^\dagger D_{i,a} + D_{i,b}^\dagger D_{i,b} \right) \\ &+ \frac{1}{6} \sum_{i < j} \left(N_i N_j + F_{ij}^\dagger F_{ij} \right) + \frac{1}{2} \sum_i N_i + \mathbb{I}. \end{aligned} \quad (\text{E3})$$

Equation (E3) is an operator identity in the two-boson algebra of the four sites, obtained by normal ordering all contractions of $\mathcal{B} \mathcal{B}^\dagger$. We have additionally verified it exactly by evaluating all Fock-space matrix elements with occupations up to three bosons per site; this evaluation is exhaustive because every term in the identity is

of degree at most two in the mode operators of each site. Restricting this identity to $\mathcal{H}^{S-\frac{1}{2}}$, where $N_i = m$, gives

$$\begin{aligned} \mathcal{B} \mathcal{B}^\dagger &= (m+1)^2 \mathbb{I} + \mathcal{B}^\dagger \mathcal{B} + \frac{1}{6} \sum_{i < j} F_{ij}^\dagger F_{ij} \\ &+ \sum_i \left(D_{i,a}^\dagger D_{i,a} + D_{i,b}^\dagger D_{i,b} \right). \end{aligned} \quad (\text{E4})$$

All terms following the first are positive semidefinite. Hence

$$\mathcal{B} \mathcal{B}^\dagger \succeq (m+1)^2 \mathbb{I} = 4S^2 \mathbb{I}. \quad (\text{E5})$$

We now show that the bound is saturated. Let

$$|F_m\rangle = \prod_{i=1}^4 \frac{(a_i^\dagger)^m}{\sqrt{m!}} |0\rangle \quad (\text{E6})$$

be the fully polarized state in $\mathcal{H}^{S-\frac{1}{2}}$. From Eqs. (4.2) and (4.5), each of F_{ij} , $\mathcal{B}$, $D_{i,a}$, and $D_{i,b}$ contains at least one b -boson annihilation operator. Therefore

$$F_{ij} |F_m\rangle = \mathcal{B} |F_m\rangle = D_{i,a} |F_m\rangle = D_{i,b} |F_m\rangle = 0. \quad (\text{E7})$$

Equation (E4) then yields

$$\mathcal{B} \mathcal{B}^\dagger |F_m\rangle = (m+1)^2 |F_m\rangle = 4S^2 |F_m\rangle. \quad (\text{E8})$$

Together with Eq. (E5), this proves

$$\lambda_{\min} \left(\mathcal{B} \mathcal{B}^\dagger \Big|_{\mathcal{H}^{S-\frac{1}{2}}} \right) = 4S^2. \quad (\text{E9})$$

The state $|F_m\rangle$ is the highest-weight state of the unique maximal-total-spin multiplet in $\mathcal{H}^{S-\frac{1}{2}}$, with $S_{\text{tet}} = 4S - 2$. Since $\mathcal{B} \mathcal{B}^\dagger$ is an $\text{SU}(2)$ scalar, the entire multiplet has eigenvalue $4S^2$. Together with the spectral pairing between $\mathcal{B} \mathcal{B}^\dagger$ and $\mathcal{B}^\dagger \mathcal{B}$, this establishes the exact local gap

$$\Delta_{\text{loc}}(S) = 4S^2, \quad (\text{E10})$$

and proves Eq. (4.35).

Appendix F: Loop zero modes: reduction and counting

Here we collect the technical details of the loop analysis: the polarized-loop reduction and transfer-matrix count quoted in Sec. VII B 1, together with the geometric and quantum-counting arguments for four-coloring backgrounds used in Sec. VII B 3.

1. Uniform-color backgrounds.

Consider a loop of length L whose sites are shared by L tetrahedra, each containing two consecutive loop sites

and two environment sites polarized along the quantization axis. Each term of Eq. (4.10) contains a pair of b annihilation operators. Every term with a b operator on an environment site vanishes, and the single surviving contribution has both b operators on the two loop sites. Up to a nonzero scalar factor and an irrelevant phase, the local zero-mode condition is therefore Eq. (7.1).

Grouping the $2S + 1$ states on each loop site into the fully polarized state U and the $2S$ deviated states $\bar{U}$, the allowed cyclic sequences are counted by

$$N_L(S) = \text{Tr } T^L = \lambda_+^L + \lambda_-^L, \quad (\text{F1})$$

where

$$\lambda_{\pm} = \frac{1 \pm \sqrt{1 + 8S}}{2}. \quad (\text{F2})$$

are the eigenvalues of the

$$T = \begin{pmatrix} 1 & 2S \\ 1 & 0 \end{pmatrix} \quad (\text{F3})$$

transfer matrix. Expanding this expression for $L = 4$ and 6 gives Eq. (7.2). For $S = \frac{1}{2}$ the same count is the monomer-dimer partition function of the cycle graph, $N_4 = 1 + 4 + 2 = 7$ and $N_6 = 1 + 6 + 9 + 2 = 18$.

2. Four-coloring backgrounds

For the four-sublattice pyrochlore coloring, an elementary hexagon contains three distinct colors, each appearing twice. Up to relabeling, cyclic permutation, and reversal, its color sequence is therefore $XYZXYZ$. To determine its holonomy explicitly, we label the three colors on the loop as A, B, C , and denote by D the fourth color, which is absent from the loop. The sequence may then be written as

$$ABCABC. \quad (\text{F4})$$

a. Identity holonomy for the pyrochlore hexagon

The first three completion maps around the hexagon are

$$T_{C,D}, \quad T_{A,D}, \quad T_{B,D}, \quad (\text{F5})$$

and on the reference coloring they propagate

$$A \xrightarrow{T_{C,D}} B \xrightarrow{T_{A,D}} C \xrightarrow{T_{B,D}} A. \quad (\text{F6})$$

Their product

$$P = T_{B,D} \circ T_{A,D} \circ T_{C,D} \quad (\text{F7})$$

therefore fixes A . Moreover, D is a fixed point of each of the three completion maps, and hence

$$P(A) = A, \quad P(D) = D. \quad (\text{F8})$$

Using Eq. (5.13), the completion rule may equivalently be written as

$$\frac{T_{z_1, z_2}(z) - z_2}{T_{z_1, z_2}(z) - z_1} = (-\omega)^{-1} \frac{z - z_2}{z - z_1}. \quad (\text{F9})$$

Thus each of the three maps has multiplier $(-\omega)^{-1}$ at their common fixed point D . The multiplier of their product is therefore

$$P'(D) = (-\omega)^{-3} = -1. \quad (\text{F10})$$

A Möbius transformation with two distinct fixed points and multiplier -1 is an involution, so

$$P^2 = \text{id}. \quad (\text{F11})$$

The second half of the hexagon contains the same three completion maps. Consequently, the full six-step holonomy is

$$\mathcal{M}_6 = P^2 = \text{id}, \quad (\text{F12})$$

which proves Eq. (7.5).

This argument depends only on the color sequence and on the local completion rule. In particular, the four color directions need not form the regular tetrahedral frame. Any four distinct directions z_A, z_B, z_C, z_D satisfying $p_-(z_A, z_B, z_C, z_D) = 0$ give the same identity holonomy, provided the corresponding local completion maps are nondegenerate.

3. Quantum kernel of an identity-holonomy loop

We now show that identity holonomy has a stronger quantum consequence. Consider an embedded loop of length L , with the environment spins fixed in coherent states, and assume that every local completion rule is a well-defined invertible Möbius transformation.

For a tetrahedron containing two neighboring loop sites i and $i+1$, fixing the other two spins reduces the four-site annihilation operator $\mathcal{B}$ to a bilinear annihilation constraint on the two loop spins. At the coherent-state level, its zero-energy condition is precisely

$$z_{i+1} = T_i(z_i). \quad (\text{F13})$$

The key observation is that these local Möbius twists can be removed by a site-dependent choice of stereographic coordinates. Introduce

$$z'_i = R_i(z_i), \quad (\text{F14})$$

with $R_1 = \text{id}$, and choose the remaining transformations recursively as

$$R_{i+1} = R_i \circ T_i^{-1}. \quad (\text{F15})$$

Equation (F13) then becomes simply

$$z'_{i+1} = z'_i. \quad (\text{F16})$$

For an open chain this change of coordinates can always be carried out. Around a closed loop, after one circuit the transported coordinate system is

$$R_{L+1} = \mathcal{M}_L^{-1}. \quad (\text{F17})$$

It therefore returns to the original coordinate system precisely when

$$\mathcal{M}_L = \text{id}. \quad (\text{F18})$$

Identity holonomy thus means that all local Möbius twists can be removed simultaneously.

Each Möbius change of coordinate can be represented by an invertible linear change of the two Schwinger-boson components at the corresponding site. Under these changes the reduced two-site annihilation constraint remains bilinear, while its coherent-state zero condition becomes $z'_i = z'_{i+1}$.

A general bilinear constraint produces, on a pair of coherent states, a factor of the form

$$c_0 + c_1 z'_i + c_2 z'_{i+1} + c_3 z'_i z'_{i+1}. \quad (\text{F19})$$

Requiring this expression to vanish for every $z'_i = z'_{i+1} = z$ gives $c_0 = c_3 = 0$ and $c_2 = -c_1$. Hence, up to a nonzero overall factor, the only such bilinear constraint is proportional to $z'_{i+1} - z'_i$. This is precisely the coherent-state zero condition of the singlet pair-annihilation operator $F_{i,i+1}$ introduced in Eq. (4.2). Thus the loop zero-energy conditions reduce to

$$F_{i,i+1}|\psi\rangle = 0, \quad i = 1, \dots, L, \quad L+1 \equiv 1. \quad (\text{F20})$$

The local transformations used above are generally nonunitary, so this change of variables is not a physical site-dependent spin rotation. It is an invertible change of basis in the local spin Hilbert spaces and therefore gives a one-to-one correspondence between the original loop kernel and the common kernel of Eq. (F20).

The latter has a simple physical interpretation. On two spin- S sites,

$$F_{ij}^\dagger F_{ij} = 2 \left(S^2 - \hat{\mathbf{S}}_i \cdot \hat{\mathbf{S}}_j \right), \quad (\text{F21})$$

and in the pair-spin- J sector its eigenvalue is

$$\lambda_J = 2S(2S+1) - J(J+1). \quad (\text{F22})$$

This vanishes only for $J = 2S$. Thus every neighboring pair on the untwisted loop must lie in its maximal-spin sector.

To determine the simultaneous kernel, one may regard each spin S as $2S$ symmetrized spin-1/2 constituents. Maximal spin on the bond $(i, i+1)$ requires the constituents belonging to these two sites to be fully symmetric. Since neighboring bonds share sites and the loop is connected, these conditions propagate around the entire loop. All $2SL$ spin-1/2 constituents are therefore fully

symmetric, so the untwisted kernel is the single maximal-spin multiplet

$$J_{\text{tot}} = LS. \quad (\text{F23})$$

Its dimension is $2SL+1$. Since the change of basis is invertible, the original embedded-loop kernel has the same dimension,

$$\dim \mathcal{K}_{\text{loop}} = 2SL+1. \quad (\text{F24})$$

It remains to identify the coherent states within this kernel. In the untwisted variables all sites have the same stereographic coordinate, so the propagated family becomes

$$|\Omega(z)\rangle = \bigotimes_{i=1}^L |z; S\rangle_i. \quad (\text{F25})$$

These are precisely the coherent states of the maximal-spin $J_{\text{tot}} = LS$ multiplet: they are obtained by applying the same global spin rotation to every site. Since the spin- LS representation is irreducible, the linear span of this coherent-state family is the entire multiplet.

Transforming back to the original coordinates gives $z_i = R_i^{-1}(z)$. Equation (F15) then implies

$$z_{i+1} = T_i(z_i), \quad (\text{F26})$$

so these are precisely the coherent states obtained by propagating an arbitrary initial spin direction around the original loop. Consequently,

$$\mathcal{K}_{\text{loop}} = \text{span} \left\{ \left| \tilde{\psi}(z) \right\rangle \left| z \in \hat{\mathbb{C}} \right. \right\}, \quad \dim \mathcal{K}_{\text{loop}} = 2SL+1. \quad (\text{F27})$$

Appendix G: Linearized coherent-state constraints and harmonic zero modes

We now characterize the infinitesimal coherent deformations of the four-sublattice zero-energy state. The analysis is analogous to the Maxwell counting of local constraints in classical frustrated magnets [3] and to the constraint-matrix formulation of classical spin liquids [48, 49]. In momentum space it is also closely related to the analysis of zero-energy manifolds in a Luttinger-Tisza spectrum. The important distinction is that here $\mathbf{k}$ labels fluctuations about a fixed four-sublattice ground state, rather than candidate ordering wavevectors of the original spins.

For a coherent product state the energy can be written as a sum of positive local contributions,

$$E = \sum_t W_t(\{z_i, \bar{z}_i\}) |p_t(\{z_i\})|^2, \quad (\text{G1})$$

where $W_t > 0$ and every coherent zero-energy state satisfies $p_t = 0$ on each tetrahedron t . Consider a small deformation of the four-sublattice solution,

$$z_i = z_i^{(0)} + \delta z_i. \quad (\text{G2})$$

Since p_t is holomorphic in the stereographic coordinates,

$$p_t = \sum_i R_{ti} \delta z_i + O(\delta z^2), \quad R_{ti} = \left. \frac{\partial p_t}{\partial z_i} \right|_{\{z^{(0)}\}}, \quad (\text{G3})$$

where R is the linearized constraint matrix. Because $p_t = 0$ on the reference state, variations of the positive factors W_t do not contribute at quadratic order. The harmonic change in the coherent-state energy is therefore

$$E^{(2)} = \delta \mathbf{z}^\dagger R^\dagger W^{(0)} R \delta \mathbf{z}, \quad (\text{G4})$$

where $W^{(0)}$ is diagonal in the local constraints and has positive entries. Hence

$$E^{(2)} = 0 \quad \Longleftrightarrow \quad R \delta \mathbf{z} = 0. \quad (\text{G5})$$

The kernel of R therefore gives the infinitesimal coherent zero modes of the four-sublattice state.

For a periodic background, R separates into Bloch matrices $R(\mathbf{k})$. The dimension

$$\nu(\mathbf{k}) = \dim \ker R(\mathbf{k}) \quad (\text{G6})$$

counts the independent complex zero modes at momentum $\mathbf{k}$. The geometry of the set on which $\nu(\mathbf{k}) > 0$ provides a useful classification of the coherent flexibility of the ordered state: a zero mode throughout the Brillouin zone gives an extensive number of infinitesimal deformations, zero modes restricted to lines give a subextensive number proportional to the linear system size, while zero modes only at isolated momenta give an $O(1)$ tangent space.

This reciprocal-space picture resembles the codimension classification of degenerate minima in Luttinger–Tisza analyses, but its meaning is different. Here the zero set describes soft deformations around one particular coherent ground-state component. Moreover, a zero mode of R is only guaranteed to satisfy the local constraints to first order; it need not extend to a finite continuous family of exact zero modes. This distinction will be important for the checkerboard lattice.

1. Linearizing the local constraint

Consider first a single tetrahedron in the four-sublattice state, with the sites ordered according to their colors A, B, C, D . Its coherent zero-energy condition is

$$p_-(z_A, z_B, z_C, z_D) = 0, \quad (\text{G7})$$

with p_- given in Eq. (5.4b). We now make an infinitesimal deformation

$$z_X \longrightarrow z^X + \delta z_X, \quad X = A, B, C, D, \quad (\text{G8})$$

about the tetrahedral solution of Eq. (5.9). To first order,

$$\delta p_- = g_A \delta z_A + g_B \delta z_B + g_C \delta z_C + g_D \delta z_D, \quad (\text{G9})$$

where

$$\begin{aligned} g_A &= \left. \frac{\partial p_-}{\partial z_A} \right|_{\text{tet}} = \frac{1}{\sqrt{6}} (z^B + \omega z^C + \omega^2 z^D), \\ g_B &= \frac{1}{\sqrt{6}} (z^A + \omega z^D + \omega^2 z^C), \\ g_C &= \frac{1}{\sqrt{6}} (z^D + \omega z^A + \omega^2 z^B), \\ g_D &= \frac{1}{\sqrt{6}} (z^C + \omega z^B + \omega^2 z^A). \end{aligned} \quad (\text{G10})$$

Using $z^B = -z^A$ and $z^D = -z^C$, these coefficients satisfy

$$g_B = -g_A, \quad g_D = -g_C, \quad \frac{g_C}{g_A} = i(2 - \sqrt{3}). \quad (\text{G11})$$

It is therefore convenient to define

$$\gamma \equiv \frac{g_C}{g_A} = i(2 - \sqrt{3}), \quad (\text{G12})$$

and divide Eq. (G9) by g_A . The linearized constraint on every tetrahedron then takes the simple form

$$\delta z_A - \delta z_B + \gamma (\delta z_C - \delta z_D) = 0. \quad (\text{G13})$$

Equation (G13) is the basic equation used below. The lattice enters only through the way in which the four colors belonging to different tetrahedra are shared between magnetic unit cells.

For the full lattice, collect all infinitesimal deformations into a vector $\delta \mathbf{z}$. The linearized constraints can then be written as

$$R \delta \mathbf{z} = 0, \quad (\text{G14})$$

where R has one row for each local tetrahedral constraint and one column for each spin. Thus

$$\ker R \quad (\text{G15})$$

is precisely the space of coherent deformations that satisfy all local constraints to first order.

This is only a linear statement. If $R \delta \mathbf{z} = 0$, then

$$p_- \left(z^{(0)} + \epsilon \delta \mathbf{z} \right) = O(\epsilon^2), \quad (\text{G16})$$

but the infinitesimal deformation need not extend to an exact finite-amplitude family. The checkerboard lattice below provides an important example of such an obstruction.

2. Bloch form

Because the four-sublattice background is periodic, the linearized problem separates into momentum sectors. Let α label the sites of the magnetic unit cell and write

$$\delta z_\alpha(\mathbf{R}) = u_\alpha(\mathbf{k}) e^{i\mathbf{k} \cdot \mathbf{R}}. \quad (\text{G17})$$

Equation (G14) then becomes

$$\mathbf{R}(\mathbf{k}) \mathbf{u}(\mathbf{k}) = 0. \quad (\text{G18})$$

If a magnetic cell contains n_s sites, the number of independent complex infinitesimal deformations at momentum $\mathbf{k}$ is

$$\nu(\mathbf{k}) = n_s - \text{rank } \mathbf{R}(\mathbf{k}). \quad (\text{G19})$$

For a finite periodic system the total number of complex tangent directions is therefore

$$N_{\text{lin}} = \sum_{\mathbf{k}} \nu(\mathbf{k}). \quad (\text{G20})$$

The Bloch matrices are not unique: changing the origin of a site or a local constraint multiplies a column or row by a nonzero Bloch phase, while changing the ordering permutes rows or columns. None of these operations changes the rank. We therefore use below the conventions that make the matrices simplest.

There are always three zero modes at $\mathbf{k} = 0$. Indeed, a common Möbius transformation preserves the coherent zero-mode condition. Infinitesimally it acts as

$$\delta z_i = a + bz_i + cz_i^2, \quad (\text{G21})$$

with three complex parameters a, b, c . These are spatially uniform and therefore occur at Γ . Physical global spin rotations form a three-real-dimensional subset of these global Möbius deformations.

3. Corner-sharing lattices

Checkerboard lattice. Choose a 2×2 magnetic cell with sites

$$A = (0, 0), \quad B = (1, 0), \quad C = (0, 1), \quad D = (1, 1), \quad (\text{G22})$$

and magnetic translations $(2, 0)$ and $(0, 2)$. Define the Bloch phases

$$X = e^{i\mathbf{k} \cdot \mathbf{A}_1}, \quad Y = e^{i\mathbf{k} \cdot \mathbf{A}_2}. \quad (\text{G23})$$

There are two crossed plaquettes per magnetic cell. In the basis (u_A, u_B, u_C, u_D) their linearized constraints are represented by

$$\mathbf{R}_{\text{cb}}(\mathbf{k}) = \begin{pmatrix} 1 & -1 & \gamma & -\gamma \\ XY & -Y & \gamma X & -\gamma \end{pmatrix}. \quad (\text{G24})$$

Since this is a 2×4 matrix,

$$\nu(\mathbf{k}) \geq 2 \quad (\text{G25})$$

at every momentum. The two rows are independent for $\mathbf{k} \neq 0$, so

$$\nu(\mathbf{k}) = 2, \quad \mathbf{k} \neq 0, \quad (\text{G26})$$

whereas at Γ the two rows coincide and $\nu(0) = 3$. On an $L \times L$ torus of magnetic cells,

$$N_{\text{lin}} = 2(L^2 - 1) + 3 = 2L^2 + 1 = \frac{N}{2} + 1. \quad (\text{G27})$$

Thus the checkerboard lattice has an extensive tangent space around the four-sublattice state.

Pyrochlore lattice. The same counting becomes equally transparent on the pyrochlore lattice. Use the usual four-site primitive cell with an “up” tetrahedron inside the cell and a “down” tetrahedron connecting neighboring cells. With primitive-cell Bloch phases X, Y, Z , a convenient form is

$$\mathbf{R}_{\text{pyr}}(\mathbf{k}) = \begin{pmatrix} 1 & -1 & \gamma & -\gamma \\ 1 & -X & \gamma Y & -\gamma Z \end{pmatrix}. \quad (\text{G28})$$

Again the rank is two everywhere except at $X = Y = Z = 1$, where it is one. Hence

$$\nu(\mathbf{k}) = 2 \quad (\mathbf{k} \neq 0), \quad \nu(0) = 3, \quad (\text{G29})$$

and on an L^3 torus

$$N_{\text{lin}} = 2(L^3 - 1) + 3 = \frac{N}{2} + 1. \quad (\text{G30})$$

The linear counts are therefore identical for the checkerboard and pyrochlore lattices. Their nonlinear behavior is nevertheless different. The compact empty-square modes of the checkerboard lattice close only to first order: their exact Möbius holonomy is parabolic rather than the identity. The pyrochlore hexagons, by contrast, have identity holonomy (Appendix F 2 a), so the corresponding infinitesimal modes extend to exact continuous weathervane families.

4. fcc lattice

For the fcc lattice we use the cubic magnetic cell of Fig. 6, with translations

$$\mathbf{A}_1 = (2, 0, 0), \quad \mathbf{A}_2 = (0, 2, 0), \quad \mathbf{A}_3 = (0, 0, 2), \quad (\text{G31})$$

and basis sites

$$A = (0, 0, 0), \quad B = (0, 1, 1), \quad C = (1, 0, 1), \quad D = (1, 1, 0). \quad (\text{G32})$$

Let

$$X = e^{i\mathbf{k} \cdot \mathbf{A}_1}, \quad Y = e^{i\mathbf{k} \cdot \mathbf{A}_2}, \quad Z = e^{i\mathbf{k} \cdot \mathbf{A}_3}. \quad (\text{G33})$$

The eight elementary tetrahedra in the cubic magnetic cell give

$$\mathbf{R}_{\text{fcc}}(\mathbf{k}) = \begin{pmatrix} 1 & -1 & \gamma & -\gamma \\ Z & -1 & \gamma & -\gamma Z \\ Y & -1 & \gamma Y & -\gamma \\ YZ & -1 & \gamma Y & -\gamma Z \\ X & -X & \gamma & -\gamma \\ XZ & -X & \gamma & -\gamma Z \\ XY & -X & \gamma Y & -\gamma \\ XYZ & -X & \gamma Y & -\gamma Z \end{pmatrix}. \quad (\text{G34})$$

The momentum dependence of its rank follows analytically from simple 4×4 minors. For example,

$$\det \mathbf{R}_{\{1,2,3,4\}} = \gamma^2(Y-1)^2(Z-1)^2, \quad (\text{G35a})$$

$$\det \mathbf{R}_{\{1,2,5,6\}} = -\gamma^2(X-1)^2(Z-1)^2, \quad (\text{G35b})$$

$$\det \mathbf{R}_{\{1,3,5,7\}} = \gamma^2(X-1)^2(Y-1)^2, \quad (\text{G35c})$$

where the subscripts specify the four selected rows of Eq. (G34). At least one of these minors is nonzero unless two of the three Bloch phases are equal to unity. Thus a nontrivial kernel can occur only on

$$Y = Z = 1, \quad X = Z = 1, \quad X = Y = 1, \quad (\text{G36})$$

which are the three $\langle 100 \rangle$ axes through Γ .

Direct substitution into Eq. (G34) gives rank two on each axis away from Γ , and rank one at Γ . Consequently,

$$\nu(\mathbf{k}) = 2 \quad (\text{G37})$$

on the three axes away from their intersection. On an L^3 torus each axis contains $L-1$ nonzero momenta, and therefore

$$N_{\text{lin}} = 3 \times 2(L-1) + 3 = 6L - 3. \quad (\text{G38})$$

A Fourier superposition of modes along, for example, the X axis can be localized in the x direction while remaining uniform in the transverse directions. It therefore produces a deformation supported on a single (100) plane. The three momentum-space axes are thus precisely the Bloch representation of the planar zero modes constructed in Fig. 6(b).

5. Square and honeycomb lattices

Square lattice. The square-lattice model contains one local tetrahedral constraint on every elementary square. Using the same 2×2 four-color cell as for the checkerboard lattice gives

$$\mathbf{R}_{\text{sq}}(\mathbf{k}) = \begin{pmatrix} 1 & -1 & \gamma & -\gamma \\ X & -1 & \gamma X & -\gamma \\ Y & -Y & \gamma & -\gamma \\ XY & -Y & \gamma X & -\gamma \end{pmatrix}. \quad (\text{G39})$$

Its determinant factorizes completely:

$$\det \mathbf{R}_{\text{sq}}(\mathbf{k}) = \gamma^2(X-1)^2(Y-1)^2. \quad (\text{G40})$$

The matrix therefore has a kernel only on the two lines

$$X = 1 \quad \text{or} \quad Y = 1. \quad (\text{G41})$$

Away from Γ its rank on either line is two, giving $\nu = 2$, while at Γ the rank is one and $\nu = 3$. Hence

$$N_{\text{lin}} = 2 \times 2(L-1) + 3 = 4L - 1. \quad (\text{G42})$$

These two null lines are the momentum-space counterparts of the two families of real-space line defects in Fig. 10(a).

Honeycomb lattice. For the honeycomb lattice let $\mathbf{a}_1, \mathbf{a}_2$ be primitive translations of the two-site honeycomb cell, with geometric sublattices u and v . We choose

$$u_{\mathbf{n}} \sim \{v_{\mathbf{n}}, v_{\mathbf{n}-\mathbf{a}_1}, v_{\mathbf{n}-\mathbf{a}_2}\} \quad (\text{G43})$$

for the three nearest neighbors of $u_{\mathbf{n}}$. The four-color state has a 2×2 enlargement of this cell. A convenient color assignment is

$$\begin{aligned} (u_{00}, u_{01}, u_{10}, u_{11}) &= (A, B, C, D), \\ (v_{00}, v_{01}, v_{10}, v_{11}) &= (D, C, B, A). \end{aligned} \quad (\text{G44})$$

With magnetic translations $2\mathbf{a}_1, 2\mathbf{a}_2$ and Bloch phases X, Y , order the amplitudes as

$$\mathbf{u} = (u_{00}, u_{01}, u_{10}, u_{11}, v_{00}, v_{01}, v_{10}, v_{11})^T. \quad (\text{G45})$$

The linearized matrix can be written in block form,

$$\mathbf{R}_{\text{hc}}(\mathbf{k}) = \begin{pmatrix} D_u & B_{uv} \\ B_{vu} & D_v \end{pmatrix}, \quad (\text{G46})$$

where

$$D_u = \text{diag}(1, -1, \gamma, -\gamma), \quad (\text{G47})$$

$$D_v = \text{diag}(-\gamma, \gamma, -1, 1), \quad (\text{G48})$$

and

$$B_{uv} = \begin{pmatrix} -\gamma & \gamma Y^{-1} & -X^{-1} & 0 \\ -\gamma & \gamma & 0 & X^{-1} \\ -\gamma & 0 & -1 & Y^{-1} \\ 0 & \gamma & -1 & 1 \end{pmatrix}, \quad (\text{G49})$$

$$B_{vu} = \begin{pmatrix} 1 & -1 & \gamma & 0 \\ Y & -1 & 0 & -\gamma \\ X & 0 & \gamma & -\gamma \\ 0 & -X & \gamma Y & -\gamma \end{pmatrix}. \quad (\text{G50})$$

Its determinant is

$$\det \mathbf{R}_{\text{hc}}(\mathbf{k}) = \frac{\gamma^4}{X^2 Y^2} (X-1)^2 (Y-1)^2 (X-Y)^2. \quad (\text{G51})$$

Thus the null set consists of the three symmetry-related lines

$$X = 1, \quad Y = 1, \quad X = Y. \quad (\text{G52})$$

On each line away from Γ the rank is six, so the nullity is two. At Γ the rank is five, leaving the three global Möbius modes. Therefore

$$N_{\text{lin}} = 3 \times 2(L - 1) + 3 = 6L - 3. \quad (\text{G53})$$

The three null lines are the reciprocal-space representation of the three C_6 -related families of line deformations shown in Fig. 10(b).

6. Triangular lattice: rhombus covering

For the triangular lattice let $\mathbf{a}_1, \mathbf{a}_2$ be primitive translations and define

$$\mathbf{d}_1 = \mathbf{a}_1, \quad \mathbf{d}_2 = \mathbf{a}_2, \quad \mathbf{d}_3 = \mathbf{a}_2 - \mathbf{a}_1. \quad (\text{G54})$$

The four-sublattice state is represented by the parities of the coordinates along $\mathbf{a}_1, \mathbf{a}_2$,

$$A = (0, 0), \quad B = (1, 0), \quad C = (0, 1), \quad D = (1, 1), \quad (\text{G55})$$

with magnetic translations $2\mathbf{a}_1, 2\mathbf{a}_2$.

The three rhombus orientations are generated by the pairs $(\mathbf{d}_1, \mathbf{d}_2)$, $(\mathbf{d}_2, \mathbf{d}_3)$, and $(\mathbf{d}_3, \mathbf{d}_1)$. The full 12×4 constraint matrix can be written as

$$\mathbf{R}_{\text{rh}}(\mathbf{k}) = \begin{pmatrix} \mathbf{R}_{12} \\ \mathbf{R}_{23} \\ \mathbf{R}_{31} \end{pmatrix}, \quad (\text{G56})$$

with

$$\mathbf{R}_{12} = \begin{pmatrix} 1 & -1 & \gamma & -\gamma \\ X & -1 & \gamma X & -\gamma \\ Y & -Y & \gamma & -\gamma \\ XY & -Y & \gamma X & -\gamma \end{pmatrix}, \quad (\text{G57})$$

$$\mathbf{R}_{23} = \begin{pmatrix} 1 & -Y/X & \gamma & -\gamma/X \\ Y & -1 & \gamma & -\gamma \\ Y & -Y/X & \gamma & -\gamma Y/X \\ Y & -Y & \gamma Y & -\gamma \end{pmatrix}, \quad (\text{G58})$$

and

$$\mathbf{R}_{31} = \begin{pmatrix} 1 & -1 & \gamma & -\gamma/X \\ X & -1 & \gamma & -\gamma \\ Y & -Y/X & \gamma & -\gamma \\ Y & -Y & \gamma X & -\gamma \end{pmatrix}. \quad (\text{G59})$$

The determinants of the three 4×4 blocks are

$$\det \mathbf{R}_{12} = \gamma^2(X - 1)^2(Y - 1)^2, \quad (\text{G60a})$$

$$\det \mathbf{R}_{23} = \frac{\gamma^2}{X^2}(X - Y)^2(Y - 1)^2, \quad (\text{G60b})$$

$$\det \mathbf{R}_{31} = \frac{\gamma^2}{X^2}(X - 1)^2(X - Y)^2. \quad (\text{G60c})$$

For any $(X, Y) \neq (1, 1)$ at least one of these determinants is nonzero. Hence

$$\text{rank } \mathbf{R}_{\text{rh}}(\mathbf{k}) = 4, \quad \mathbf{k} \neq 0, \quad (\text{G61})$$

and there are no nonuniform infinitesimal coherent deformations. At Γ all local constraints reduce to the same color equation, the rank drops to one, and the kernel consists of the three global Möbius modes:

$$N_{\text{lin}} = 3. \quad (\text{G62})$$

7. Triangular lattice: claw covering

The alternative triangular-lattice construction contains two four-site motifs centered on every site. In the notation above, the three outer sites of the two motifs are displaced from the center by

$$\begin{aligned} \mathcal{C}_+ &= \{\mathbf{d}_1, \mathbf{d}_3, -\mathbf{d}_2\}, \\ \mathcal{C}_- &= \{\mathbf{d}_2, -\mathbf{d}_1, -\mathbf{d}_3\}. \end{aligned} \quad (\text{G63})$$

These are the two alternating sets of nearest neighbors shown in Fig. 13.

Ordering first the four $\mathcal{C}_+$ constraints and then the four $\mathcal{C}_-$ constraints gives

$$\mathbf{R}_{\text{claw}}(\mathbf{k}) = \begin{pmatrix} 1 & -1 & \gamma/Y & -\gamma/X \\ X & -1 & \gamma & -\gamma/Y \\ 1 & -Y/X & \gamma & -\gamma \\ Y & -1 & \gamma X & -\gamma \\ 1 & -1/X & \gamma & -\gamma/Y \\ 1 & -1 & \gamma X/Y & -\gamma \\ Y & -1 & \gamma & -\gamma/X \\ X & -Y & \gamma & -\gamma \end{pmatrix}. \quad (\text{G64})$$

The absence of nonzero-momentum zero modes follows from three 4×4 minors. With the row numbering of Eq. (G64),

$$\det \mathbf{R}_{\{1,4,6,7\}} = -\frac{\gamma^2}{XY}(X - 1)^2(Y - 1)^2, \quad (\text{G65a})$$

$$\det \mathbf{R}_{\{2,4,6,8\}} = \frac{\gamma^2}{Y^2}(X^2 - Y)(Y - 1)^3, \quad (\text{G65b})$$

$$\det \mathbf{R}_{\{3,4,7,8\}} = \frac{\gamma^2}{X^2}(X - 1)^3(X - Y^2). \quad (\text{G65c})$$

If $X \neq 1$ and $Y \neq 1$, the first minor is nonzero. If $X = 1$ but $Y \neq 1$, the second becomes $-\gamma^2(Y - 1)^4/Y^2$, while if $Y = 1$ but $X \neq 1$, the third becomes $\gamma^2(X - 1)^4/X^2$. Thus

$$\text{rank } \mathbf{R}_{\text{claw}}(\mathbf{k}) = 4 \quad \text{for every } \mathbf{k} \neq 0. \quad (\text{G66})$$

At Γ the rank is one and only the three global Möbius modes remain:

$$N_{\text{lin}} = 3. \quad (\text{G67})$$

It is noteworthy that the rhombus and claw coverings have the same linearized coherent zero-mode space around the four-sublattice state, even though their quantum zero-energy kernels are very different. The number of infinitesimal coherent deformations and the dimension of the full quantum kernel are therefore independent characteristics of the model.

8. Summary

The results are summarized in Table VIII. The dimension and geometry of the null set of $\mathbf{R}(\mathbf{k})$ provide a direct reciprocal-space measure of the coherent flexibility of the four-sublattice state.

The hierarchy is therefore particularly transparent. On the corner-sharing checkerboard and pyrochlore lattices the kernel exists throughout the Brillouin zone, producing an extensive number of infinitesimal deformations. On the fcc, square, and honeycomb lattices it survives only on one-dimensional momentum-space sets, giving $O(L)$ modes and corresponding real-space plane or line deformations. For both triangular-lattice constructions, only the three global Möbius modes remain.

The linearized analysis only describes the tangent space around a given coherent state; it does not determine the full nonlinear zero-mode variety or the quantum kernel. This distinction is already visible on the checkerboard lattice, where the empty-square modes satisfy the constraints to first order but are obstructed at higher orders. On the pyrochlore lattice, by contrast, the identity holonomy of a hexagon allows the corresponding modes to develop into exact continuous zero-mode families. The triangular models provide the opposite example: although the four-sublattice coherent state has no nonuniform infinitesimal deformations, the quantum kernel still contains additional polarized and entangled zero-energy states. Thus the constraint matrix characterizes the infinitesimal coherent degrees of freedom around a chosen ordered state, not the full ground-state degeneracy.

Appendix H: Matching bounds for the polarized sector

This appendix derives the bounds quoted in Sec. VIIC by applying known results from graph theory to the dual graph G . Under the correspondence established there, the polarized product zero modes are counted by the monomer–dimer partition function at unit activity,

$$\mathcal{D}_{\text{match}}(G) = Z_G(1) = \sum_{k \geq 0} m_k(G), \quad (\text{H1})$$

where $m_k(G)$ is the number of matchings containing k edges. For the checkerboard and pyrochlore lattices, G is a 4-regular bipartite graph with $N_t = N/2$ vertices. We can therefore directly apply rigorous lower and upper bounds known for the matching partition function of regular bipartite graphs.

The lower-matching bounds of Friedland–Gurvits and Csikvári [76, 77] give

$$Z_G(1) \geq \left(\frac{19 + 13\sqrt{13}}{34} \right)^{N_t} \simeq 1.3919^N. \quad (\text{H2})$$

Conversely, among 4-regular graphs the matching count is maximized by disjoint unions of $K_{4,4}$ [78]. Since

$$Z_{K_{4,4}}(1) = \sum_{k=0}^4 \binom{4}{k}^2 k! = 1 + 16 + 72 + 96 + 24 = 209, \quad (\text{H3})$$

and each $K_{4,4}$ block contains eight dual-graph vertices,

$$Z_G(1) \leq 209^{N_t/8} = 209^{N/16} \simeq 1.3964^N. \quad (\text{H4})$$

Together, Eqs. (H2) and (H4) give Eq. (7.10).

For the periodic 16-site checkerboard cluster, the dual graph is precisely $K_{4,4}$. The five terms $1 + 16 + 72 + 96 + 24$ in Eq. (H3) are the numbers m_k of matchings with $k = 0, \dots, 4$ occupied edges and therefore count polarized product states with k down spins. These are fixed- J^z states rather than total-spin multiplets. For example, the 16 one-down-spin states consist of the $J = 8$ descendant of the fully polarized multiplet and 15 highest-weight states with $J = 7$, in agreement with the 15 zero-energy $J = 7$ multiplets in Table II.

Appendix I: High-polarization counting of zero modes

Here we explain why different chirality assignments have identical zero-mode counts close to full polarization, as observed in Table II. We consider spin 1/2 and work in a fixed quantization axis. Let k be the number of down spins, so that

$$J^z = \frac{N}{2} - k. \quad (\text{I1})$$

The dimension of this fixed- J^z sector is $\binom{N}{k}$.

For spin 1/2, each local CCI parent term projects onto a single forbidden chiral singlet $|\phi_t\rangle$ on tetrahedron t . Since a four-spin singlet has $J = 0$, $|\phi_t\rangle$ lies entirely in the sector with two up and two down spins. Its contribution to the global k -down-spin sector therefore generates constraint vectors of the form

$$|\phi_t\rangle \otimes |\eta\rangle, \quad (\text{I2})$$

where $|\eta\rangle$ contains the remaining $k - 2$ down spins on the $N - 4$ sites outside tetrahedron t .

TABLE VIII. Linearized coherent deformations around the four-sublattice ordered state. The column “cell” gives the number of spin variables and local constraints in the magnetic unit cell. The “null set” specifies the momenta for which the linearized constraint matrix $\mathbf{R}(\mathbf{k})$ has a nontrivial kernel, and “nullity” gives its dimension away from Γ . The last column gives the total number of complex infinitesimal deformations on a periodic system of L^d magnetic cells. All lattices have three global Möbius modes at Γ .

lattice	sharing	cell	null set	nullity	linear modes
checkerboard	corner	4/2	full zone	2	$N/2 + 1$
pyrochlore	corner	4/2	full zone	2	$N/2 + 1$
fcc	edge	4/8	three $\langle 100 \rangle$ axes	2	$6L - 3$
square	edge	4/4	two axes	2	$4L - 1$
honeycomb	edge	8/8	three C_6 -related lines	2	$6L - 3$
triangular, rhombi	face	4/12	Γ only	—	3
triangular, claws	edge	4/8	Γ only	—	3

For each tetrahedron there are $\binom{N-4}{k-2}$ such vectors. Let $\mathcal{R}_k$ denote their linear span over all tetrahedra and all choices of $|\eta\rangle$. Since the Hamiltonian is a sum of local projectors, the zero-energy subspace in the k -down-spin sector is the orthogonal complement of $\mathcal{R}_k$. Hence

$$\mathcal{N}_0(k) = \binom{N}{k} - \dim \mathcal{R}_k. \quad (I3)$$

The number of vectors generating $\mathcal{R}_k$ is

$$M_k = N_{\text{tet}} \binom{N-4}{k-2}. \quad (I4)$$

Therefore

$$\mathcal{N}_0(k) \geq \binom{N}{k} - N_{\text{tet}} \binom{N-4}{k-2}, \quad (I5)$$

with equality whenever the constraint vectors are linearly independent:

$$\mathcal{N}_0(k) = \binom{N}{k} - N_{\text{tet}} \binom{N-4}{k-2}. \quad (I6)$$

This immediately explains the insensitivity to the chirality pattern in the independent-constraint regime. Changing the local chirality changes the individual forbidden singlets $|\phi_t\rangle$, but not the number of independent constraints. The chirality assignment can affect the zero-mode count only once linear relations develop among constraint vectors belonging to different tetrahedra.

Table II is organized in total-spin multiplets rather than fixed- J^z sectors. If n_J denotes the number of zero-energy multiplets with total spin J , then, for the high-polarization sectors considered here,

$$\mathcal{N}_0(k) = \sum_{J \geq N/2-k} n_J, \quad (I7)$$

because each spin- J multiplet contributes exactly one state at a given J^z whenever $J \geq J^z$.

For the periodic $N = 16$ clusters, the constraints are independent in the two-down-spin sector. Using

$$N_{\text{tet}} = 8, 16, 16, 48 \quad (I8)$$

for the checkerboard, square, honeycomb, and triangular clusters, respectively, Eq. (I6) gives

$$\mathcal{N}_0(2) = 112, 104, 104, 72. \quad (I9)$$

These values agree with the sums of the $J \geq 6$ multiplet counts in Table II and are independent of the chirality assignment.

For $k = 3$, the constraints remain independent for the checkerboard, square, and honeycomb clusters, giving

$$\mathcal{N}_0(3) = 464, 368, 368, \quad (I10)$$

in agreement with the sums over $J \geq 5$. For the triangular cluster, by contrast,

$$N_{\text{tet}} \binom{N-4}{k-2} = 48 \times 12 > \binom{16}{3}, \quad (I11)$$

so the constraint vectors must already be linearly dependent.

As the polarization is lowered further, nontrivial linear relations among constraints become increasingly important. Their rank can then depend on the relative chiralities of the local parent terms, and the uniform and alternating spectra begin to separate. For the 16-site square cluster this first occurs at $J = 4$, where the alternating and uniform Hamiltonians have 453 and 452 zero-energy multiplets, respectively. For the honeycomb cluster the spectra remain identical through $J = 4$ and first differ at $J = 3$, with 478 and 476 multiplets. Thus the chirality dependence of the low-spin spectrum can be understood as a rank-deficiency effect among overlapping local constraints.

This counting should be distinguished from the matching construction of Sec. VII C. A matching imposes the stronger condition that no tetrahedron contain two down spins and therefore produces individual product zero

modes. The rank counting above instead implements the full quantum constraint: configurations with two down spins on the same tetrahedron are allowed as long as their local wave function is orthogonal to the forbidden chiral singlet. The matching states therefore span only a subset of the high-polarization zero-energy sector. For example, on the 16-site checkerboard cluster, the matching construction gives $m_2 = 72$ two-down-spin product states, whereas the complete $k = 2$ zero-energy sector has dimension $\mathcal{N}_0(2) = 112$.

Appendix J: Anderson towers for tetrahedral magnetic order

We determine the symmetry content of the Anderson tower associated with tetrahedral magnetic order, following the finite-size symmetry analysis of Refs. [58, 59]. A finite-spin realization contains one additional ingredient: the quantum reference state may transform with a one-dimensional phase under the combined lattice and spin rotations that leave the classical tetrahedral configuration invariant. This phase does not change the tower multiplicities, but it can shift their discrete lattice-symmetry labels.

1. Single-chirality A_4 tower

A chirality-selecting interaction leaves the proper tetrahedral group $T \simeq A_4$ of the 12 even permutations as the relevant discrete symmetry of a single tetrahedral orientation. Its conjugacy classes are

$$\{e\}, \quad 4C_3^+, \quad 4C_3^-, \quad 3C_2,$$

where the two sets of threefold rotations are distinct conjugacy classes in A_4 . We choose C_3^+ to contain the cycle $\hat{P}_{(123)}$ used in Eq. (3.10). The irreducible representations are the singlets $A, E_{\pm}$, with $E_{\pm}$ a complex-conjugate pair, and the triplet T . Their characters are listed in Table IX(a).

Let $\{\mathcal{S}_a^{(0)}\}_{a=1}^4$ be a reference tetrahedral configuration. For every $g \in T$, the permutation of the four sublattices can be compensated by a global spin rotation R_g . The combined operation therefore leaves the classical ordered configuration invariant.

For the corresponding quantum coherent reference state, however, the same operation need only return the state to the same ray,

$$\hat{U}(R_g)\hat{P}_g|\Psi\rangle = \eta(g)|\Psi\rangle, \quad g \in T, \quad (J1)$$

where η is a one-dimensional representation of A_4 ,

$$\eta \in \{A, E_+, E_-\}. \quad (J2)$$

The origin of this phase is purely quantum mechanical. The operation (R_g, g) returns every classical spin direction to itself, but the quantum state can acquire a phase.

TABLE IX. Character tables relevant to tetrahedral order. (a) The proper tetrahedral group $T \simeq A_4$, with $\omega = e^{2\pi i/3}$ and C_3^+ chosen to contain the cycle (123). (b) The full permutation group $S_4 \simeq T_d$. In passing from A_4 to S_4 , the classes $4C_3^{\pm}$ merge into the single class of eight three-cycles; the last two columns contain the odd permutations, which exchange the two chiralities.

(a) $A_4 \simeq T$					
	e	$4C_3^+$	$4C_3^-$	$3C_2$	
A	1	1	1	1	
E_+	1	ω	ω^2	1	
E_-	1	ω^2	ω	1	
T	3	0	0	-1	

(b) $S_4 \simeq T_d$					
	e	8(123)	3(12)(34)	6(1234)	6(12)
A_1	1	1	1	1	1
A_2	1	1	1	-1	-1
E	2	-1	2	0	0
T_1	3	0	-1	1	-1
T_2	3	0	-1	-1	1

Equivalently, η specifies a twisted boundary condition for the collective rotor under the discrete tetrahedral identifications of the classical order-parameter manifold.

The collective-coordinate Hilbert space carries the induced representation

$$\mathcal{H}_{\text{tower}}^{(\eta)} = \text{Ind}_{T_{\text{diag}}}^{SO(3)_{\text{spin}} \times T_{\text{lattice}}} \eta, \quad (J3)$$

where

$$T_{\text{diag}} = \{(R_g, g) \mid g \in T\}. \quad (J4)$$

At fixed integer total spin J , Peter–Weyl decomposition, or equivalently Frobenius reciprocity, gives

$$\mathcal{H}_J^{(\eta)} = \mathcal{V}_J \otimes \mathcal{M}_J^{(\eta)}, \quad (J5)$$

where $\mathcal{V}_J$ is the $(2J+1)$ -dimensional spin- J representation and

$$\mathcal{M}_J^{(\eta)} \simeq (D^J \downarrow_T) \otimes \eta. \quad (J6)$$

Here we used the equivalence of the integer-spin representation D^J and its dual. A nontrivial stabilizer character therefore only tensors the tetrahedral lattice representation by a one-dimensional irrep; it cannot change the number of spin- J multiplets.

It is useful first to consider the untwisted case $\eta = A$. Using the standard $SO(3)$ character

$$\chi_J(\theta) = \frac{\sin[(J + \frac{1}{2})\theta]}{\sin(\theta/2)}, \quad (J7)$$

and evaluating it at the tetrahedral rotation angles $\theta = 0, 2\pi/3$, and π , one obtains

$$\chi_J(e) = 2J + 1, \quad (\text{J8a})$$

$$\chi_J(C_3^\pm) = u_J, \quad (\text{J8b})$$

$$\chi_J(C_2) = (-1)^J, \quad (\text{J8c})$$

where

$$u_J = \frac{\sin[(2J+1)\pi/3]}{\sin(\pi/3)} = \begin{cases} 1, & J = 0 \pmod{3}, \\ 0, & J = 1 \pmod{3}, \\ -1, & J = 2 \pmod{3}. \end{cases} \quad (\text{J9})$$

The character projection

$$\nu_J(J) = \frac{1}{12} \sum_{\mathcal{C}} |\mathcal{C}| \chi_{\Gamma}^*(\mathcal{C}) \chi_J(\mathcal{C}) \quad (\text{J10})$$

then yields

$$\nu_A(J) = \frac{2J+1+8u_J+3(-1)^J}{12}, \quad (\text{J11})$$

$$\nu_{E_+}(J) = \nu_{E_-}(J) = \frac{2J+1-4u_J+3(-1)^J}{12}, \quad (\text{J12})$$

$$\nu_T(J) = \frac{2J+1-(-1)^J}{4}. \quad (\text{J13})$$

The dimension identity

$$\nu_A(J) + \nu_{E_+}(J) + \nu_{E_-}(J) + 3\nu_T(J) = 2J+1 \quad (\text{J14})$$

shows that the tower contains $2J+1$ spin- J multiplets, or $(2J+1)^2$ states at fixed J , as required for a rigid rotor on $\text{SO}(3)$.

The untwisted decomposition is listed in Table X. For a general stabilizer character η , every irrep in the untwisted tower is simply tensored by η . Equivalently,

$$\nu_{\Gamma}^{(\eta)}(J) = \nu_{\Gamma \otimes \eta^*}(J). \quad (\text{J15})$$

The relevant tensor products are

$$\mathbf{E}_+ \otimes \mathbf{E}_+ = \mathbf{E}_-, \quad \mathbf{E}_+ \otimes \mathbf{E}_- = \mathbf{A}, \quad \mathbf{E}_{\pm} \otimes \mathbf{T} = \mathbf{T}. \quad (\text{J16})$$

Thus a twist cyclically permutes $\mathbf{A}, \mathbf{E}_+, \mathbf{E}_-$ while leaving the tetrahedral triplet unchanged.

The class characters are periodic in J with period six, giving

$$D^{J+6} \downarrow_T = D^J \downarrow_T \oplus \mathbf{A} \oplus \mathbf{E}_+ \oplus \mathbf{E}_- \oplus 3\mathbf{T}, \quad (\text{J17})$$

where the added term is the regular representation of T .

2. Tower of the full color-ice subspace and S_4 symmetry

Without explicit chirality selection, both enantiomeric tetrahedral configurations belong to the collective order-parameter manifold. Odd permutations of the sublattices

reverse the orientation and exchange the two chiral sectors. If they are symmetries, the discrete symmetry is therefore enlarged from A_4 to $S_4 \simeq T_d$, with $A_4 \triangleleft S_4$ and $S_4/A_4 \simeq \mathbb{Z}_2$.

The A_4 quantum numbers of a single chiral sector then have to be combined into irreducible representations of S_4 . Restricting an S_4 irrep to the even permutations gives

$$\mathbf{A}_{1,2} \downarrow_{A_4} = \mathbf{A}, \quad \mathbf{E} \downarrow_{A_4} = \mathbf{E}_+ \oplus \mathbf{E}_-, \quad \mathbf{T}_{1,2} \downarrow_{A_4} = \mathbf{T}. \quad (\text{J18})$$

Thus $\mathbf{A}_1$ and $\mathbf{A}_2$ are indistinguishable if only even permutations are retained, and similarly both $\mathbf{T}_1$ and $\mathbf{T}_2$ reduce to the same tetrahedral triplet. By contrast, the two complex-conjugate one-dimensional irreps $\mathbf{E}_+$ and $\mathbf{E}_-$ combine into the two-dimensional $\mathbf{E}$ irrep once the odd permutations relating the two chiralities are restored.

Equivalently, starting from an A_4 irrep in one chiral sector and restoring the odd permutations generates the corresponding S_4 representation,

$$\text{Ind}_{A_4}^{S_4} \mathbf{A} = \mathbf{A}_1 \oplus \mathbf{A}_2, \quad (\text{J19a})$$

$$\text{Ind}_{A_4}^{S_4} \mathbf{E}_{\pm} = \mathbf{E}, \quad (\text{J19b})$$

$$\text{Ind}_{A_4}^{S_4} \mathbf{T} = \mathbf{T}_1 \oplus \mathbf{T}_2. \quad (\text{J19c})$$

The induction therefore amounts physically to adjoining the opposite-chirality partner generated by an odd sublattice permutation.

For a stabilizer character η , the lattice representation at fixed J is therefore

$$\mathcal{M}_J^{(\pm, \eta)} = \text{Ind}_{A_4}^{S_4} \mathcal{M}_J^{(\eta)} = \text{Ind}_{A_4}^{S_4} [(D^J \downarrow_{A_4}) \otimes \eta]. \quad (\text{J20})$$

Writing the twisted A_4 multiplicities as $\nu_{\Gamma}^{(\eta)}(J)$ gives

$$\begin{aligned} \mathcal{M}_J^{(\pm, \eta)} = & \nu_A^{(\eta)}(J) (\mathbf{A}_1 \oplus \mathbf{A}_2) \\ & + [\nu_{E_+}^{(\eta)}(J) + \nu_{E_-}^{(\eta)}(J)] \mathbf{E} \end{aligned}$$

TABLE X. Untwisted ($\eta = \mathbf{A}$) multiplicities of the A_4 irreducible representations in the single-chirality tetrahedral Anderson tower. For a nontrivial stabilizer character η , the irreps in each row are tensored by η according to Eq. (J6). The last column gives $\dim \mathcal{M}_J = 2J+1$, the number of spin- J multiplets.

J	ν_A	ν_{E_+}	ν_{E_-}	ν_T	$\dim \mathcal{M}_J$
0	1	0	0	0	1
1	0	0	0	1	3
2	0	1	1	1	5
3	1	0	0	2	7
4	1	1	1	2	9
5	0	1	1	3	11
6	2	1	1	3	13
7	1	1	1	4	15
8	1	2	2	4	17

$$+ \nu_T(J) (\mathbf{T}_1 \oplus \mathbf{T}_2). \quad (\text{J21})$$

Its dimension is independent of the twist,

$$\dim \mathcal{M}_J^{(\pm, \eta)} = 2(2J + 1), \quad (\text{J22})$$

so the full two-chirality tower contains twice as many spin- J multiplets as a single chiral component.

For the untwisted case $\eta = \mathbf{A}$, the first few sectors are

$$\mathcal{M}_0^{(\pm)} = \mathbf{A}_1 \oplus \mathbf{A}_2, \quad (\text{J23})$$

$$\mathcal{M}_1^{(\pm)} = \mathbf{T}_1 \oplus \mathbf{T}_2, \quad (\text{J24})$$

$$\mathcal{M}_2^{(\pm)} = 2\mathbf{E} \oplus \mathbf{T}_1 \oplus \mathbf{T}_2, \quad (\text{J25})$$

$$\mathcal{M}_3^{(\pm)} = \mathbf{A}_1 \oplus \mathbf{A}_2 \oplus 2\mathbf{T}_1 \oplus 2\mathbf{T}_2. \quad (\text{J26})$$

For a nontrivial η , the corresponding S_4 content follows directly from Eq. (J20); the total multiplicity remains $2(2J + 1)$.

3. Relation to the finite-spin color-ice subspaces

The color-ice subspaces introduced above provide finite-spin realizations of the tetrahedral towers. The fixed-chirality subspace $\mathcal{V}^S$ realizes the A_4 tower, while $\mathcal{W}^S$ contains both chiralities and realizes the corresponding S_4 structure.

For $S = 1/2$, the CCI subspace $\mathcal{V}^{1/2}$ contains a unique $J = 0$ state, the chiral singlet $|- \rangle$. Under the three-cycle $C_3^+ = (123)$ it transforms as $\hat{P}_{(123)}|- \rangle = \omega|- \rangle$, and therefore carries the one-dimensional A_4 irrep $\mathbf{E}_+$. Since a spin- S coherent state is the symmetric product of $2S$ identical spin- $1/2$ spinors, its stabilizer character is $\eta_S = \mathbf{E}_+^{\otimes 2S}$. Thus

$$\eta_S = \begin{cases} \mathbf{A}, & 2S = 0 \pmod{3}, \\ \mathbf{E}_+, & 2S = 1 \pmod{3}, \\ \mathbf{E}_-, & 2S = 2 \pmod{3}. \end{cases} \quad (\text{J27})$$

The opposite chiral sector carries the conjugate character η_S^* .

Identifying $J = S_{\text{tet}}$, the fixed-chirality subspace reproduces the complete twisted rotor for $0 \leq J \leq 2S$:

$$\mathcal{V}_J^S \simeq (D^J \downarrow_{A_4}) \otimes \eta_S, \quad \dim \mathcal{V}_J^S = 2J + 1. \quad (\text{J28})$$

Thus Table X gives the rotor multiplicities for every S , while the one-dimensional irreps $\mathbf{A}, \mathbf{E}_+, \mathbf{E}_-$ are cyclically shifted according to Eq. (J27).

The full color-ice subspace is the sum of the two chiral sectors,

$$\mathcal{W}_J^S = \mathcal{V}_{+,J}^S + \mathcal{V}_{-,J}^S. \quad (\text{J29})$$

For $0 \leq J < 2S$ they are linearly independent, so that

$$\mathcal{W}_J^S = \mathcal{V}_{+,J}^S \oplus \mathcal{V}_{-,J}^S \simeq \mathcal{M}_J^{(\pm, \eta_S)}, \quad \dim \mathcal{W}_J^S = 2(2J + 1). \quad (\text{J30})$$

Equivalently, in this range the kernel-dimension relation becomes

$$\dim \mathcal{H}_J^S - \dim \mathcal{H}_J^{S-1} = 4J + 2. \quad (\text{J31})$$

At $J = 2S$ the two finite-spin chiral subspaces begin to overlap, and the $\mathcal{W}_J^S$ multiplicities fall below the rigid-rotor value. The finite-spin spaces subsequently truncate and terminate in the unique fully polarized multiplet at $J = 4S$. These deviations are finite- S effects: for any fixed J , the rotor multiplicities and symmetry content are recovered once S is sufficiently large.

For the four-sublattice triangular clusters, $2\tilde{S} = N/4$, and therefore

$$\eta_{\tilde{S}} = \mathbf{E}_+^{\otimes N/4}. \quad (\text{J32})$$

Thus the $N = 12$ cluster realizes the untwisted tower, $\eta_{3/2} = \mathbf{A}$, whereas the $N = 16$ cluster carries $\eta_2 = \mathbf{E}_+$. This produces the shifted lattice quantum numbers discussed in Sec. IX.

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